



US 20250281653A1

(19) **United States**

(12) **Patent Application Publication**
JARDINE et al.

(10) **Pub. No.: US 2025/0281653 A1**

(43) **Pub. Date: Sep. 11, 2025**

(54) **METHODS FOR TREATING PROSTATE
CANCER WITH
RADIOPHARMACEUTICALS**

on Mar. 11, 2024, provisional application No. 63/647,
921, filed on May 15, 2024, provisional application
No. 63/705,945, filed on Oct. 10, 2024.

(71) Applicant: **ARTBIO, Inc.**, Cambridge, MA (US)

Publication Classification

(72) Inventors: **Victoria Louise JARDINE**, Denham
(GB); **Aur lie BOUZELMAT**, London
(GB)

(51) **Int. Cl.**
A61K 51/04 (2006.01)
A61K 31/519 (2006.01)

(21) Appl. No.: **19/210,753**

(52) **U.S. Cl.**
CPC *A61K 51/0497* (2013.01); *A61K 31/519*
(2013.01); *A61K 51/0406* (2013.01); *A61K*
2121/00 (2013.01)

(22) Filed: **May 16, 2025**

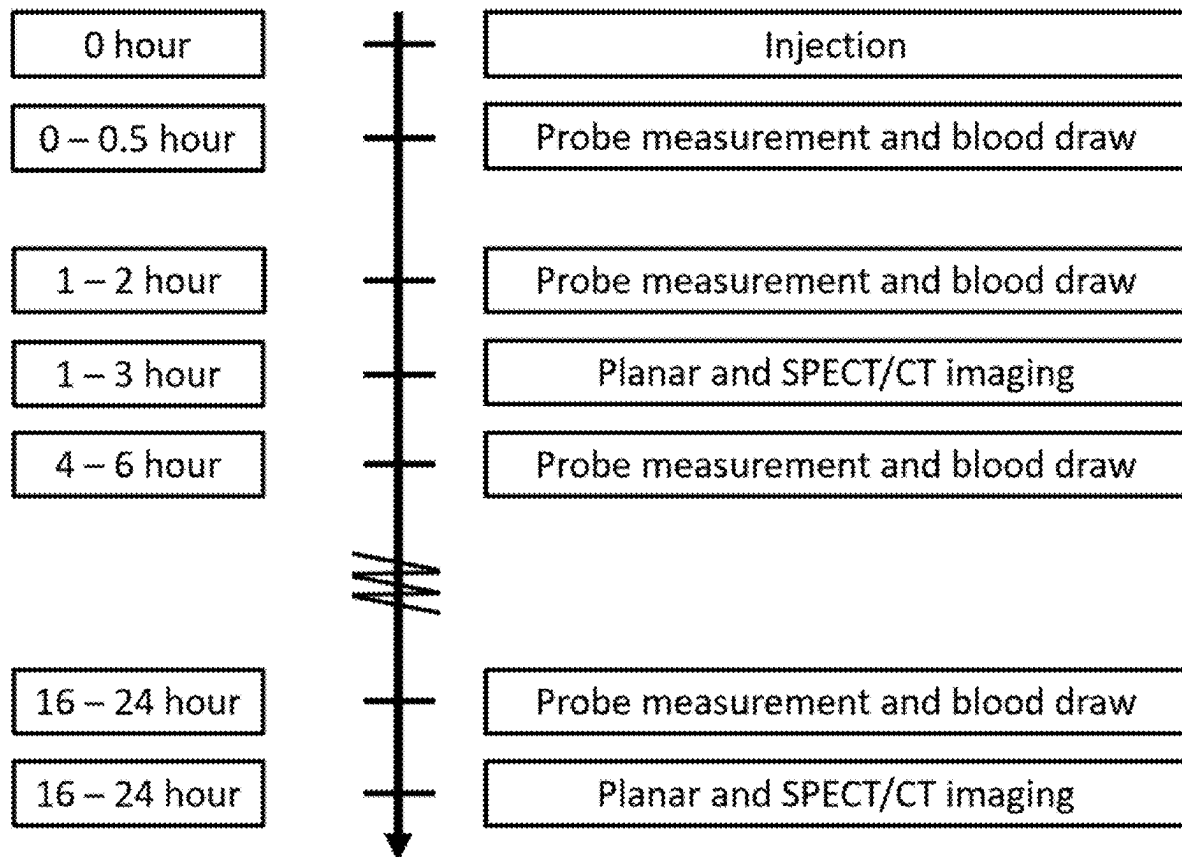
Related U.S. Application Data

(63) Continuation of application No. PCT/US2025/
018987, filed on Mar. 7, 2025.

(60) Provisional application No. 63/563,214, filed on Mar.
8, 2024, provisional application No. 63/563,921, filed

(57) **ABSTRACT**

Provided herein are methods of treating prostate cancer
using radiopharmaceuticals. The radiopharmaceuticals com-
positions can comprise a prostate specific membrane antigen
(PSMA) targeting moiety, a metal chelator, and a radionu-
clide chelated to the metal chelator.



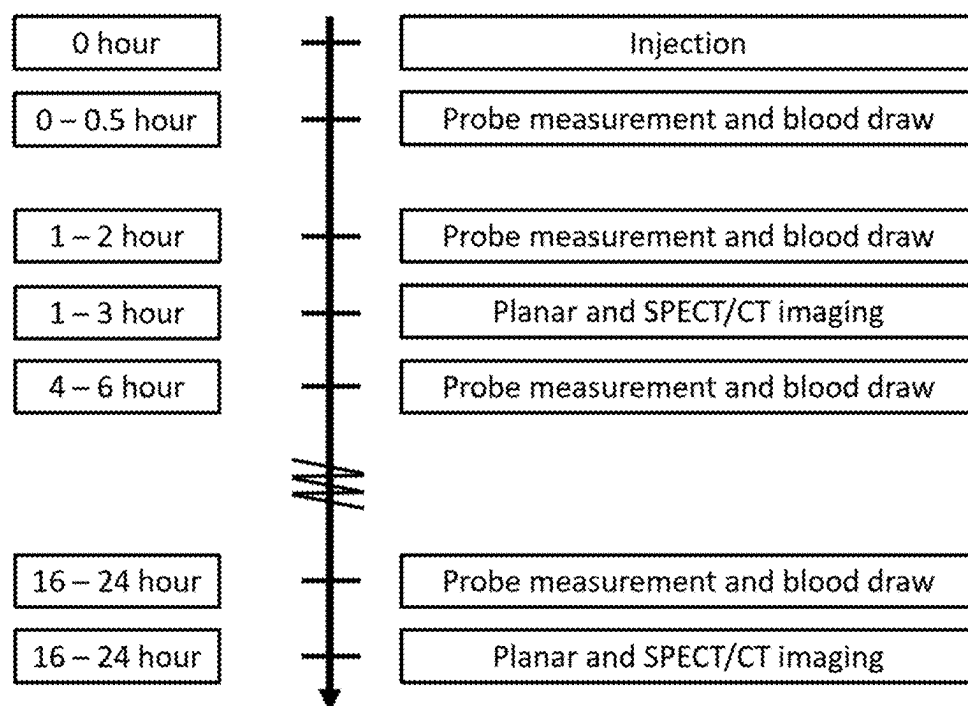


FIG. 1

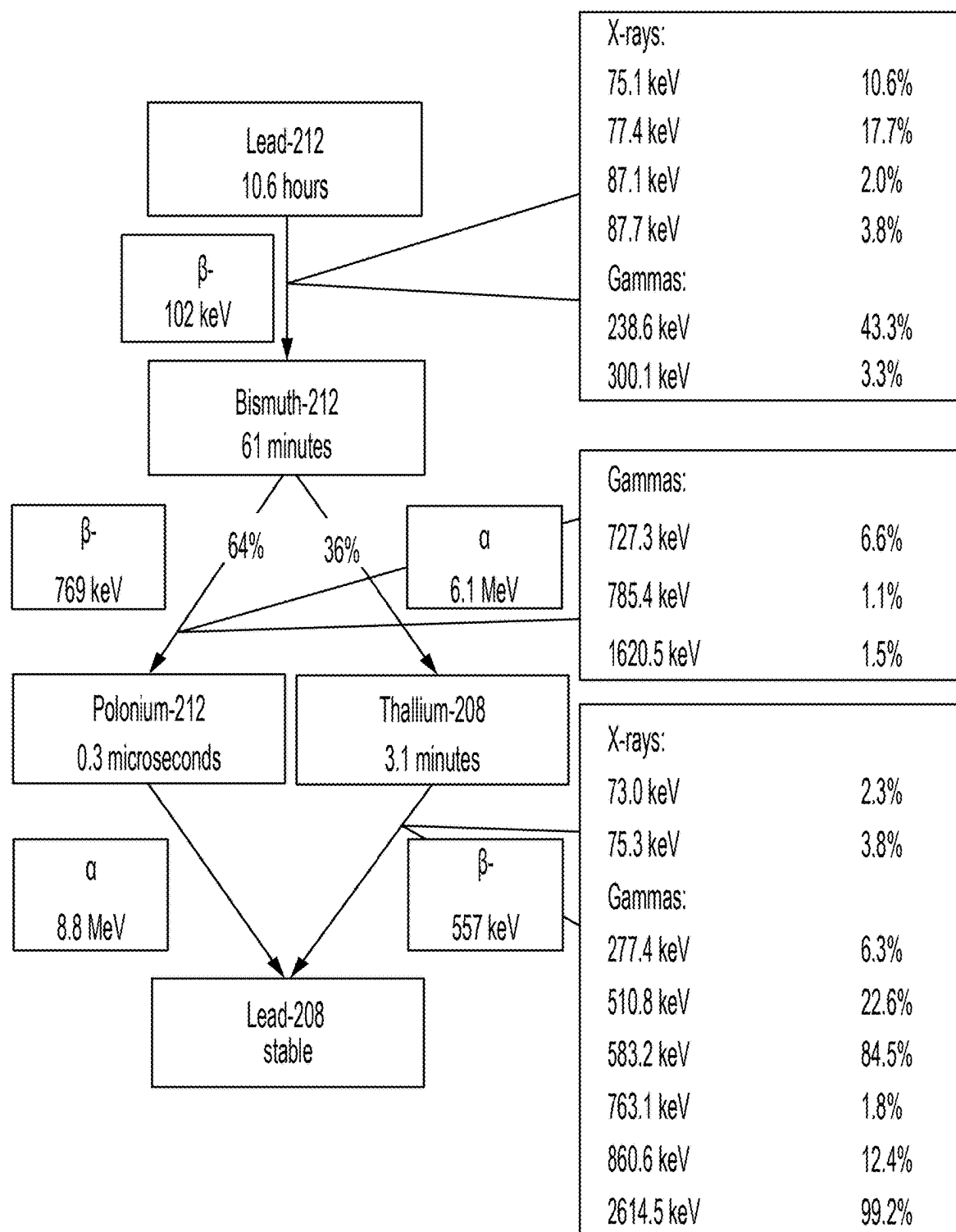


FIG. 2

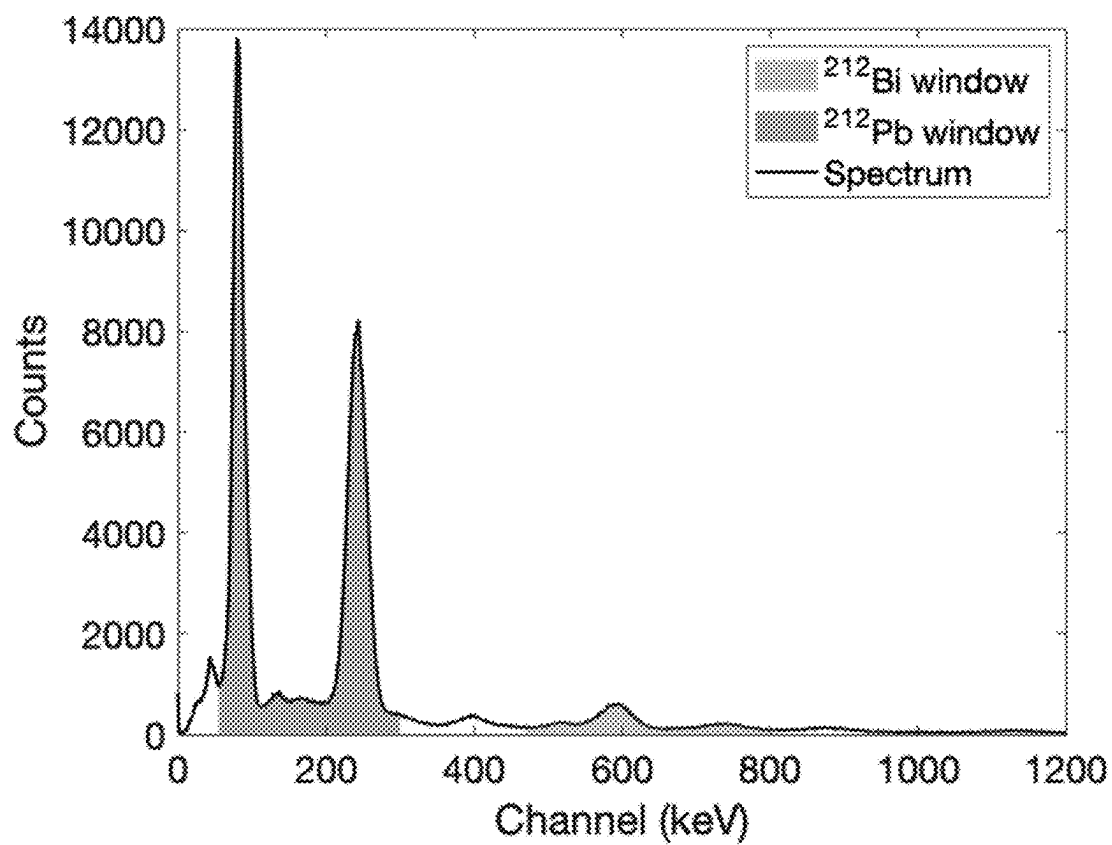


FIG. 3

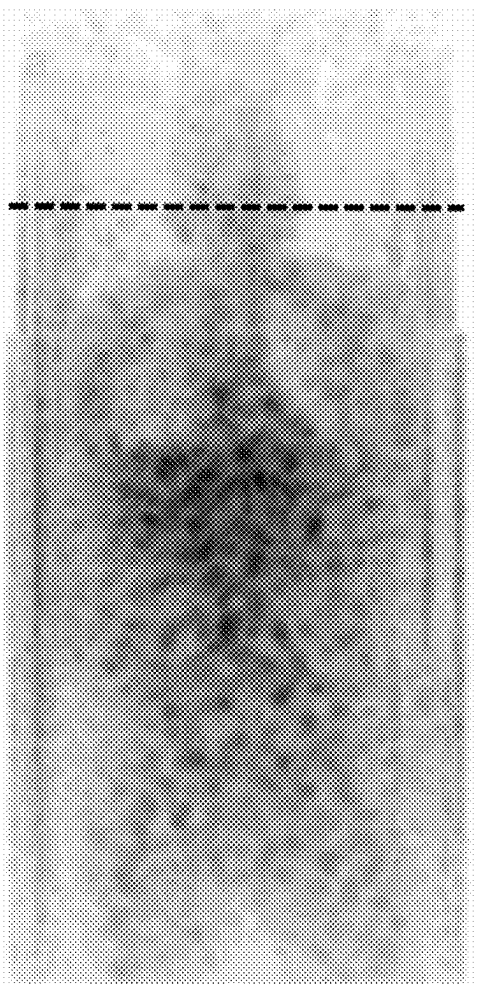


FIG. 4A

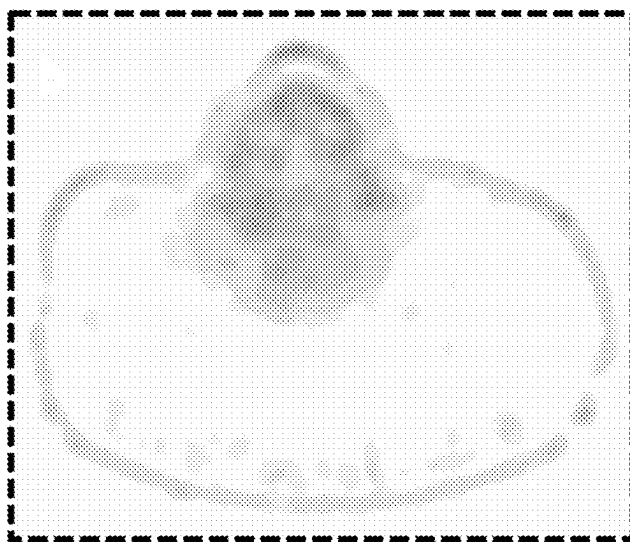
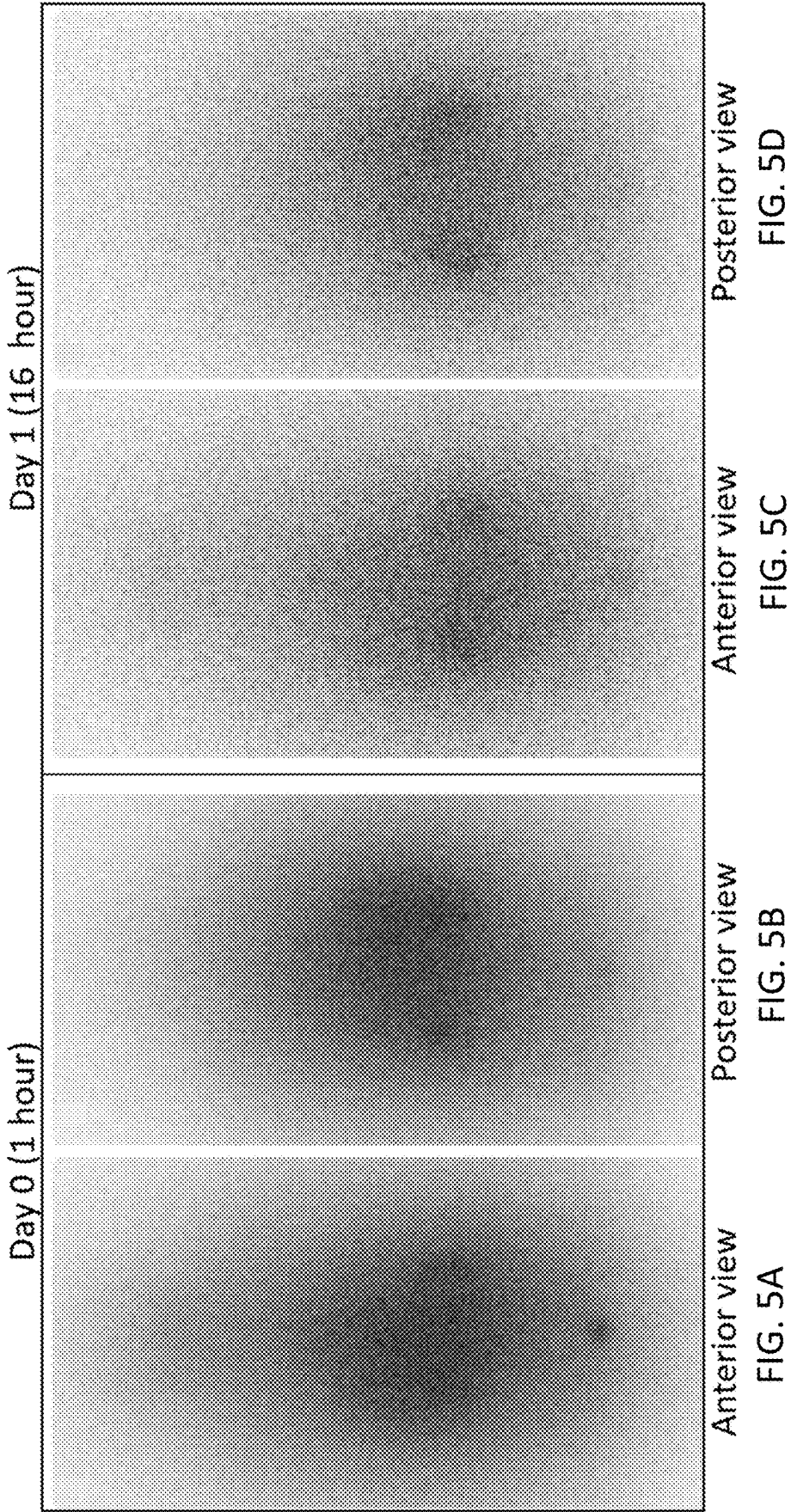
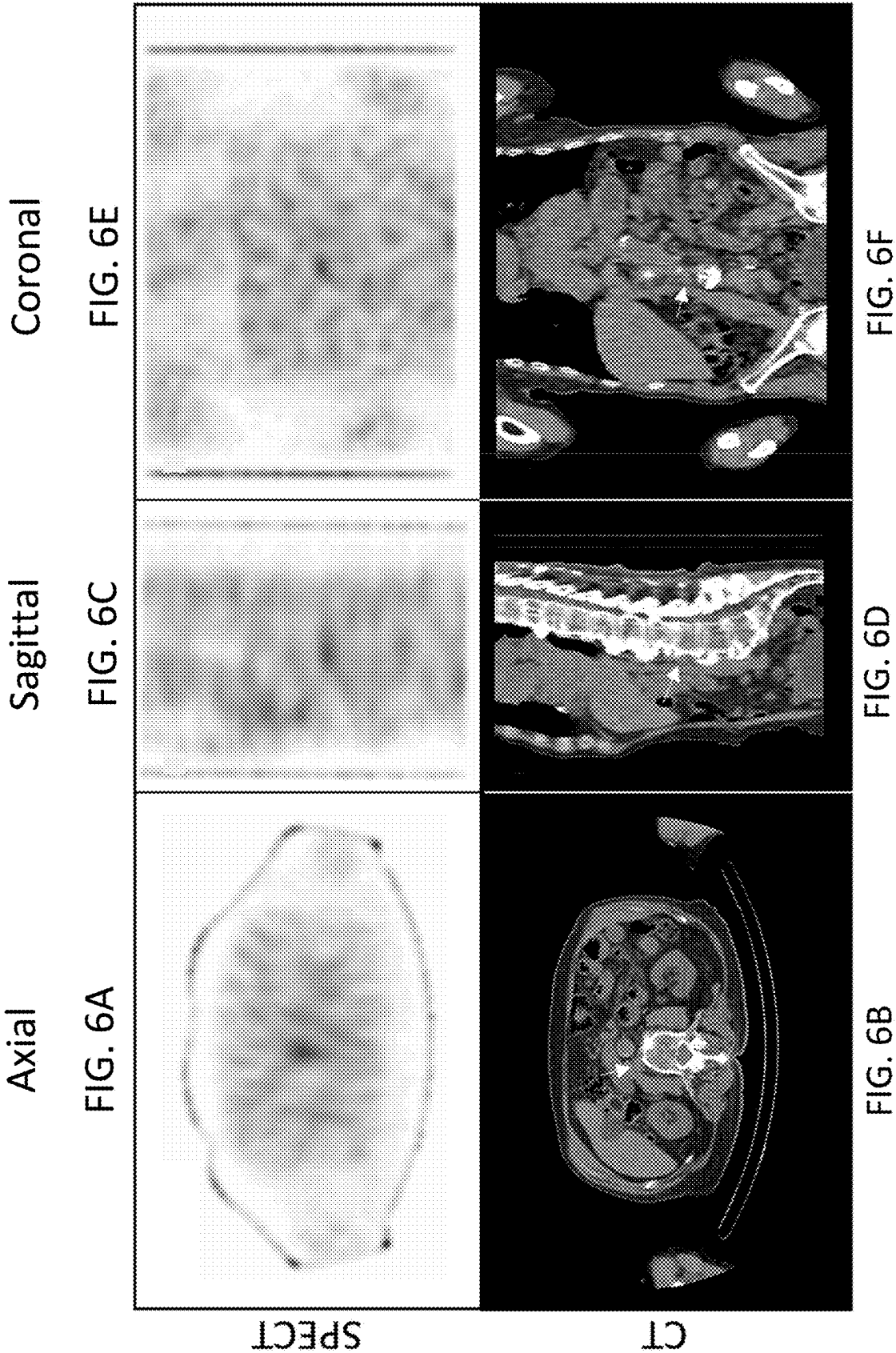
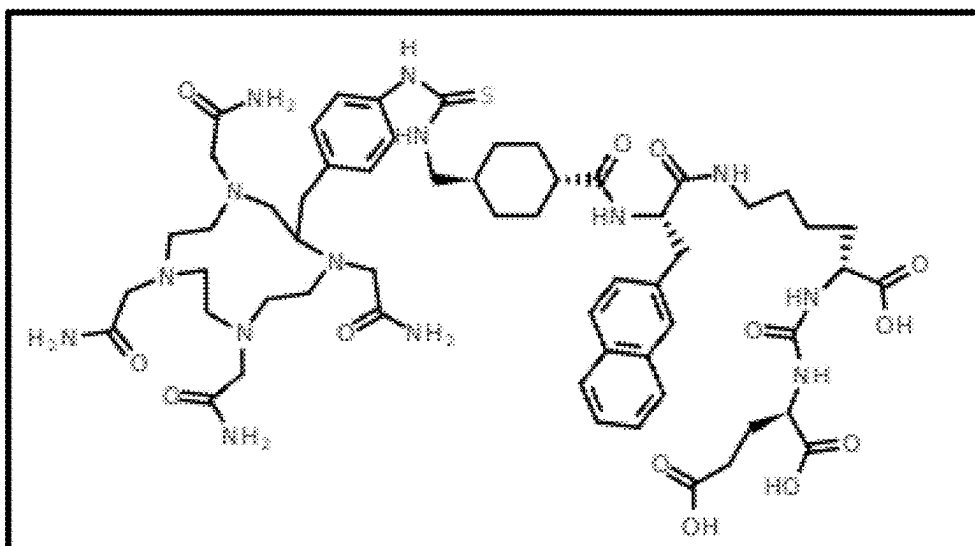


FIG. 4B

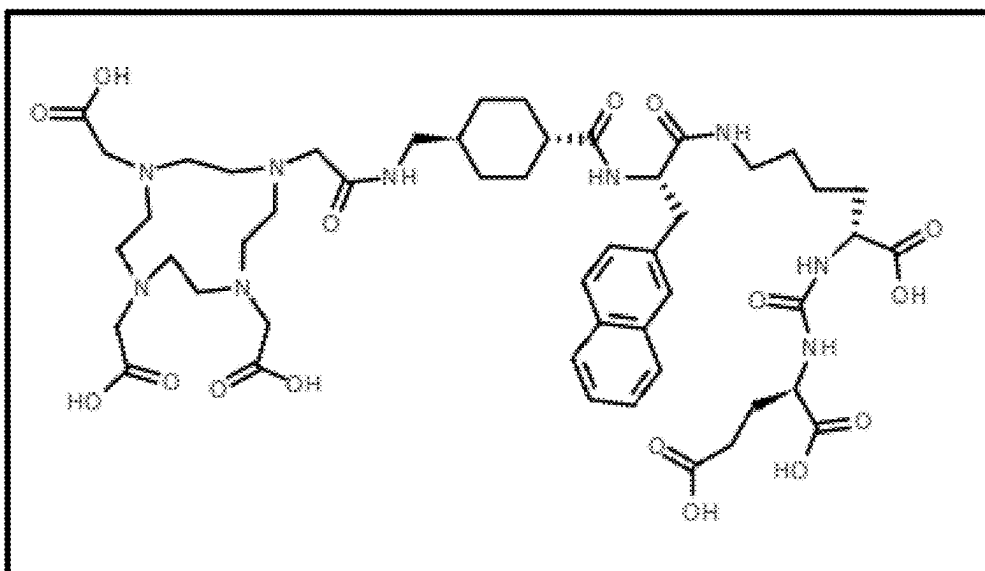






NAME	Compound 1
P450	0.09
MolLogPERME	-6.65
MolEffluxRatioCACO2	0.69
MolLogPERMECACO2	-5.96
bbbScore	0.92
MolPGP	0.99
MolPGPinhibitor	0.25
MolPGPsubstrate	0.99
Formula	C57H82N14O13S
Molecular_weight	1202.59
Num_heavy_atoms	85
Num_rotatable_bonds	36
Num_H_bond_acceptors	29
Num_H_bond_donors	17
TPSA	648.04
Log_P	-4.47
Log_D	-8.03
Log_S	-2.44
Solubility (mg/ml)	4.4
Solubility_Class	Soluble
GI_Adsorption	Moderate
Active_Efflux	Low
BBB_Permearable	No
PGP_Substrate	Yes
PAINS	No
Synthetic_Accessibility	0.80

FIG. 7



NAME	PSMA-617
P450	0.01
MolLogPERME	-6.08
MolEffluxRatioCACO2	0.51
MolLogPERMECACO2	-6.69
bbbScore	1
MolPGP	0.76
MolPGPinhibitor	0.00
MolPGPsubstrate	0.76
Formula	C49H71N9O16
Molecular_weight	1041.50
Num_heavy_atoms	74
Num_rotatable_bonds	32
Num_H_bond_acceptors	30
Num_H_bond_donors	11
TPSA	505.72
Log_P	-6.80
Log_D	-10.36
Log_S	-2.24
Solubility (mg/ml)	5.95
Solubility_Class	Soluble
GI Adsorption	Low
Active_Efflux	Low
BBB_Permearable	No
PGP_Substrate	Yes
PAINS	No
Synthetic Accessibility	1.00

FIG. 8

METHODS FOR TREATING PROSTATE CANCER WITH RADIOPHARMACEUTICALS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of International Application No. PCT/US2025/018987, filed Mar. 7, 2025, which claims the benefit of U.S. Provisional Application No. 63/563,214 filed Mar. 8, 2024, U.S. Provisional Application No. 63/563,921 filed Mar. 11, 2024, U.S. Provisional Application No. 63/647,921 filed May 15, 2024, and U.S. Provisional Application No. 63/705,945 filed Oct. 10, 2024, each of which is incorporated by reference herein in their entirety.

BACKGROUND

[0002] Prostate-specific membrane antigen (PSMA) is a transmembrane glycoprotein found in more than 90% of prostate cancer cells at levels 100-1000 greater than normal cells. Targeted cancer therapy that is highly sensitive and specific to PSMA expressing tissues has the potential to induce tumor cell death while sparing normal tissues. PSMA-targeted radiopharmaceutical therapies are being developed due to the sensitivity of prostate cancer to radiation. The PSMA targeted radiopharmaceutical ^{225}Ac -PSMA-617 was reported to have the potential to achieve a complete prostate cancer remission; however, dose limiting toxicities resulted from undesired salivary gland uptake (Kratochwil et al. *J. Nucl. Med.* 2017; 58:1624-1631). Salivary glands physiologically express PSMA, and targeted therapy accumulation in the salivary glands can be attributed to a combination of both PSMA specific and non-specific uptake mechanisms.

SUMMARY

[0003] Undesired salivary gland uptake of PSMA-targeted radiotherapies can damage the tissue leading to development of xerostomia, a dry mouth, as a critical side effect. Salivary gland damage leading to xerostomia can depend on several factors a) the type of radiation therapy (external beam radiation therapy vs radiotherapeutic agent) b) the radionuclide used (alpha vs beta emission, properties of radionuclide rate of decay) and c) total salivary gland uptake of the PSMA-targeted radiotherapy. Factors b) and c) of these can influence how the administered radioactivity in MBq or GBq relates to the absorbed dose to the salivary glands in Gy, or for alpha emitters in Sv as a relative biological effect (RBE) also needs to be applied to account for the greater potency. For external beam radiation therapy (EBRT), minimal functional impairment occurs at mean doses of <10-15 Gy. Increasing doses in the range of 20-30Gy lead to progressive deterioration. (Gupta et al. *Radiation Oncology*, 2015;

10:67). For radiotherapeutic agents (e.g., with ^{177}Lu or alpha emitters) the full dose/toxicity profile for the salivary glands is still being established, however it is noted as a common or dose limiting toxicity in all small molecule-based agents.

[0004] Administration of 7.4 GBq (200 mCi) of ^{177}Lu -PSMA-617 once every 6 weeks (up to 6 cycles) is estimated to provide 4.5 Gy per cycle (i.e., 27 Gy over 6 cycles) of radiation to the salivary glands. Dry mouth (i.e., xerostomia) occurred in 38.8% of the population of the phase 3 VISION study; all grade 1 or 2 severity (Sartor et al. *NEJM* 2021; 385: 12 1091-1103).

[0005] Multiple early phase/compassionate use studies with ^{225}Ac -PSMA-617 have been published, typically reporting xerostomia as a key toxicity occurring in most/all patients. Severe xerostomia was identified as a dose-limiting above 100 kBq/kg, i.e., approximate 7 MBq for a 70 kg male patient (Kratochwil et al. *J. Nucl. Med.* 2017; 58:1624-1631). Salivary gland absorbed dose estimate for 1 MBq of ^{215}Ac -PSMA-617 was 2.3 Sv (using a relative biological effect (RBE) of 5), for a single dose of 7.4 MBq this equates to 17.24 Sv. Typically 4 to 6 doses given in 6 to 8 week cycles are proposed; 4 cycles would equate to 68.96 Sv and 6 cycles to 103.44 Sv.

[0006] Accordingly, there remains a need for the development of PSMA-targeted radiotherapies with reduced or eliminated uptake into salivary glands.

[0007] In one aspect, provided herein is a method of treating a cancer in a subject in need thereof, comprising administering to the subject a radiopharmaceutical conjugate, wherein the radiopharmaceutical conjugate comprises,

[0008] (a) a metal chelator;

[0009] (b) a target-binding moiety; and

[0010] (c) a radionuclide bound to the metal chelator, wherein the radionuclide is ^{212}Pb ; wherein the radiopharmaceutical conjugate is administered to the subject about every 1 to 7 weeks. In some embodiments, the radiopharmaceutical conjugate comprises a structure of Formula (I):



Formula (I)

[0011] or a pharmaceutically acceptable salt thereof, wherein:

[0012] TL is a PSMA targeting ligand comprising a urea;

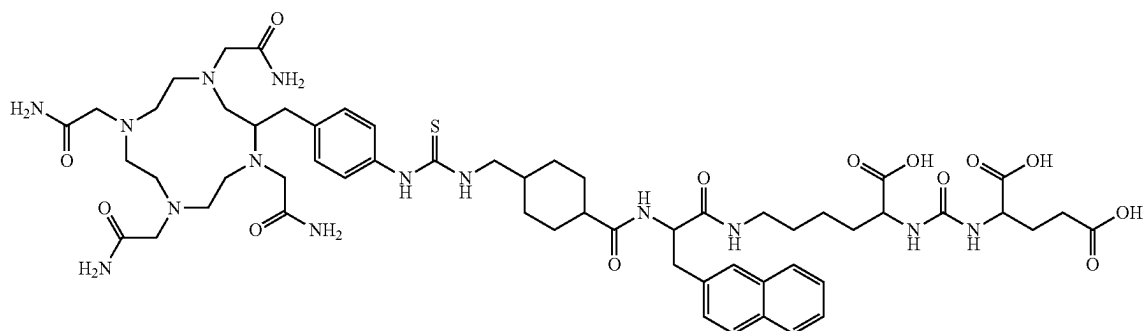
[0013] L is a bivalent linking moiety; and

[0014] R^M is a metal chelator.

[0015] In some embodiments, the cancer is a prostate cancer.

[0016] In one aspect, provided herein is a method of treating a prostate cancer in a subject in need thereof, comprising administering to the subject a radiopharmaceutical conjugate, wherein the radiopharmaceutical conjugate comprises,

[0017] (a) a compound having the structure:



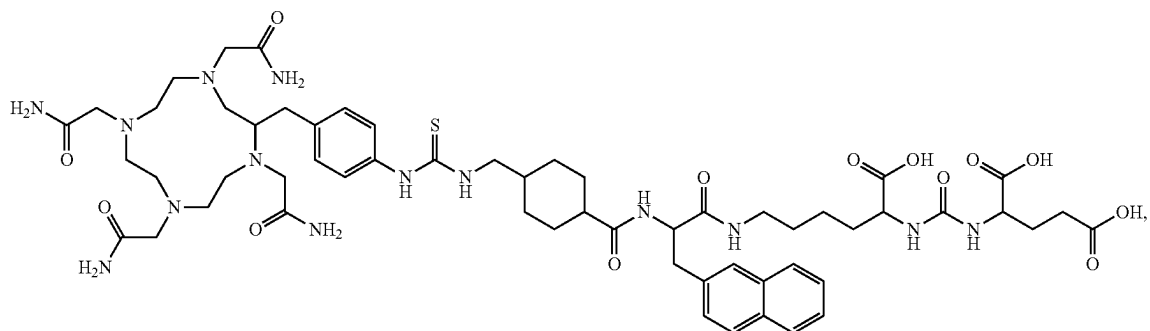
or a pharmaceutically acceptable salt thereof; and

[0018] (b) a radionuclide bound to the metal chelator, wherein the radionuclide is ^{212}Pb ; wherein the radiopharmaceutical conjugate is administered to the subject in an amount between 50 kBq to 500 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 100 kBq to 500 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 1 MBq to 500 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 10 MBq to 500 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 50 MBq to 500 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 10 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 100 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 75 MBq to about 300 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 75 MBq to about 300 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 70 MBq per dose about 75 MBq per dose, about 80 MBq per dose, about 85 MBq per dose, about 90 MBq per dose, about 95 MBq per dose, about 100 MBq per dose, about 105 MBq per dose, about 110 MBq per dose, about 115 MBq per dose, about 120 MBq per dose, about 125 MBq per dose, about 130 MBq per dose, about 140 MBq per dose, about 145 MBq per dose, about 150 MBq per dose, about 155 MBq per dose, about 160 MBq per dose, about 165 MBq per dose, about 170 MBq per dose, about 175 MBq per dose, about 180 MBq per dose, about 185 MBq per dose, about 190 MBq per dose, about 195 MBq per dose, about 200 MBq per dose, about 205 MBq per dose, about 210 MBq per dose, about 215 MBq per dose, about 220 MBq per dose, about 225 MBq per dose, about 230 MBq per dose, about 235 MBq per dose, about 240 MBq per dose, about 245 MBq per dose, about 250

MBq per dose, about 255 MBq per dose, about 260 MBq per dose, about 265 MBq per dose, about 270 MBq per dose, about 275 MBq per dose, about 280 MBq per dose, about 285 MBq per dose, about 290 MBq per dose, about 295 MBq per dose, or about 300 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 75 MBq per dose, about 100 MBq per dose, about 125 MBq per dose, about 150 MBq per dose, about 175 MBq per dose, about 200 MBq per dose, about 225 MBq per dose, about 250 MBq per dose, about 275 MBq per dose, or about 300 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 75 MBq per dose, about 100 MBq per dose, about 150 MBq per dose, about 200 MBq per dose, about 250 MBq per dose, or about 300 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 75 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 100 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 125 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 150 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 175 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 200 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 225 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 250 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 275 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 300 MBq per dose.

[0019] In one aspect, provided herein is a method of treating a prostate cancer in a subject in need thereof, comprising administering to the subject a radiopharmaceutical conjugate, wherein the radiopharmaceutical conjugate comprises,

[0020] (a) a compound having the structure:

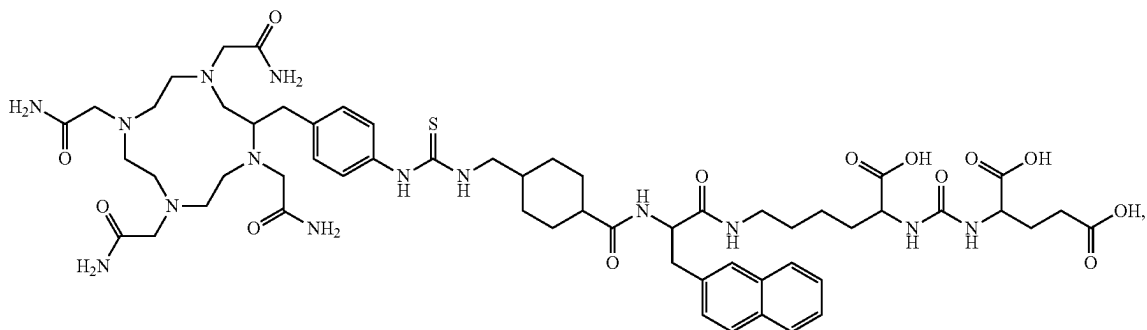


or a pharmaceutically acceptable salt thereof; and

[0021] (b) a radionuclide bound to the metal chelator, wherein the radionuclide is ^{212}Pb ; wherein the radiopharmaceutical conjugate is administered without an agent for treating xerostomia.

[0022] In one aspect, provided herein is a method of treating a prostate cancer in a subject in need thereof, comprising administering to the subject a radiopharmaceutical conjugate, wherein the radiopharmaceutical conjugate comprises,

[0023] (a) a compound having the structure:

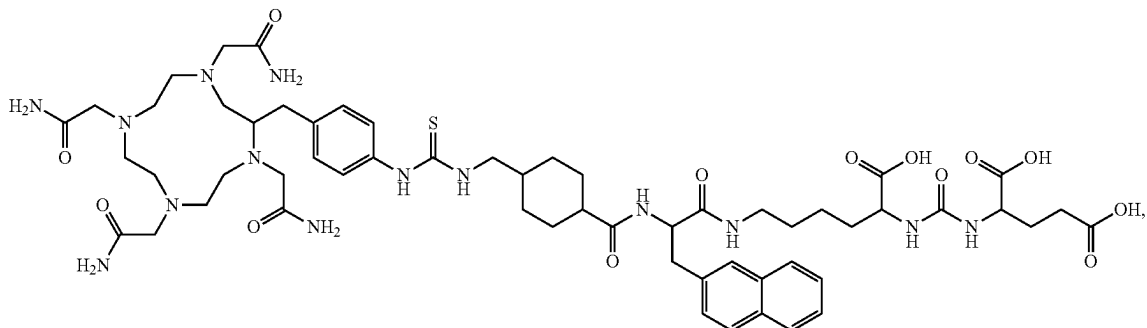


or a pharmaceutically acceptable salt thereof; and

[0024] (b) a radionuclide bound to the metal chelator, wherein the radionuclide is ^{212}Pb ; wherein the subject has previously received an anti-cancer agent before the administering of the radiopharmaceutical conjugate

[0025] In one aspect, provided herein is a method of treating a prostate cancer in a subject in need thereof, comprising administering to the subject a radiopharmaceutical conjugate, wherein the radiopharmaceutical conjugate comprises,

[0026] (a) a compound having the structure:



or a pharmaceutically acceptable salt thereof; and

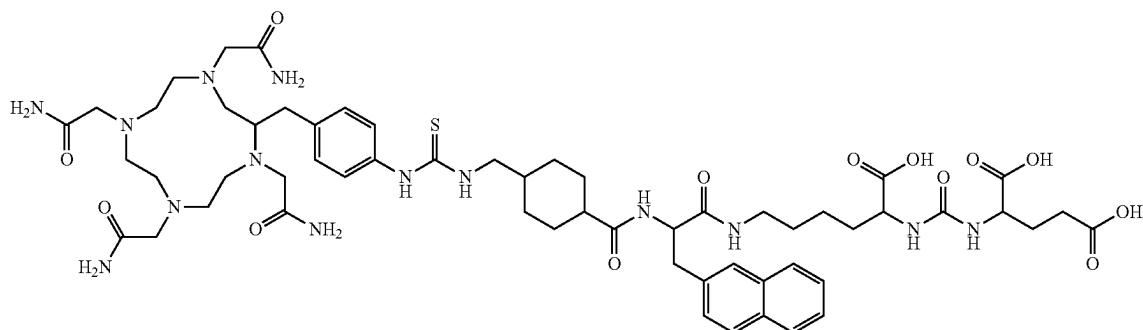
[0027] (b) a radionuclide bound to the metal chelator, wherein the radionuclide is ^{212}Pb ; wherein the radiopharmaceutical conjugate is administered in combination with a second anti-cancer agent.

[0028] In one aspect, provided herein is a method of treating a prostate cancer in a subject in need thereof, comprising administering

[0029] (a) an imaging radiopharmaceutical agent comprising a first radionuclide; and

[0030] (b) a radiopharmaceutical conjugate comprising:

[0031] (i) a compound having the structure:

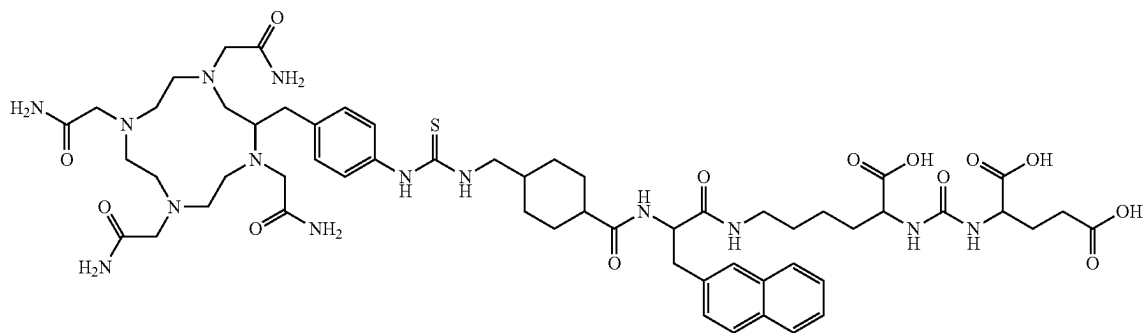


or a pharmaceutically acceptable salt thereof; and

[0032] (ii) a second radionuclide bound to the metal chelator, wherein the second radionuclide is ^{212}Pb ; wherein the imaging radiopharmaceutical agent is administered to the patient prior to the administering of the radiopharmaceutical conjugate.

[0033] In one aspect, provided herein is a liquid radiopharmaceutical composition comprising

[0034] (a) a compound having the structure:



or a pharmaceutically acceptable salt thereof; and

[0035] (b) a radionuclide bound to the metal chelator, wherein the radionuclide is ^{212}Pb , wherein the liquid radiopharmaceutical composition has a radioactivity of about 75 MBq to about 300 MBq.

INCORPORATION BY REFERENCE

[0036] All publications, patents, and patent applications mentioned in this specification are herein incorporated by reference for the specific purposes identified herein.

BRIEF DESCRIPTION OF THE DRAWINGS

[0037] The novel features of the disclosure are set forth with particularity in the appended claims. A better understanding of the features and advantages of the present disclosure will be obtained by reference to the following detailed description that sets forth illustrative embodiments, in which the principles of the disclosure are utilized, and the accompanying drawing (also “figure” and “FIG.” herein), of which:

[0038] FIG. 1 illustrates timepoints of planar and SPECT imaging, whole-body probe measurements and blood sample draws.

[0039] FIG. 2 illustrates the decay chain of ^{212}Pb including energy of X-rays and Gamma emissions. See From: Kvassheim, M., Revheim, ME. R. & Stokke, C. Quantitative SPECT/CT imaging of lead-212: a phantom study. EJNMMI Phys 9, 52 (2022).

[0040] FIG. 3 illustrates the energy spectra indicating ^{212}Bi and ^{212}Pb windows.

[0041] FIG. 4A-4B illustrates the SPECT/CT imaging in participant NO-01-001: normal tissue. SPECT imaging following ^{212}Pb -Compound 1 administration illustrating normal tissue for participant NO-01-001 on Day 0 (1 hour timepoint): FIG. 4A illustrates whole body MIP (anterior view); and FIG. 4B illustrates axial section through the cranium at the level of the parotid glands. ^{212}Pb -Compound 1 uptake could not be visualized in the parotid glands.

[0042] FIG. 5A-5D illustrates the planar gamma imaging in participant N0-01-002: normal tissue. Planar gamma

imaging following ^{212}Pb -Compound 1 administration illustrating normal tissue for participant NO-01-002 on Day 0 (FIG. 5A and FIG. 5B illustrate anterior and posterior views, respectively, at the 1 hour timepoint post injection) and Day 1 (FIG. 5C and FIG. 5D illustrate anterior and posterior views respectively at the 16 hour timepoint post injection).

[0043] FIG. 6A-6F illustrates the SPECT/CT imaging in participant NO-01-003; abdominal retrocaval lymph node. Paired SPECT (summed) (FIG. 6A, FIG. 6C, and FIG. 6E) and non-contrast enhanced CT scans (FIG. 6B, FIG. 6D, and FIG. 6F) from the SPECT/CT for participant NO-01-003 on Day 0 (1 hour timepoint): FIG. 6A and FIG. 6B illustrate axial section; FIG. 6C and FIG. 6D illustrate sagittal section; [0044] FIG. 6E and FIG. 6F illustrate coronal section. The retrocaval lymph node metastasis (11 mm short axis) is indicated with white arrows on the CT images, with anatomical location corresponding with the area of increased ^{212}Pb -Compound 1 uptake (dark) on each of the SPECT sections.

[0045] FIG. 7 depicts the predicted DMPK properties for Compound 1.

[0046] FIG. 8 depicts the predicted DMPK properties for PSMA-617.

DETAILED DESCRIPTION

[0047] The following description and examples illustrate embodiments of the present disclosure in detail. It is to be understood that this present disclosure is not limited to the particular embodiments described herein and as such can vary. Those of skill in the art will recognize that there are numerous variations and modifications of this present disclosure, which are encompassed within its scope.

[0048] Although various features of the present disclosure may be described in the context of a single embodiment, the features may also be provided separately or in any suitable combination. Conversely, although the present disclosure may be described herein in the context of separate embodiments for clarity, the present disclosure may also be implemented in a single embodiment.

[0049] The section headings used herein are for organizational purposes only and are not to be construed as limiting the subject matter described.

[0050] All terms are intended to be understood as they would be understood by a person skilled in the art. Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which the disclosure pertains.

[0051] The following definitions supplement those in the art and are directed to the current application and are not to be imputed to any related or unrelated case, e.g., to any commonly owned patent or application. Although any methods and materials similar or equivalent to those described herein can be used in the practice for testing of the present disclosure, the preferred materials and methods are described herein. Accordingly, the terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting.

I. Definitions

[0052] As used in the specification and appended claims, unless specified to the contrary, the following terms have the meaning indicated below.

[0053] As used herein and in the appended claims, the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “an agent” includes a plurality of such agents.

[0054] The term “about” or “approximately” can mean within an acceptable error range for the particular value as determined by one of ordinary skill in the art, which will depend in part on how the value is measured or determined, i.e., the limitations of the measurement system. For example, “about” can mean within 1 or more than 1 standard deviation, per the practice in the art. Alternatively, “about” can mean a range of up to 20%, up to 15%, up to 10%, up to 5%, or up to 1% of a given value. Alternatively, particularly with respect to biological systems or processes, the term can mean within an order of magnitude, within 5-fold, or within 2-fold, of a value.

[0055] The term “comprising” (and related terms such as “comprise” or “comprises” or “having” or “including”) is not intended to exclude that in other certain embodiments, for example, an embodiment of any composition of matter, composition, method, or process, or the like, described herein, “consist of” or “consist essentially of” the described features.

[0056] “Amino” refers to the $-\text{NH}_2$ radical.

[0057] “Nitro” refers to the $-\text{NO}_2$ radical.

[0058] “Oxo” refers to the $=\text{O}$ radical.

[0059] “Imino” refers to the $=\text{N}-\text{H}$ radical.

[0060] “Hydroxy” or “hydroxyl” refers to the $-\text{OH}$ radical.

[0061] “Alkyl” refers to an optionally substituted straight-chain, or optionally substituted branched-chain saturated hydrocarbon monoradical. An alkyl group can have from one to about twenty carbon atoms, from one to about ten carbon atoms, or from one to six carbon atoms. Examples include, but are not limited to, methyl, ethyl, n-propyl, isopropyl, 2-methyl-1-propyl, 2-methyl-2-propyl, 2-methyl-1-butyl, 3-methyl-1-butyl, 2-methyl-3-butyl, 2,2-dimethyl-1-propyl, 2-methyl-1-pentyl, 3-methyl-1-pentyl, 4-methyl-1-pentyl, 2-methyl-2-pentyl, 3-methyl-2-pentyl, 4-methyl-2-pentyl, 2,2-dimethyl-1-butyl, 3,3-dimethyl-1-butyl, 2-ethyl-1-butyl, n-butyl, isobutyl, sec-butyl, t-butyl, n-pentyl, isopentyl, neopentyl, tert-amyl, and hexyl, and longer alkyl groups, such as heptyl, octyl, and the like. Whenever it appears herein, a numerical range such as “ $\text{C}_1\text{-C}_6$ alkyl” means that the alkyl group consists of 1 carbon atom, 2 carbon atoms, 3 carbon atoms, 4 carbon atoms, 5 carbon atoms or 6 carbon atoms, although the present definition also covers the occurrence of the term “alkyl” where no numerical range is designated. In some embodiments, the alkyl is a $\text{C}_1\text{-C}_{10}$ alkyl, a $\text{C}_1\text{-C}_9$ alkyl, a $\text{C}_1\text{-C}_8$ alkyl, a $\text{C}_1\text{-C}_7$ alkyl, a $\text{C}_1\text{-C}_6$ alkyl, a $\text{C}_1\text{-C}_5$ alkyl, a $\text{C}_1\text{-C}_4$ alkyl, a $\text{C}_1\text{-C}_3$ alkyl, a $\text{C}_1\text{-C}_2$ alkyl, or a C_1 alkyl. Unless stated otherwise specifically in the specification, an alkyl group is optionally substituted, for example, with oxo, halogen, amino, nitrile, nitro, hydroxyl, haloalkyl, alkoxy, aryl, cycloalkyl, heterocycloalkyl, heteroaryl, and the like. In some embodiments, the alkyl is optionally substituted with oxo, halogen, $-\text{CN}$, $-\text{CF}_3$, $-\text{OH}$, $-\text{OMe}$, $-\text{NH}_2$, $-\text{NO}_2$, or $-\text{C}\equiv\text{CH}$. In some embodiments, the alkyl is optionally substituted with oxo, halogen, $-\text{CN}$, $-\text{CF}_3$, $-\text{OH}$, or $-\text{OMe}$. In some embodiments, the alkyl is optionally substituted with halogen. The term “alkylene” refers to a bivalent alkyl.

[0062] The term “aryl” refers to a radical comprising at least one aromatic ring wherein each of the atoms forming the ring is a carbon atom. Aryl groups can be optionally substituted. Examples of aryl groups include, but are not limited to phenyl, and naphthyl. In some embodiments, the aryl is phenyl. Depending on the structure, an aryl group can be a monoradical or a diradical (i.e., an arylene group). Unless stated otherwise specifically in the specification, the term “aryl” or the prefix “ar-” (such as in “aralkyl”) is meant to include aryl radicals that are optionally substituted. In some embodiments, an aryl group comprises a partially reduced cycloalkyl group defined herein (e.g., 1,2-dihydronaphthalene). In some embodiments, an aryl group comprises a fully reduced cycloalkyl group defined herein (e.g., 1,2,3,4-tetrahydronaphthalene). When aryl comprises a cycloalkyl group, the aryl is bonded to the rest of the molecule through an aromatic ring carbon atom. An aryl radical can be a monocyclic or polycyclic (e.g., bicyclic, tricyclic, or tetracyclic) ring system, which may include fused, spiro or bridged ring systems. Unless stated otherwise specifically in the specification, an aryl may be optionally substituted, for example, with halogen, amino, alkylamino, aminoalkyl, nitrile, nitro, hydroxyl, alkyl, alkenyl, alkynyl, haloalkyl, heteroalkyl, alkoxy, aryl, cycloalkyl, heterocycloalkyl, heteroaryl, $-\text{S}(\text{O})_2\text{NH}-\text{C}_1-\text{C}_6\text{alkyl}$, and the like. In some embodiments, an aryl is optionally substituted with halogen, methyl, ethyl, $-\text{CN}$, $-\text{CF}_3$, $-\text{OH}$, $-\text{OMe}$, $-\text{NH}_2$, $-\text{NO}_2$, $-\text{S}(\text{O})_2\text{NH}_2$, $-\text{S}(\text{O})_2\text{NHCH}_3$, $-\text{S}(\text{O})_2\text{NHCH}_2\text{CH}_3$, $-\text{S}(\text{O})_2\text{NHCH}(\text{CH}_3)_2$, $-\text{S}(\text{O})_2\text{N}(\text{CH}_3)_2$, or $-\text{S}(\text{O})_2\text{NHC}(\text{CH}_3)_3$. In some embodiments, an aryl is optionally substituted with halogen, methyl, ethyl, $-\text{CN}$, $-\text{CF}_3$, $-\text{OH}$, or $-\text{OMe}$. In some embodiments, the aryl is optionally substituted with halogen. In some embodiments, the aryl is substituted with alkyl, alkenyl, alkynyl, haloalkyl, or heteroalkyl, wherein each alkyl, alkenyl, alkynyl, haloalkyl, heteroalkyl is independently unsubstituted, or substituted with halogen, methyl, ethyl, $-\text{CN}$, $-\text{CF}_3$, $-\text{OH}$, $-\text{OMe}$, $-\text{NH}_2$, or $-\text{NO}_2$. The term “arylene” refers to a bivalent aryl.

[0063] The term “cycloalkyl” refers to a monocyclic or polycyclic non-aromatic radical, wherein each of the atoms forming the ring (i.e., skeletal atoms) is a carbon atom. In some embodiments, cycloalkyls are saturated or partially unsaturated. In some embodiments, cycloalkyls are spirocyclic or bridged compounds. In some embodiments, cycloalkyls are fused with an aromatic ring (in which case the cycloalkyl is bonded through a non-aromatic ring carbon atom). Cycloalkyl groups include groups having from 3 to 10 ring atoms. Representative cycloalkyls include, but are not limited to, cycloalkyls having from three to ten carbon atoms, from three to eight carbon atoms, from three to six carbon atoms, or from three to five carbon atoms. Monocyclic cycloalkyl radicals include, for example, cyclopropyl, cyclobutyl, cyclopentyl, cyclohexyl, cycloheptyl, and cyclooctyl. In some embodiments, the monocyclic cycloalkyl is cyclopentyl. In some embodiments, the monocyclic cycloalkyl is cyclopentenyl or cyclohexenyl. In some embodiments, the monocyclic cycloalkyl is cyclopentenyl. Polycyclic radicals include, for example, adamantyl, 1,2-dihydronaphthalenyl, 1,4-dihydronaphthalenyl, tetralinyl, decalinyl, 3,4-dihydronaphthalenyl-1(2H)-one, spiro[2.2]pentyl, norbornyl and bicyclo[1.1.1]pentyl. Unless otherwise stated specifically in the specification, a cycloalkyl group may be optionally substituted. Representative cycloalkyls include, but are not limited to, cycloalkyls having from three to fifteen carbon atoms (C_3-C_{15} cycloalkyl), from three to ten carbon atoms (C_3-C_{10} cycloalkyl), from three to eight

carbon atoms (C_3-C_8 cycloalkyl), from three to six carbon atoms (C_3-C_6 cycloalkyl), from three to five carbon atoms (C_3-C_5 cycloalkyl), or three to four carbon atoms (C_3-C_4 cycloalkyl). A cycloalkyl can comprise a fused, spiro or bridged ring system. In some embodiments, the cycloalkyl comprises a fused ring system. In some embodiments, the cycloalkyl comprises a spiro ring system. In some embodiments, the cycloalkyl comprises a bridged ring system. In some embodiments, the cycloalkyl is a 3- to 6-membered cycloalkyl. In some embodiments, the cycloalkyl is a 5- to 6-membered cycloalkyl. Monocyclic cycloalkyls include, for example, cyclopropyl, cyclobutyl, cyclopentyl, cyclohexyl, cycloheptyl, and cyclooctyl. Polycyclic cycloalkyls or carbocycles include, for example, adamantyl, norbornyl, decalinyl, bicyclo[3.3.0]octane, bicyclo[4.3.0]nonane, cis-decalin, trans-decalin, bicyclo[2.1.1]hexane, bicyclo[2.2.1]heptane, bicyclo[2.2.2]octane, bicyclo[3.2.2]nonane, and bicyclo[3.3.2]decane, and 7,7-dimethyl-bicyclo[2.2.1]heptanyl. Partially saturated cycloalkyls include, for example, cyclopentenyl, cyclohexenyl, cycloheptenyl, and cyclooctenyl. Unless stated otherwise specifically in the specification, a cycloalkyl is optionally substituted, for example, with oxo, halogen, amino, nitrile, nitro, hydroxyl, alkyl, alkenyl, alkynyl, haloalkyl, alkoxy, aryl, cycloalkyl, heterocycloalkyl, heteroaryl, and the like. In some embodiments, a cycloalkyl is optionally substituted with oxo, halogen, methyl, ethyl, $-\text{CN}$, $-\text{CF}_3$, $-\text{OH}$, $-\text{OMe}$, $-\text{NH}_2$, or $-\text{NO}_2$. In some embodiments, a cycloalkyl is optionally substituted with oxo, halogen, methyl, ethyl, $-\text{CN}$, $-\text{CF}_3$, $-\text{OH}$, or $-\text{OMe}$. In some embodiments, the cycloalkyl is optionally substituted with halogen. The term “cycloalkylene” refers to a divalent cycloalkyl.

[0064] “Heteroalkyl” refers to an alkyl group in which one or more skeletal atoms of the alkyl are selected from an atom other than carbon, e.g., oxygen, nitrogen (e.g., $-\text{NH}-$, $-\text{N}(\text{alkyl})-$), sulfur, or combinations thereof. A heteroalkyl is attached to the rest of the molecule at a carbon atom of the heteroalkyl. In one aspect, a heteroalkyl is a C_1-C_6 heteroalkyl wherein the heteroalkyl is comprised of 1 to 6 carbon atoms and one or more atoms other than carbon, e.g., oxygen, nitrogen (e.g., $-\text{NH}-$, $-\text{N}(\text{alkyl})-$), sulfur, or combinations thereof wherein the heteroalkyl is attached to the rest of the molecule at a carbon atom of the heteroalkyl. Examples of such heteroalkyl are, for example, $-\text{CH}_2-\text{O}-\text{CH}_2-$, $-\text{CH}_2-\text{N}(\text{alkyl})-\text{CH}_2-$, $-\text{CH}_2-\text{N}(\text{aryl})-\text{CH}_2-$, $-\text{OCH}_2\text{CH}_2\text{O}-$, $-\text{OCH}_2\text{CH}_2\text{OCH}_2\text{CH}_2\text{O}-$, or $-\text{OCH}_2\text{CH}_2\text{OCH}_2\text{CH}_2\text{OCH}_2\text{CH}_2\text{O}-$. Unless stated otherwise specifically in the specification, a heteroalkyl is optionally substituted for example, with oxo, halogen, amino, nitrile, nitro, hydroxyl, alkyl, alkenyl, alkynyl, haloalkyl, alkoxy, aryl, cycloalkyl, heterocycloalkyl, heteroaryl, and the like. In some embodiments, a heteroalkyl is optionally substituted with oxo, halogen, methyl, ethyl, $-\text{CN}$, $-\text{CF}_3$, $-\text{OH}$, $-\text{OMe}$, $-\text{NH}_2$, or $-\text{NO}_2$. In some embodiments, a heteroalkyl is optionally substituted with oxo, halogen, methyl, ethyl, $-\text{CN}$, $-\text{CF}_3$, $-\text{OH}$, or $-\text{OMe}$. In some embodiments, the heteroalkyl is optionally substituted with halogen. The term “heteroalkylene” refers to a bivalent heteroalkyl.

[0065] The term “heterocycloalkyl” refers to a cycloalkyl group that includes at least one heteroatom selected from nitrogen, oxygen, and sulfur. Unless stated otherwise specifically in the specification, the heterocycloalkyl radical may be a monocyclic, or bicyclic ring system, which may include fused (when fused with an aryl or a heteroaryl ring, the heterocycloalkyl is bonded through a non-aromatic ring atom) or bridged ring systems. The nitrogen, carbon or

sulfur atoms in the heterocyclyl radical may be optionally oxidized. The nitrogen atom may be optionally quaternized. The heterocycloalkyl radical is partially or fully saturated. Examples of heterocycloalkyl radicals include, but are not limited to, dioxolanyl, thienyl[1,3]dithianyl, tetrahydroquinolyl, tetrahydroisoquinolyl, decahydroquinolyl, decahydroisoquinolyl, imidazolynyl, imidazolidynyl, isothiazolidynyl, isoxazolidynyl, morpholynyl, octahydroindolyl, octahydroisindolyl, 2-oxopiperazinyl, 2-oxopiperidinyl, 2-oxopyrrolidinyl, oxazolidynyl, piperidinyl, piperazinyl, 4-piperidonyl, pyrrolidinyl, pyrazolidynyl, quinuclidynyl, thiazolidynyl, tetrahydrofuryl, trithianyl, tetrahydropyranyl, thiomorpholynyl, thiamorpholynyl, 1-oxo-thiomorpholynyl, 1,1-dioxo-thiomorpholynyl. The term heterocycloalkyl also includes all ring forms of carbohydrates, including but not limited to monosaccharides, disaccharides and oligosaccharides. Unless otherwise noted, heterocycloalkyls have from 2 to 12 carbons in the ring. In some embodiments, heterocycloalkyls have from 2 to 10 carbons in the ring. In some embodiments, heterocycloalkyls have from 2 to 10 carbons in the ring and 1 or 2 N atoms. In some embodiments, heterocycloalkyls have from 2 to 10 carbons in the ring and 3 or 4 N atoms. In some embodiments, heterocycloalkyls have from 2 to 12 carbons, 0-2 N atoms, 0-2 O atoms, 0-2 P atoms, and 0-1 S atoms in the ring. In some embodiments, heterocycloalkyls have from 2 to 12 carbons, 1-3 N atoms, 0-1 O atoms, and 0-1 S atoms in the ring. It is understood that when referring to the number of carbon atoms in a heterocycloalkyl, the number of carbon atoms in the heterocycloalkyl is not the same as the total number of atoms (including the heteroatoms) that make up the heterocycloalkyl (i.e., skeletal atoms of the heterocycloalkyl ring). Unless stated otherwise specifically in the specification, a heterocycloalkyl is optionally substituted, for example, with oxo, halogen, amino, nitrile, nitro, hydroxyl, alkyl, alkenyl, alkynyl, haloalkyl, alkoxy, aryl, cycloalkyl, heterocycloalkyl, heteroaryl, and the like. In some embodiments, a heterocycloalkyl is optionally substituted with oxo, halogen, methyl, ethyl, —CN, —CF₃, —OH, —OMe, —NH₂, or —NO₂. In some embodiments, a heterocycloalkyl is optionally substituted with oxo, halogen, methyl, ethyl, —CN, —CF₃, —OH, or —OMe. In some embodiments, the heterocycloalkyl is optionally substituted with halogen. The term “heterocycloalkylene” refers to a bivalent heterocycloalkyl.

[0066] “Heteroaryl” refers to a ring system radical comprising carbon atom(s) and one or more ring heteroatoms that selected from the group consisting of nitrogen, oxygen, phosphorous, and sulfur, and at least one aromatic ring. In some embodiments, heteroaryl is monocyclic, bicyclic or polycyclic. Illustrative examples of monocyclic heteroaryls include pyridinyl, imidazolyl, pyrimidinyl, pyrazolyl, triazolyl, pyrazinyl, tetrazolyl, furyl, thienyl, isoxazolyl, thiazolyl, oxazolyl, isothiazolyl, pyrrolyl, pyridazinyl, triazinyl, oxadiazolyl, thiadiazolyl, furazanyl, indolizine, indole, benzofuran, benzothiophene, indazole, benzimidazole, purine, quinolizine, quinoline, isoquinoline, cinnoline, phthalazine, quinazoline, quinoxaline, 1,8-naphthyridine, and pteridine. Illustrative examples of monocyclic heteroaryls include pyridinyl, imidazolyl, pyrimidinyl, pyrazolyl, triazolyl, pyrazinyl, tetrazolyl, furyl, thienyl, isoxazolyl, thiazolyl, oxazolyl, isothiazolyl, pyrrolyl, pyridazinyl, triazinyl, oxadiazolyl, thiadiazolyl, and furazanyl. Illustrative examples of bicyclic heteroaryls include indolizine, indole, benzofuran, benzothiophene, indazole, benzimidazole, purine, quinolizine, quinoline, isoquinoline, cinnoline, phthalazine, quinazoline, quinoxaline, 1,8-naphthyridine, and pteridine.

In some embodiments, heteroaryl is pyridinyl, pyrazinyl, pyrimidinyl, thiazolyl, thienyl, thiadiazolyl or furyl. In some embodiments, a heteroaryl contains 0-6 N atoms in the ring. In some embodiments, a heteroaryl contains 1-4 N atoms in the ring. In some embodiments, a heteroaryl contains 4-6 N atoms in the ring. In some embodiments, a heteroaryl contains 0-4 N atoms, 0-1 O atoms, 0-1 P atoms, and 0-1 S atoms in the ring. In some embodiments, a heteroaryl contains 1-4 N atoms, 0-1 O atoms, and 0-1 S atoms in the ring. In some embodiments, heteroaryl is a C₁-C₉ heteroaryl. In some embodiments, monocyclic heteroaryl is a C₁-C₈ heteroaryl. In some embodiments, monocyclic heteroaryl is a 5-membered or 6-membered heteroaryl. In some embodiments, a bicyclic heteroaryl is a C₆-C₉ heteroaryl. In some embodiments, a heteroaryl group comprises a partially reduced cycloalkyl or heterocycloalkyl group defined herein (e.g., 7,8-dihydroquinoline). In some embodiments, a heteroaryl group comprises a fully reduced cycloalkyl or heterocycloalkyl group defined herein (e.g., 5,6,7,8-tetrahydroquinoline). When heteroaryl comprises a cycloalkyl or heterocycloalkyl group, the heteroaryl is bonded to the rest of the molecule through a heteroaromatic ring carbon or hetero atom. A heteroaryl radical can be a monocyclic or polycyclic (e.g., bicyclic, tricyclic, or tetracyclic) ring system, which may include fused, spiro or bridged ring systems. Unless stated otherwise specifically in the specification, a heteroaryl is optionally substituted, for example, with halogen, amino, nitrile, nitro, hydroxyl, alkyl, alkenyl, alkynyl, haloalkyl, alkoxy, aryl, cycloalkyl, heterocycloalkyl, heteroaryl, and the like. In some embodiments, a heteroaryl is optionally substituted with halogen, methyl, ethyl, —CN, —CF₃, —OH, —OMe, —NH₂, or —NO₂. In some embodiments, a heteroaryl is optionally substituted with halogen, methyl, ethyl, —CN, —CF₃, —OH, or —OMe. In some embodiments, the heteroaryl is optionally substituted with halogen. The term “heteroarylene” refers to a bivalent heteroaryl.

[0067] The term “optionally substituted” or “substituted” means that the referenced group is optionally substituted with one or more additional group(s) individually and independently selected from D, halogen, —CN, oxo, —NH₂, —NH(alkyl), —N(alkyl)₂, —OH, —CO₂H, —CO₂alkyl, —C(=O)NH₂, —C(=O)NH(alkyl), —C(=O)N(alkyl)₂, —S(=O)₂NH₂, —S(=O)₂NH(alkyl), —S(=O)₂N(alkyl)₂, alkyl, cycloalkyl, fluoroalkyl, heteroalkyl, alkoxy, fluoroalkoxy, heterocycloalkyl, aryl, heteroaryl, aryloxy, alkylthio, arylthio, alkylsulfoxide, arylsulfoxide, alkylsulfone, and arylsulfone. In some embodiments, optional substituents are independently selected from D, halogen, —CN, oxo, —NH₂, —NH(CH₃), —N(CH₃)₂, —OH, —CO₂H, —CO₂(C₁-C₄alkyl), —C(=O)NH₂, —C(=O)NH(C₁-C₄alkyl), —C(=O)N(C₁-C₄alkyl)₂, —S(=O)₂NH₂, —S(=O)₂NH(C₁-C₄alkyl), —S(=O)₂N(C₁-C₄alkyl)₂, C₁-C₄alkyl, C₃-C₆cycloalkyl, C₁-C₄fluoroalkyl, C₁-C₄heteroalkyl, C₁-C₄alkoxy, C₁-C₄fluoroalkoxy, —SC₁-C₄alkyl, —S(=O)C₁-C₄alkyl, and —S(=O)₂C₁-C₄alkyl. In some embodiments, optional substituents are independently selected from D, halogen, —CN, —NH₂, —OH, —NH(CH₃), —N(CH₃)₂, —NH(cyclopropyl), —CH₃, —CH₂CH₃, —CF₃, —OCH₃, and —OCF₃. In some embodiments, substituted groups are substituted with one or two of the preceding groups. When indicating the number of substituents, the term “one or more” means from one substituent to the highest possible

number of substitutions, i.e. replacement of one hydrogen up to replacement of all hydrogens by substituents.

[0068] The term “unsubstituted” means that the specified group bears no substituents.

[0069] The term “moiety” refers to a specific segment or functional group of a molecule. Chemical moieties are often recognized chemical entities embedded in or appended to a molecule.

[0070] The terms “treat,” “prevent,” “ameliorate,” and “inhibit,” as well as words stemming therefrom, as used herein, do not necessarily imply 100% or complete treatment, prevention, amelioration, or inhibition. Rather, there are varying degrees of treatment, prevention, amelioration, and inhibition of which one of ordinary skill in the art recognizes as having a potential benefit or therapeutic effect. In this respect, the disclosed methods can provide any amount of any level of treatment, prevention, amelioration, or inhibition of the disorder in a mammal. For example, a disorder, including symptoms or conditions thereof, may be reduced by, for example, about 100%, about 90%, about 80%, about 70%, about 60%, about 50%, about 40%, about 30%, about 20%, or about 10%. Furthermore, the treatment, prevention, amelioration, or inhibition provided by the methods disclosed herein can include treatment, prevention, amelioration, or inhibition of one or more conditions or symptoms of the disorder, e.g., cancer or an inflammatory disease. Also, for purposes herein, “treatment,” “prevention,” “amelioration,” or “inhibition” encompass delaying the onset of the disorder, or a symptom or condition thereof. As used herein, “treating” includes the concepts of “alleviating,” which refers to lessening the frequency of occurrence or recurrence, or the severity, of any symptoms or other ill effects related to a disorder and/or the associated side effects. The term “treating” also encompasses the concept of “managing” which refers to reducing the severity of a particular disease or disorder in a patient or delaying its recurrence, e.g., lengthening the period of remission in a patient who had suffered from the disease.

[0071] As used herein, the term “prevent,” “preventing,” or “prevention” means prevention of the occurrence or onset of one or more symptoms associated with a particular disorder and does not necessarily mean the complete prevention of a disorder. For example, the term “prevent,” “preventing,” and “prevention” refers to the administration of therapy on a prophylactic or preventative basis to an individual who may ultimately, without treatment, manifest at least one symptom of a disease or condition but who has not yet done so. Such individuals can be identified since there are risk factors that are known to correlate with the subsequent occurrence of the disease. Alternatively, prevention therapy can be administered without prior identification of a risk factor, as a prophylactic measure. Delaying the onset of the at least one symptom can also be considered prevention or prophylaxis.

[0072] The term “therapeutically effective amount” as used herein to refer to an amount effective at the dosage and duration necessary to achieve the desired therapeutic result. A therapeutically effective amount of the composition may vary depending on factors such as the individual’s condition, age, sex, and weight, the radiopharmaceutical conjugate administered, and the route of administration. A therapeutically effective amount can also be an amount that exceeds any toxic or deleterious effect of the composition that would have a beneficial effect on the treatment.

[0073] Certain compounds described herein may exist in tautomeric forms, and all such tautomeric forms of the compounds being within the scope of the disclosure.

[0074] Unless otherwise stated, structures depicted herein are also meant to include all stereochemical forms of the structure; i.e., the R and S configurations for each asymmetric center. Therefore, single stereochemical isomers as well as enantiomeric and diastereomeric mixtures of the present compounds are within the scope of the disclosure.

[0075] Ranges provided herein are understood to be shorthand for all of the values within the range. For example, a range of 1 to 50 is understood to include any number, combination of numbers, or sub-range from the group consisting of 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, or 50, as well as all intervening decimal values between the aforementioned integers such as, for example, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, and 1.9. With respect to sub-ranges, “nested sub-ranges” that extend from either end point of the range are specifically contemplated. For example, a nested sub-range of an exemplary range of 1 to 50 may comprise 1 to 10, 1 to 20, 1 to 30, and 1 to 40 in one direction, or 50 to 40, 50 to 30, 50 to 20, and 50 to 10 in the other direction. Further, ranges described herein such as a range “between 1 and 50” encompass the end points of the ranges. For example, a range “between 10 μ g and 100 mg” encompasses both 10 μ g and 100 mg.

[0076] The term “subject” or “patient” encompasses mammals. Examples of mammals include, but are not limited to, any member of the Mammalian class: humans, non-human primates such as chimpanzees, and other apes and monkey species; farm animals such as cattle, horses, sheep, goats, swine; domestic animals such as rabbits, dogs, and cats; laboratory animals including rodents, such as rats, mice and guinea pigs, and the like. In one aspect, the mammal is a companion animal such as a dog or a cat. In one aspect, the mammal is a human. In one aspect, the subject or patient is a human.

II. Technical Overview

[0077] Targeted Radiopharmaceuticals (TRP) are a new generation of nuclear medicine for cancer treatment, imaging, or diagnosis. A TRP can selectively deliver high concentrations of radionuclide-containing molecules to the target cells such as a tumor, and no or very low concentrations to the undesired cells present in normal, healthy tissues. The process can be achieved by engineering the drug molecule with the high-affinity binder (e.g., targeting ligands) and linking it to the radioactive isotope. The biological targets of these binders are highly expressed on tumor cells and have low or no expression in healthy tissues and organs. When the radioisotope decays, it emits highly energetic ionizing radiation in form of alpha, beta, and/or gamma particles. The released energy at the target sites can cause damage or death of the target tissues or be visualized by imaging scanner to achieve therapeutic, imaging, or diagnostic purposes.

[0078] FIG. 2 shows the decay chain of a ^{212}Pb radionuclide. ^{212}Pb has a 10.6 hour half-life and decays to alpha-particle emitting daughter ions ^{212}Bi and ^{212}Po . Eventually, the decayed daughters accumulate in the form of stable ^{208}Pb . The ^{212}Bi and ^{212}Po daughter ions emit a 6.1 MeV and 8.8 MeV alpha-particle, respectively.

III. Radiopharmaceutical Conjugate

[0079] In one aspect, provided herein are radiopharmaceutical conjugates comprising:

- [0080] (a) a metal chelator;
- [0081] (b) a target-binding moiety; and
- [0082] (c) a radionuclide bound to the metal chelator, wherein the radionuclide is ^{212}Pb .

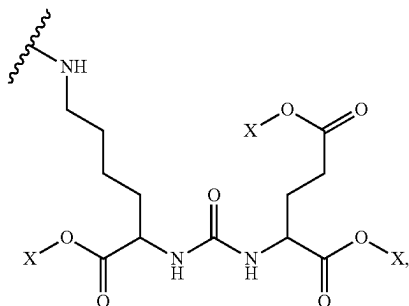
In some embodiments, the radiopharmaceutical conjugate comprises a structure of Formula (I):



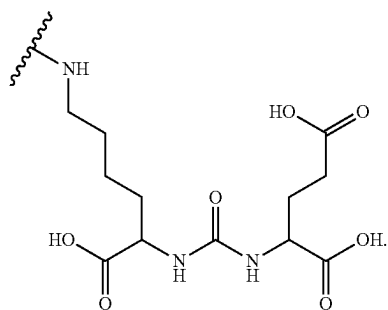
[0083] or a pharmaceutically acceptable salt thereof, wherein:

- [0084] TL is a PSMA targeting ligand comprising a urea;
- [0085] L is a bivalent linking moiety; and
- [0086] R^M is a metal chelator.

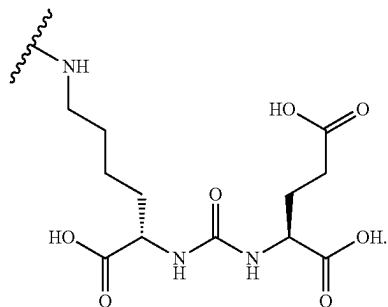
[0087] In some embodiments of Formula (I), TL is



wherein each X is independently selected from the group consisting of H and $\text{C}_1\text{-C}_4$ alkyl. In some embodiments, TL is

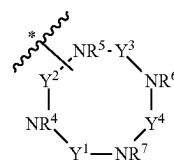


In some embodiments TL is



[0088] In some embodiments of Formula (I), the a metal chelator R^M is a metal chelator described herein. In some embodiments of Formula (I), R^M is TCMC.

[0089] In some embodiments of Formula (I), R^M has the structure of Formula (M-1):



(M-1)

[0090] wherein:

[0091] Y^1 , Y^2 , Y^3 , and Y^4 are each independently an optionally substituted -C_{1-6} alkylene-;

[0092] R^4 , R^5 , R^6 , and R^7 are each independently selected from the group consisting of -C_{1-10} alkylene- C(=O)OR^3 , -C_{1-10} alkylene- $\text{C(=O)N(R}^3)_2$, -C_{1-10} alkylene- $\text{P(=O)(OR}^3)_2$, -C_{1-10} alkylene- $\text{P(=O)OR}^3(\text{R}^3)$, and -C_{1-10} alkylene- $\text{P(=O)(R}^3)_2$; or one of R^4 and R^6 or R^5 and R^7 together from a $\text{-(CH}_2)_m\text{-}$ bridge; wherein each C_{1-10} alkylene is optionally substituted;

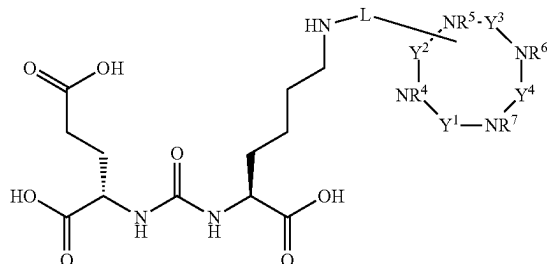
[0093] each R^3 is independently selected from the group consisting of H and C_{1-6} alkyl, wherein the C_{1-6} alkyl is optionally substituted;

[0094] m is 1 to 3; and

[0095] * is attached to L.

[0096] In some embodiments of Formula (I), the radiopharmaceutical conjugate comprises a structure of Formula (I-A):

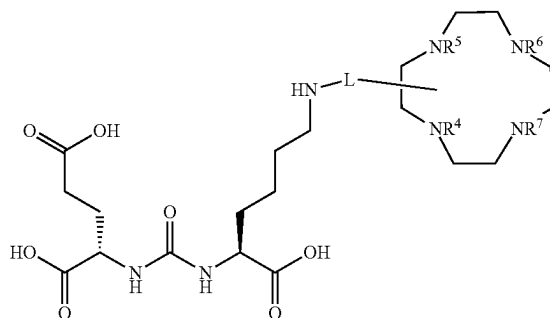
Formula (I-A)



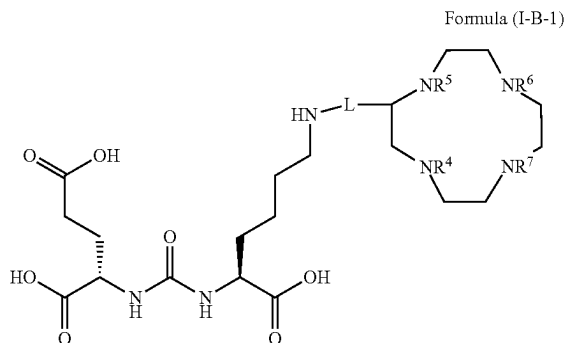
[0097] In some embodiments of Formula (I) or Formula (I-A), Y^1 , Y^2 , Y^3 , and Y^4 are each independently an optionally substituted -C_2 alkylene-.

[0098] In some embodiments of Formula (I) or Formula (I-A), the radiopharmaceutical conjugate comprises a structure of Formula (I-A-1):

Formula (I-A-1)



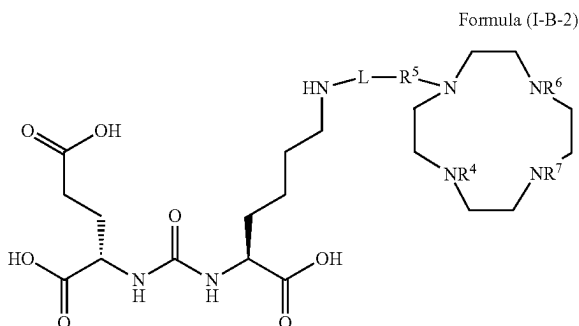
[0099] In some embodiments of Formula (I) or Formula (I-A), the radiopharmaceutical conjugate comprises a structure of Formula (I-B-1):



wherein R^4 , R^5 , R^6 , and R^7 are each independently selected from the group consisting of $-C_{1-3}$ alkylene- $C(=O)OR^3$, and $-C_{1-3}$ alkylene- $C(=O)N(R^3)_2$, wherein each C_{1-3} alkylene is optionally substituted with $-C(=O)OH$.

[0100] In some embodiments of Formula (I-B-1), R^4 , R^5 , R^6 , and R^7 are each $-C_{1-3}$ alkylene- $C(=O)NH_2$.

[0101] In some embodiments of Formula (I) or Formula (I-A), the radiopharmaceutical conjugate comprises a structure of Formula (I-B-2):



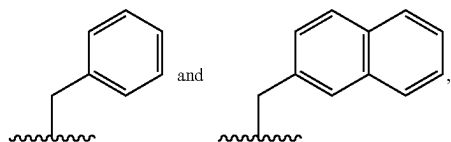
[0102] wherein R^1 is $-C_{1-3}$ alkylene- $C(=O)-$, wherein the C_{1-3} alkylene is optionally substituted with $-C(=O)OH$; and

[0103] R^4 , R^6 , and R^7 are each independently selected from the group consisting of $-C_{1-3}$ alkylene- $C(=O)OR^3$, and $-C_{1-3}$ alkylene- $C(=O)N(R^3)_2$, wherein each C_{1-3} alkylene is optionally substituted with $-C(=O)OH$.

[0104] In some embodiments of Formula (I-B-2), R^4 , R^6 , and R^7 are each $-C_{1-3}$ alkylene- $C(=O)NH_2$, such as $-CH_2-C(=O)NH_2$.

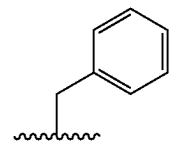
[0105] In some embodiments of Formula (I), Formula (I-A), Formula (I-A-1), Formula (I-B-1), Formula (I-B-1), or Formula (I-B-2), L comprises an optionally substituted C_{1-10} heteroalkylene. In some embodiments, the heteroalkylene comprises at least one nitrogen atom; and

[0106] wherein the heteroalkylene is optionally substituted with one or more substituents selected from

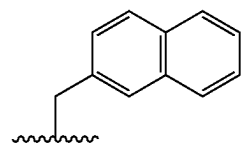


[0107] each of which is optionally substituted with C_{1-3} alkyl, halogen, and $-OH$.

In some embodiments, the heteroalkylene is substituted with

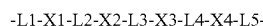


which is optionally substituted with C_{1-3} alkyl, halogen, and $-OH$. In some embodiments, the heteroalkylene is substituted with



which is optionally substituted with C_{1-3} alkyl, halogen, and $-OH$.

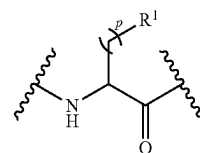
[0108] In some embodiments of Formula (I), Formula (I-A), Formula (I-A-1), Formula (I-B-1), Formula (I-B-1), or Formula (I-B-2), L has the structure of:



[0109] wherein,

[0110] L1, L2, L3, L4, and L5 are each independently selected from a bond, $-O-$, $-S-$, $-C(=O)-$, $-C(=O)NR^3-$, $-NR^3-$, $-C(=O)O-$, $-C(=O)S-$, $-S(=O)_2-$, $-S(=O)_2NR^3-$, $S(=O)NR^3-$, $-OS(=O)_2-$, $-N(R^3)C(=S)N(R^3)-$, and $-N(R^3)C(=O)N(R^3)-$;

[0111] X1, X2, X3, and X4 are each independently selected from a bond, a structure of



C_{1-10} alkylene-, $-C_{1-10}$ heteroalkylene-, $-C_{5-7}$ cycloalkylene-, $-C_{1-10}$ alkylene- C_{5-7} cycloalkylene-, $-C_6$ arylene-, $-C_{1-10}$ alkylene- C_6 arylene-, 5 to 7-membered heterocycloalkylene-, $-C_{1-10}$ alkylene-5 to 7-membered heterocycloalkylene-, 5 to 7-membered heteroarylene-, and $-C_{1-10}$ alkylene-5 to 7-membered heteroarylene-, wherein each alkylene and heteroalkylene is independently optionally substituted;

[0112] p is 0, 1, 2, 3, or 4; and

[0113] R^1 is $-H$, $-OH$, $-OMe$, $-SH$, $-SMe$, $-NH_2$, $-C(=O)OH$, $-C(=O)NH_2$, C_{6-10} aryl, or 5 to 10-membered heteroaryl, wherein each aryl and heteroaryl are independently optionally substituted with one or more substituents selected from halogen and $-OH$;

[0114] In some embodiments, the alkylene and heteroalkylene of X1, X2, X3, and/or X4 is independently, optionally substituted with $-C(=O)OH$.

[0115] In some embodiments, the radiopharmaceutical conjugate comprises a compound having the structure listed in Table 1, or a pharmaceutically acceptable salt thereof.

TABLE 1

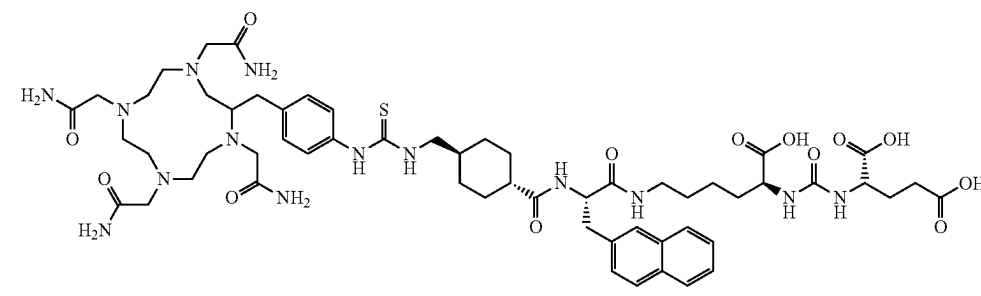
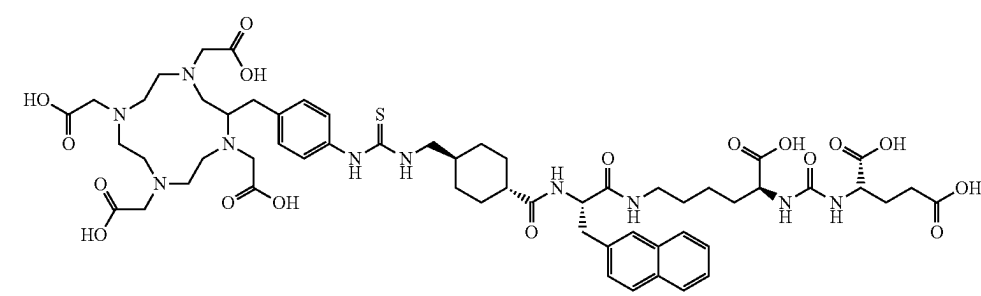
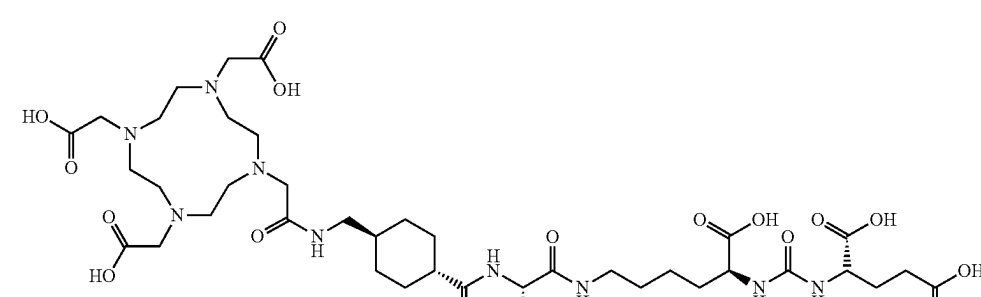
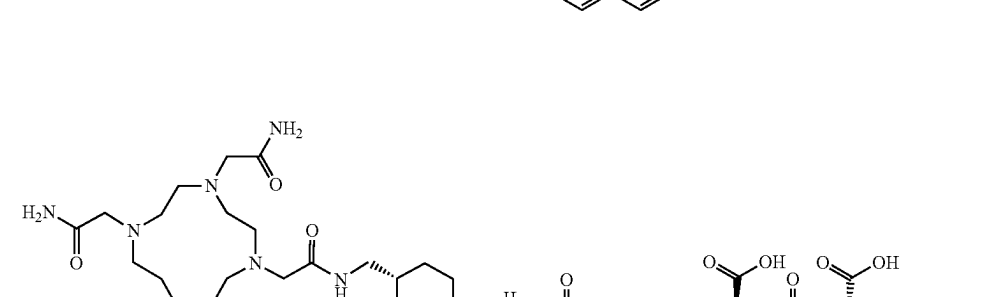
Cpd	Structure
1	
2	
3	
4	

TABLE 1-continued

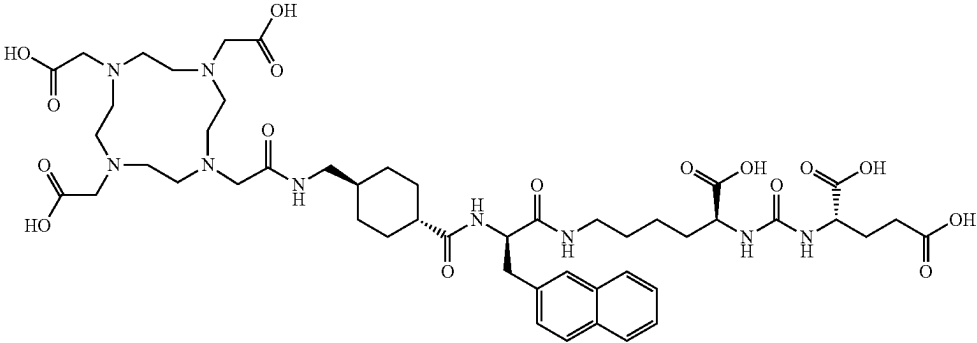
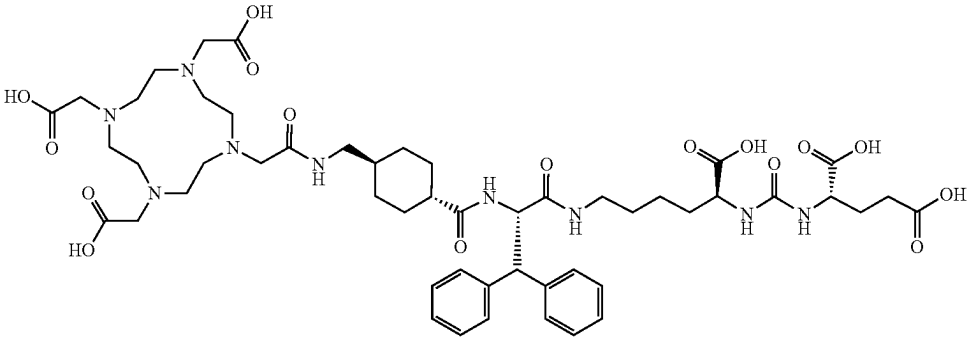
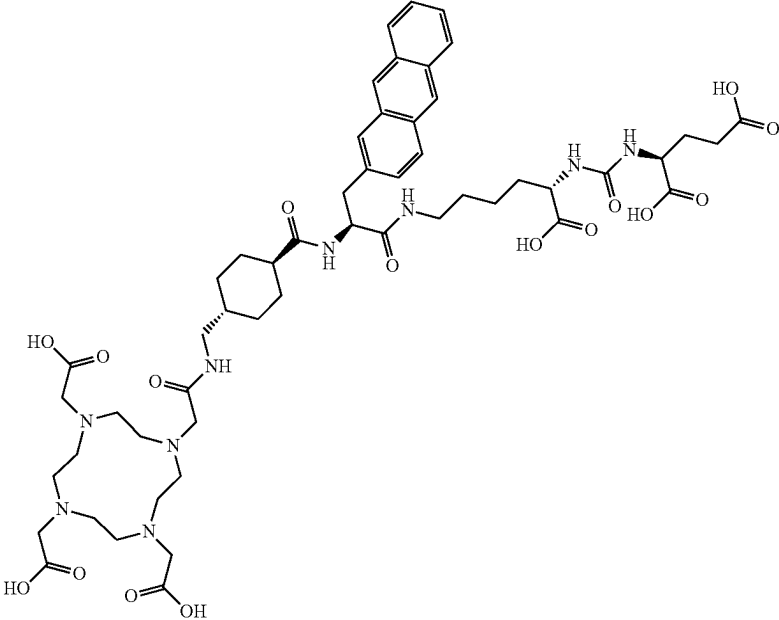
Cpd	Structure
5	
6	
7	

TABLE 1-continued

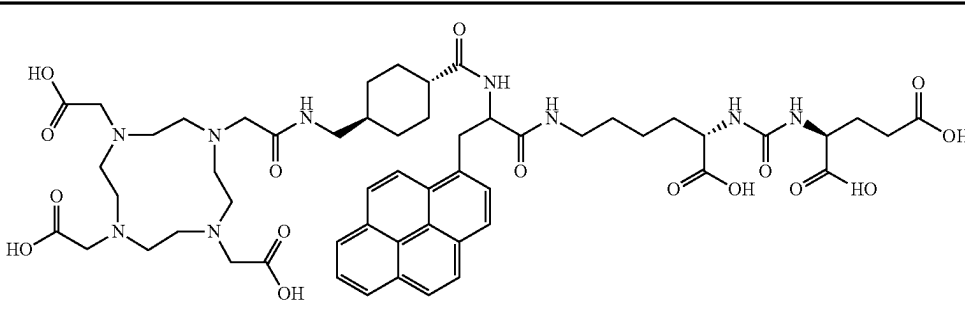
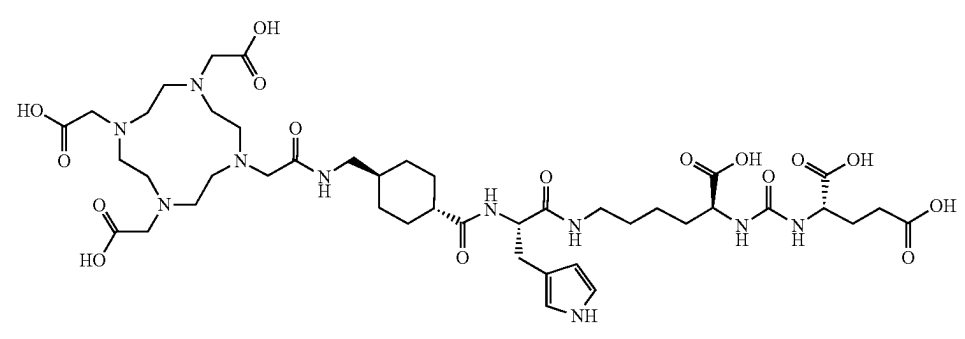
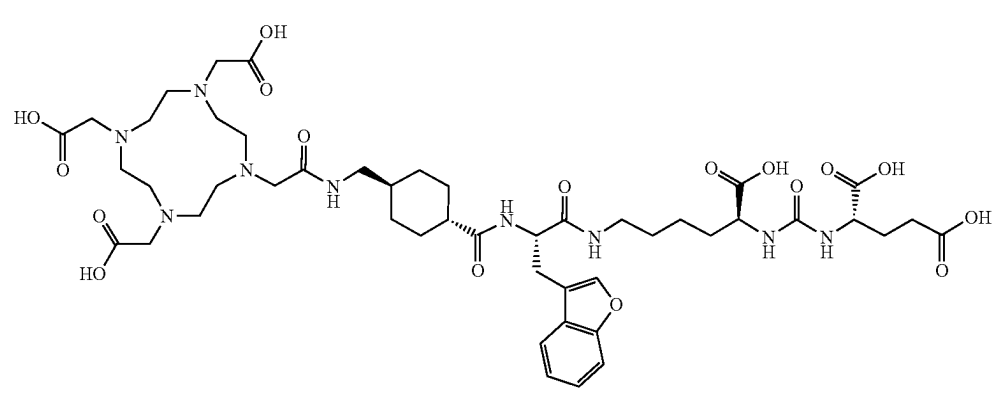
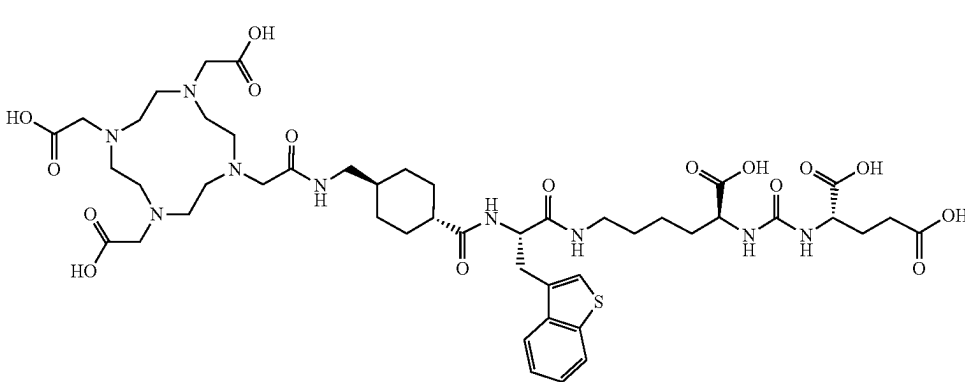
Cpd	Structure
8	
9	
10	
11	

TABLE 1-continued

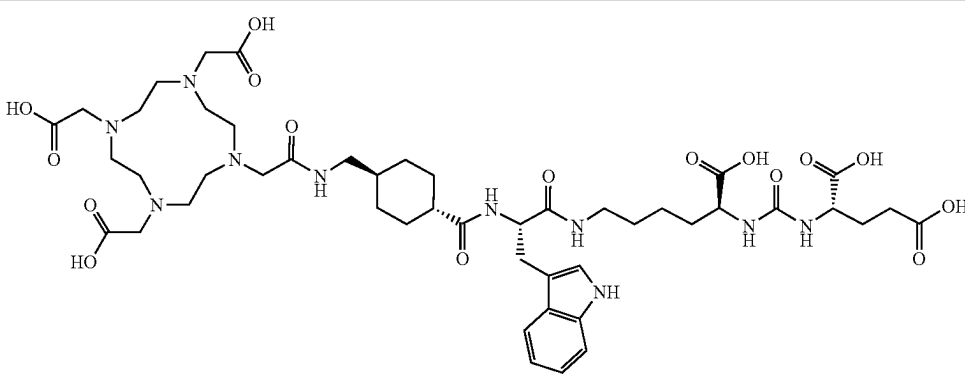
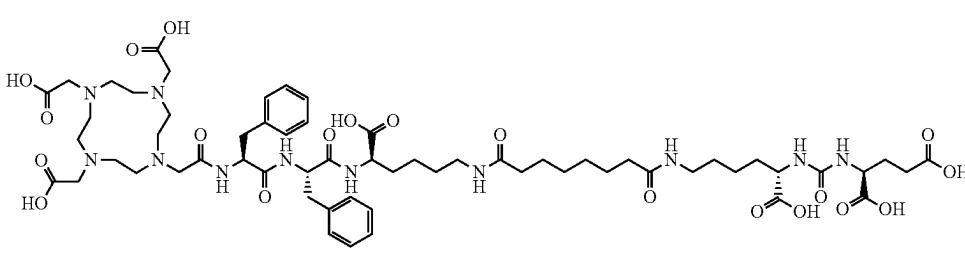
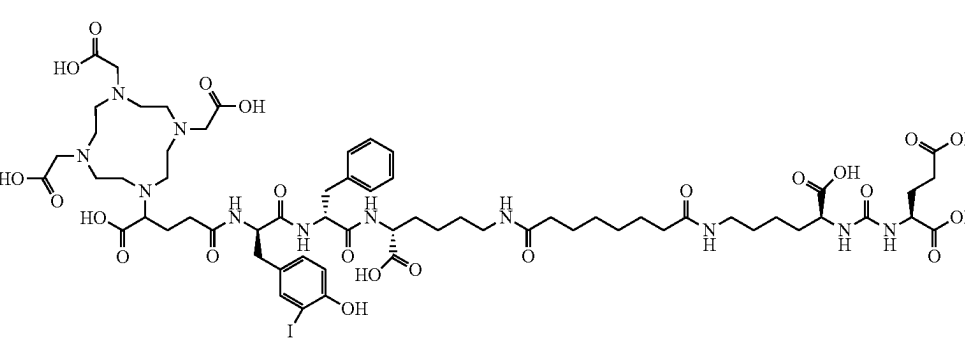
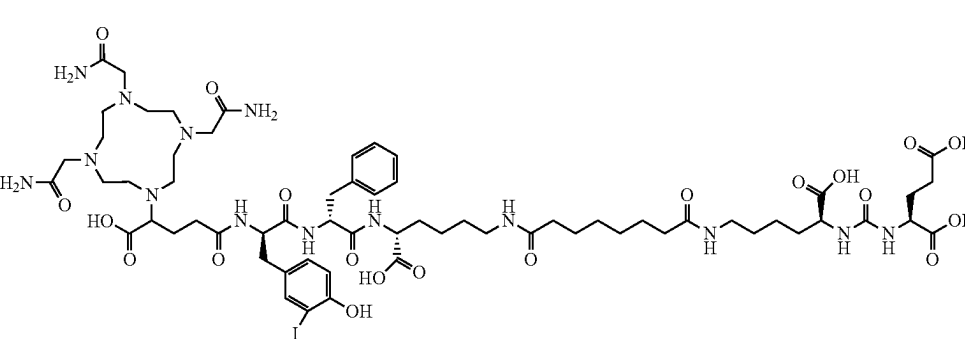
Cpd	Structure
11	
12	
13	
14	

TABLE 1-continued

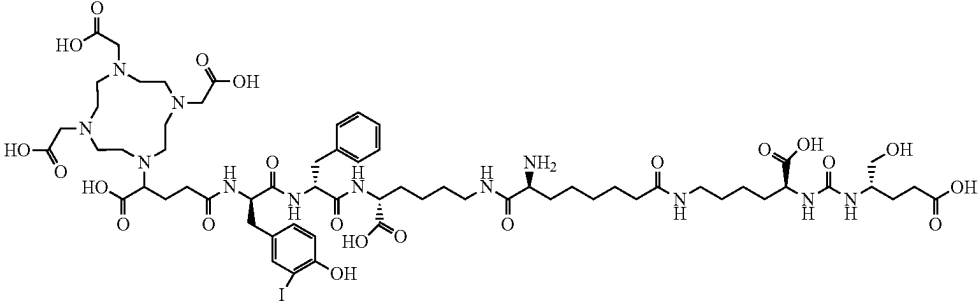
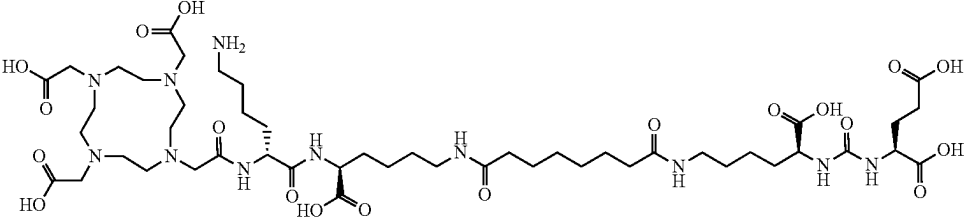
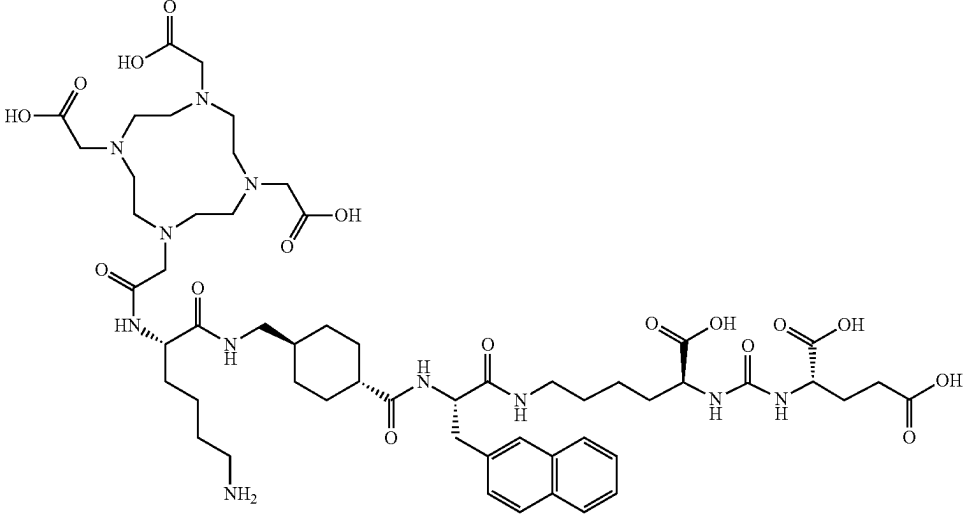
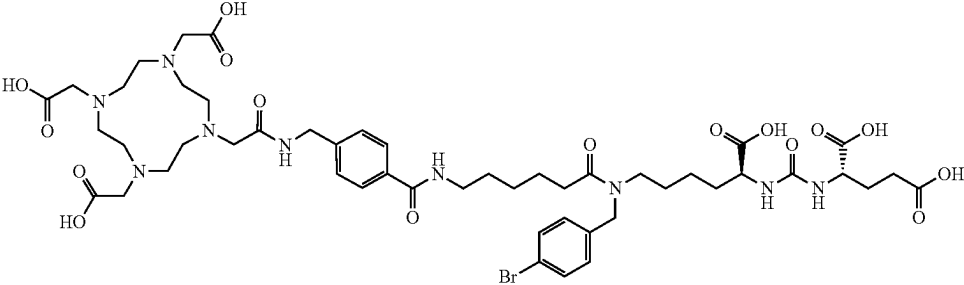
Cpd	Structure
15	
16	
17	
18	

TABLE 1-continued

Cpd	Structure
19	
20	
21	
22	
23	

TABLE 1-continued

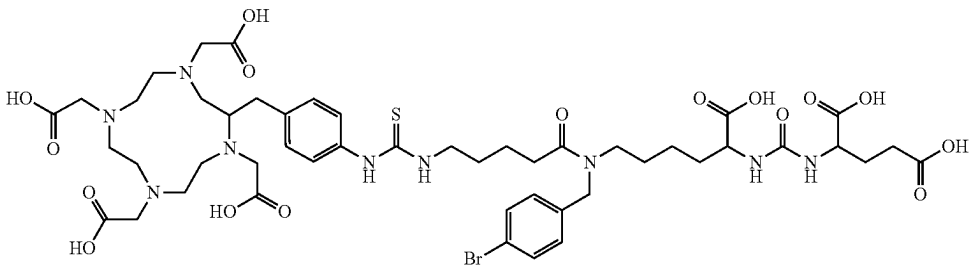
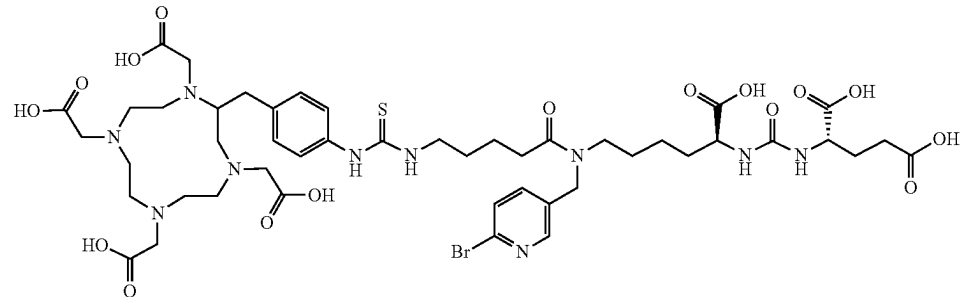
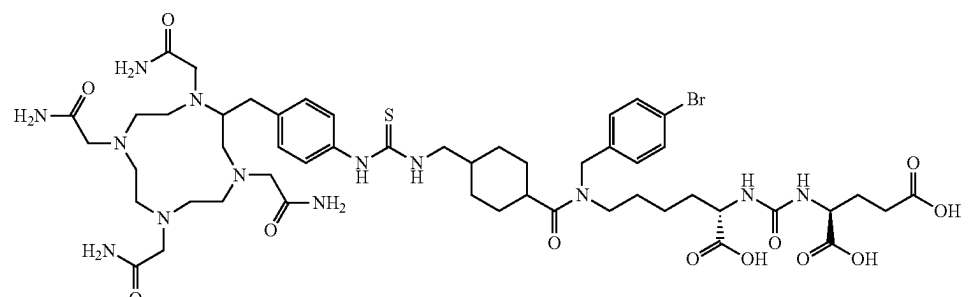
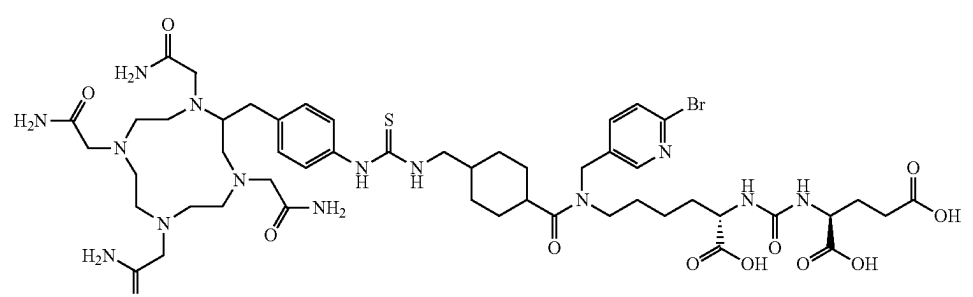
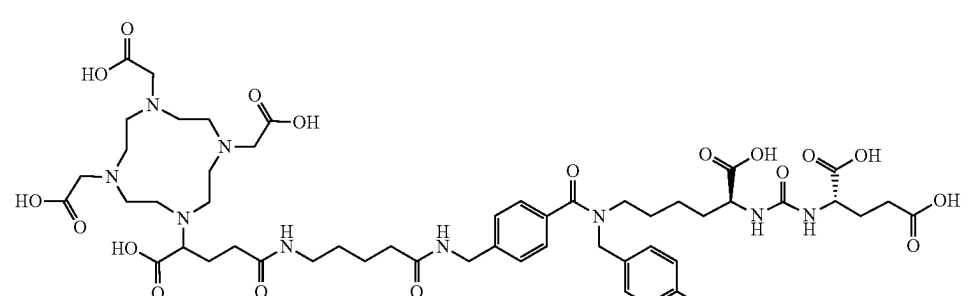
Cpd	Structure
24	
25	
26	
27	
28	

TABLE 1-continued

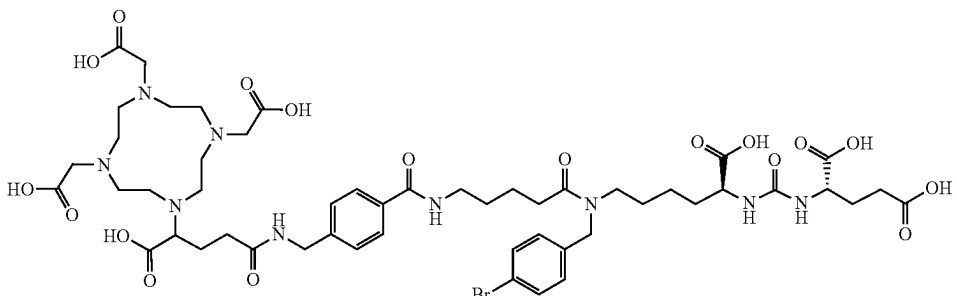
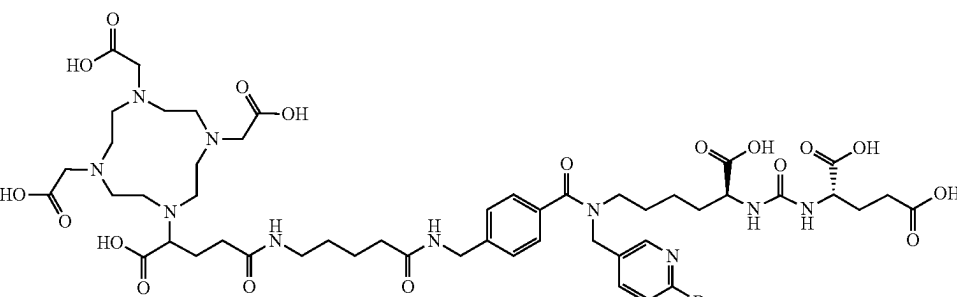
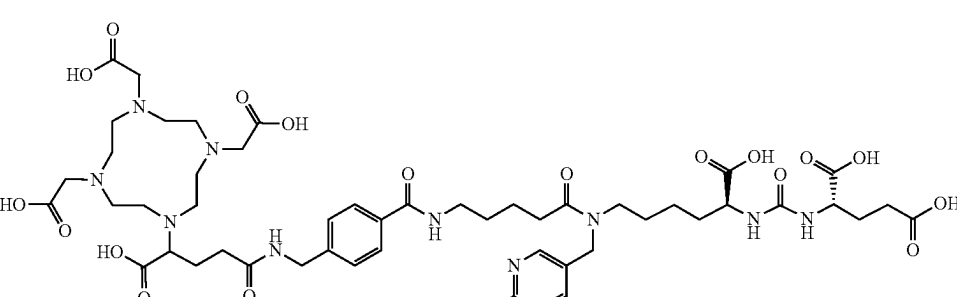
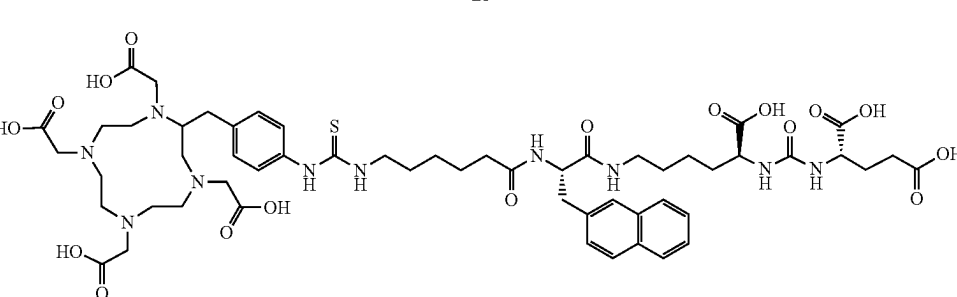
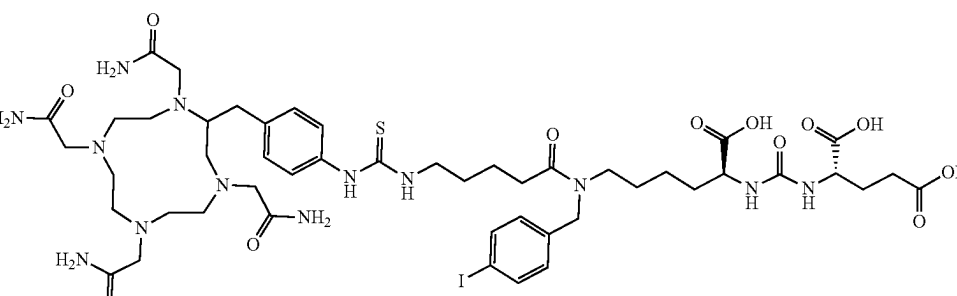
Cpd	Structure
29	
30	
31	
32	
33	

TABLE 1-continued

Cpd	Structure
34	
35	
36	
37	
38	

TABLE 1-continued

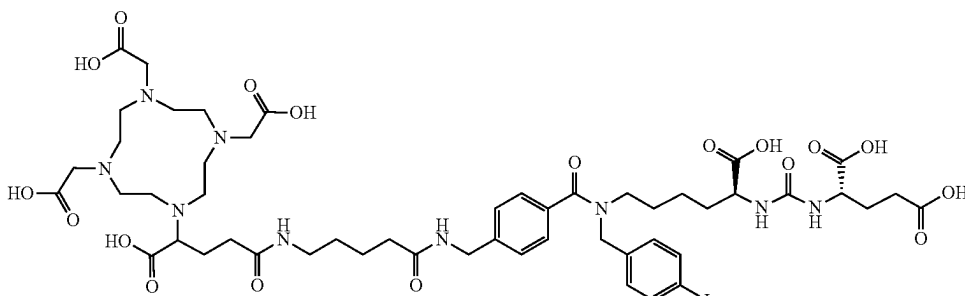
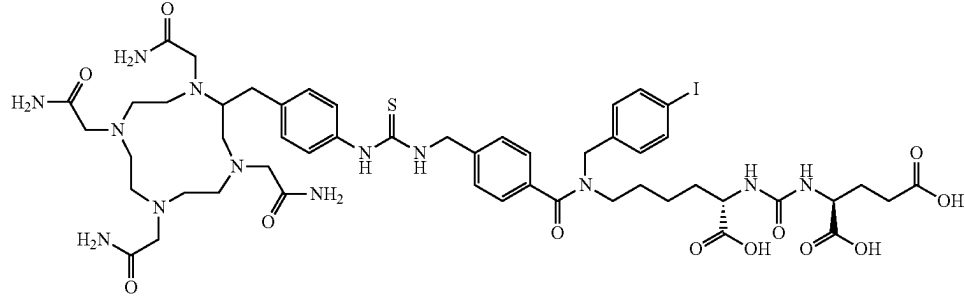
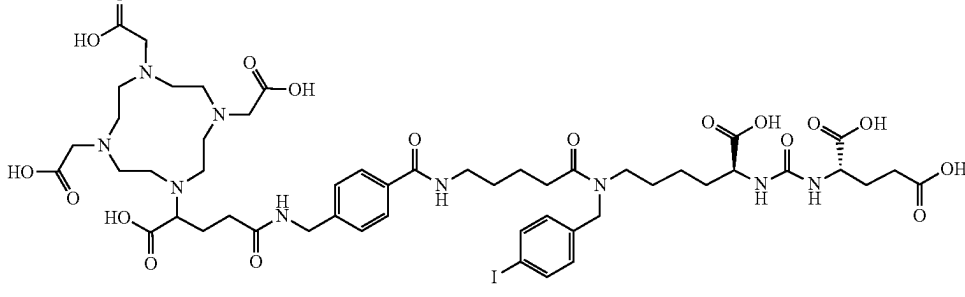
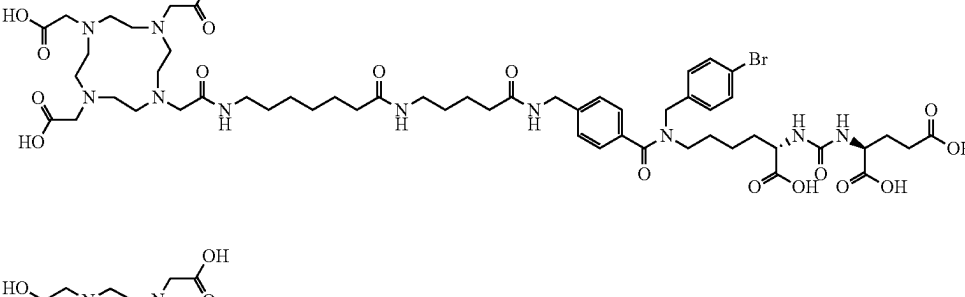
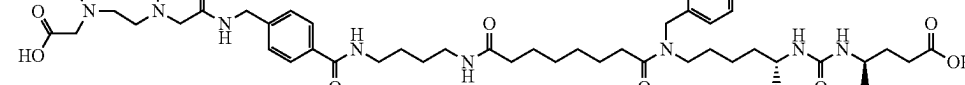
Cpd	Structure
39	
40	
41	
42	
43	

TABLE 1-continued

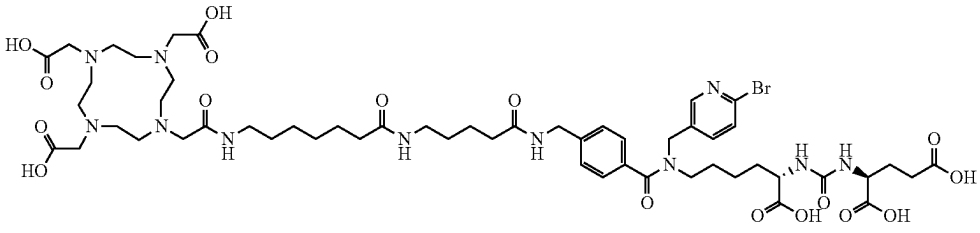
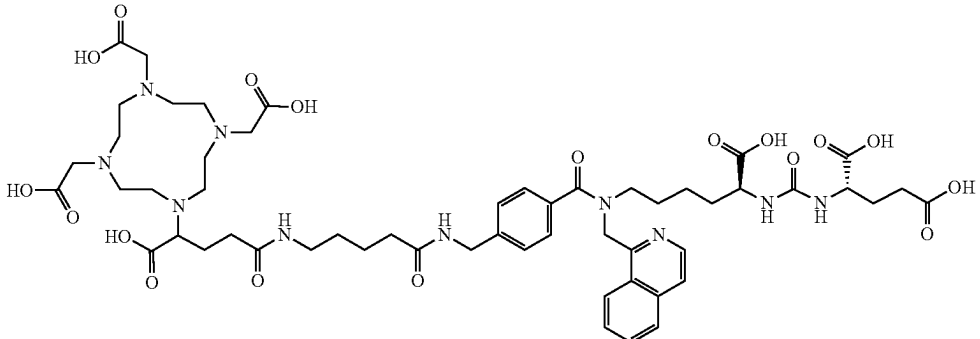
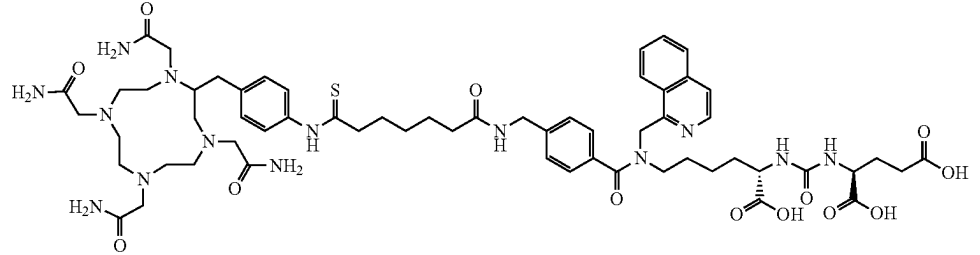
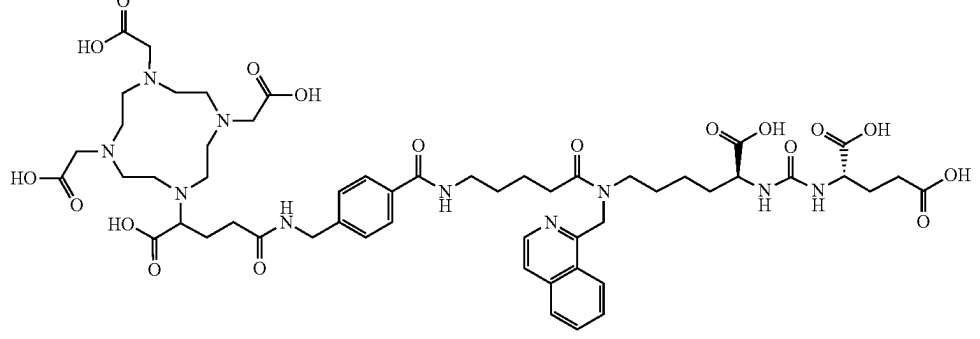
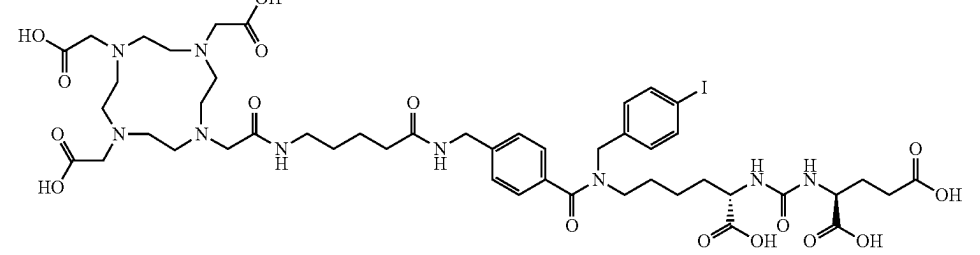
Cpd	Structure
44	
45	
46	
47	
48	

TABLE 1-continued

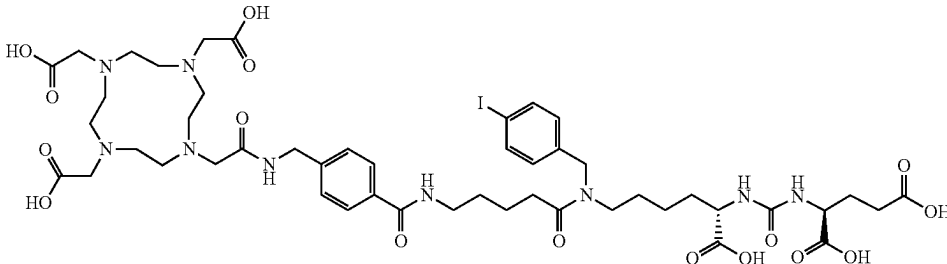
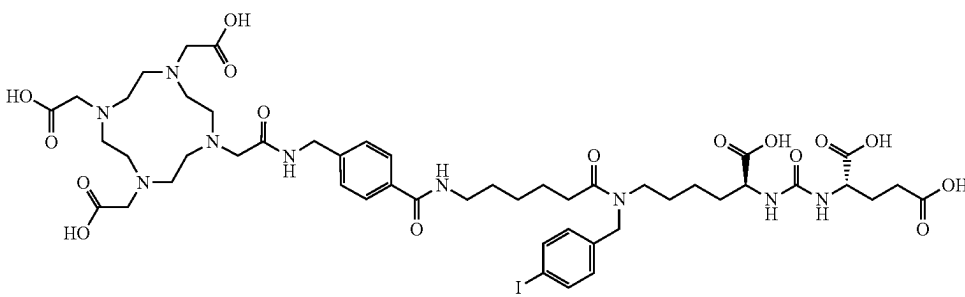
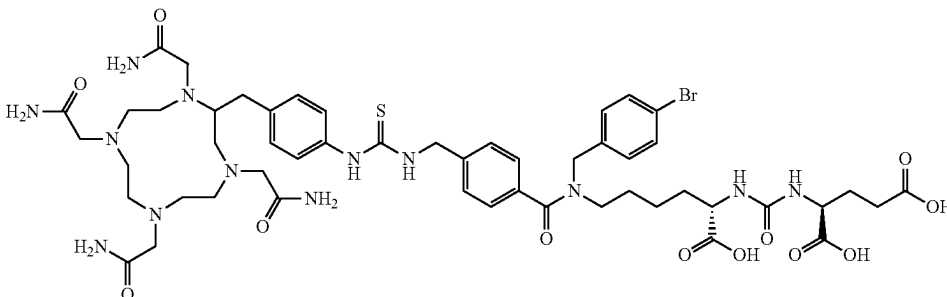
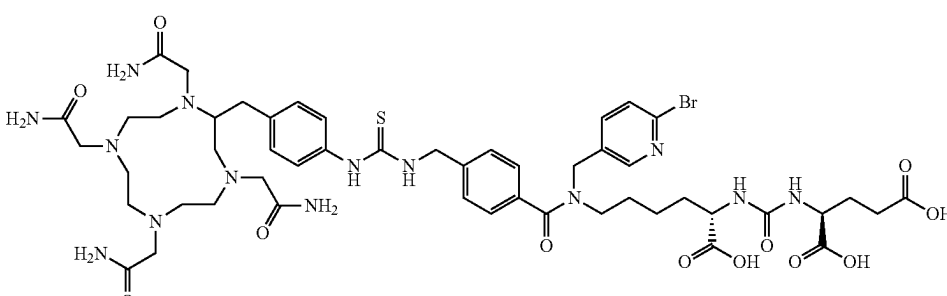
Cpd	Structure
49	
50	
51	
52	

TABLE 1-continued

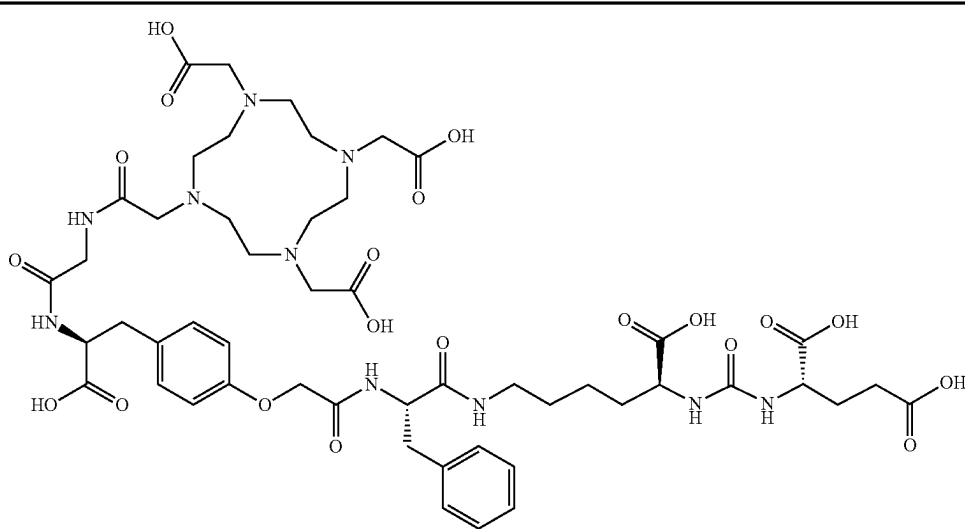
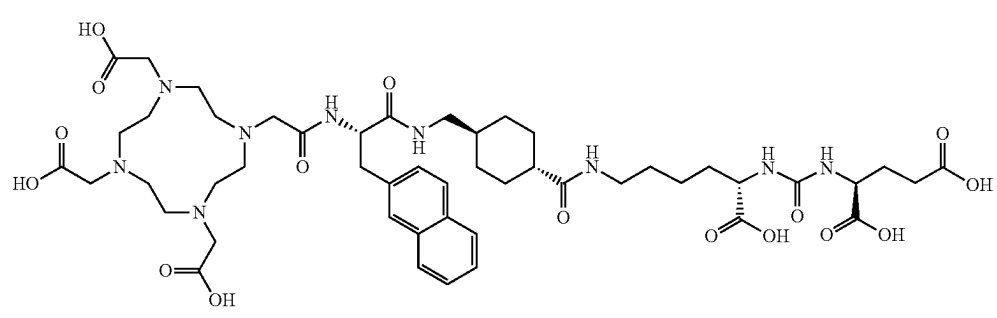
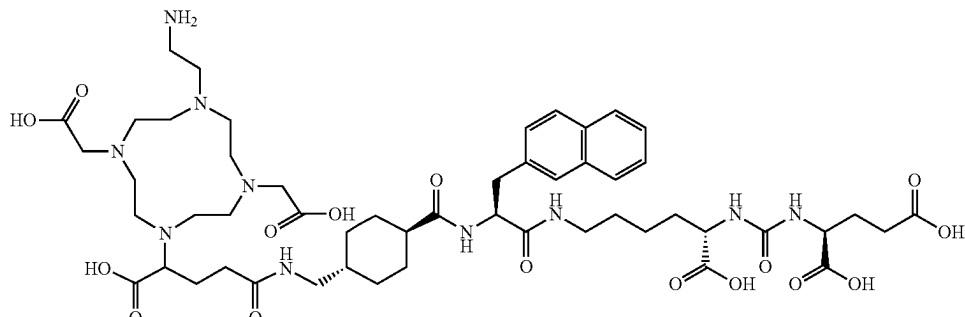
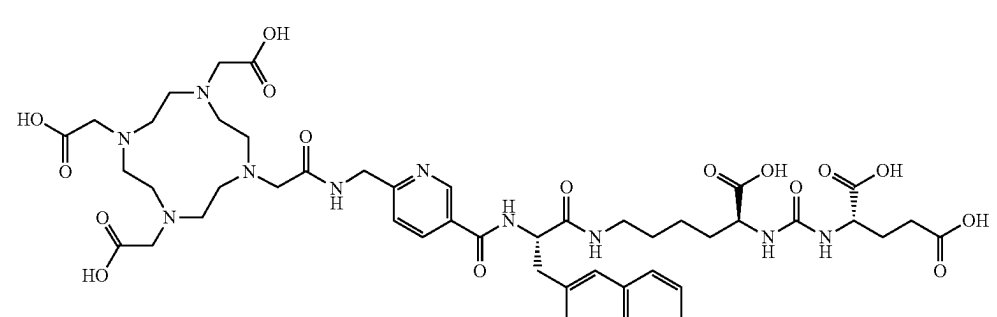
Cpd	Structure
53	
54	
55	
56	

TABLE 1-continued

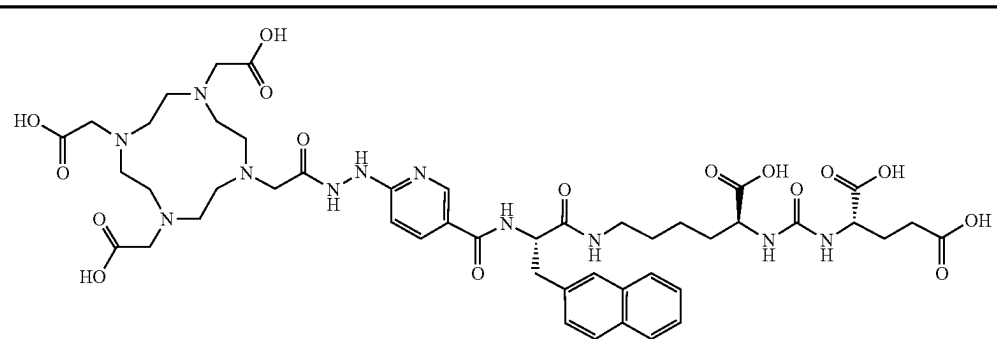
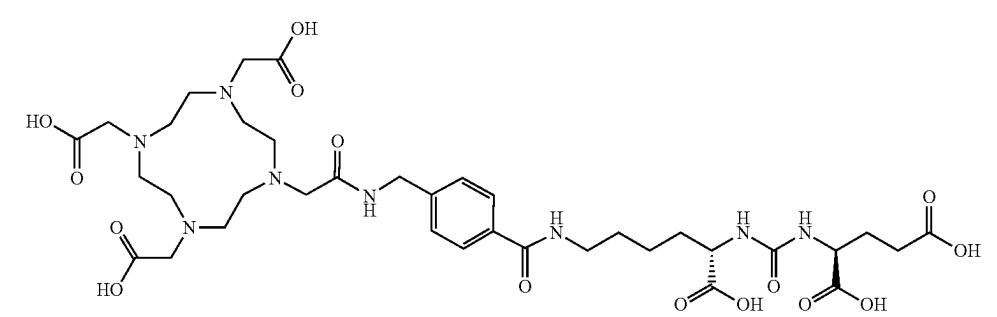
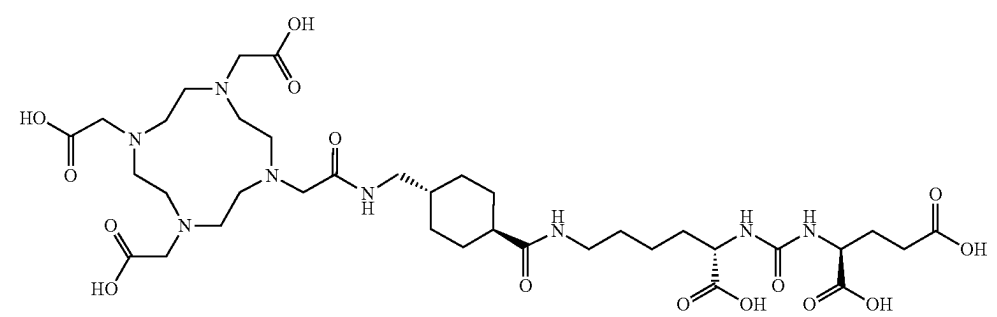
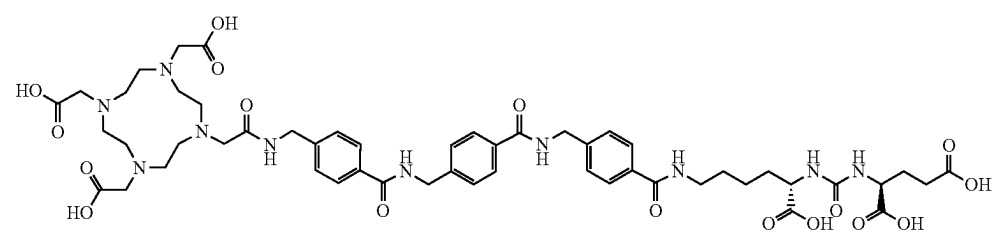
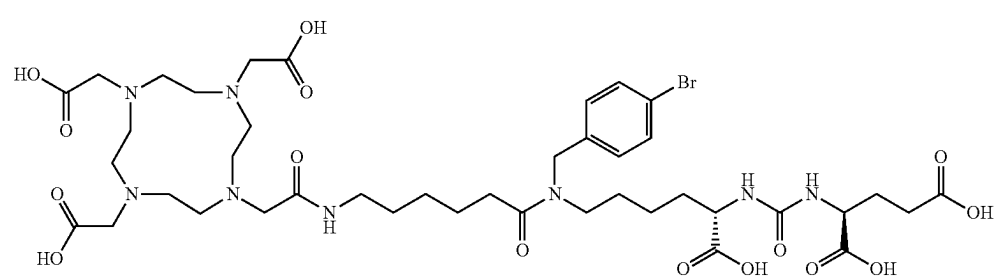
Cpd	Structure
57	
58	
59	
60	
61	

TABLE 1-continued

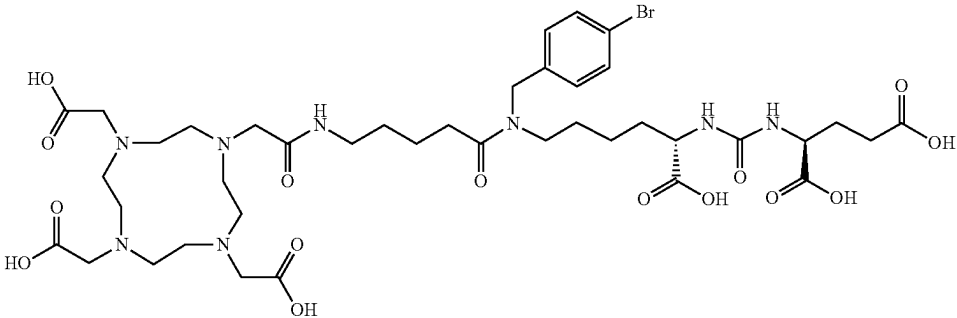
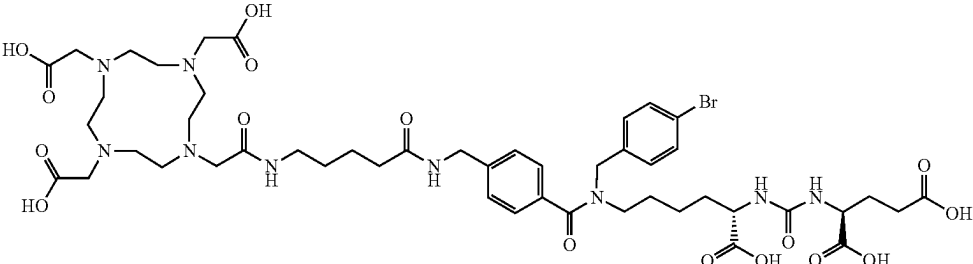
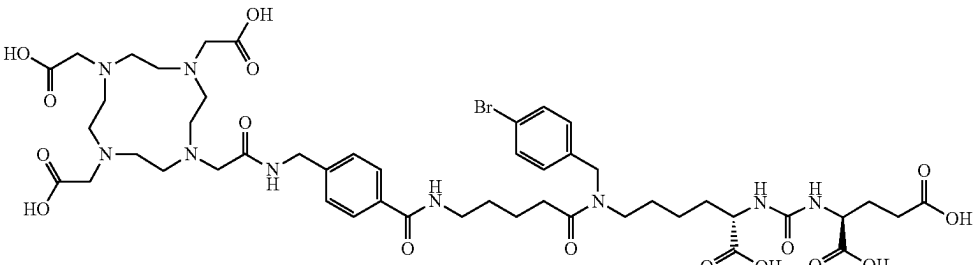
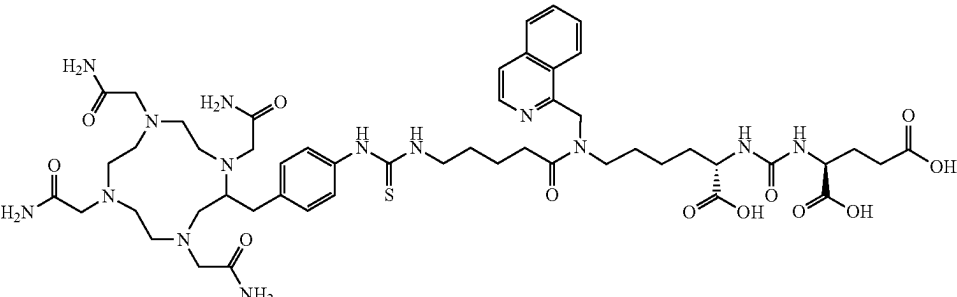
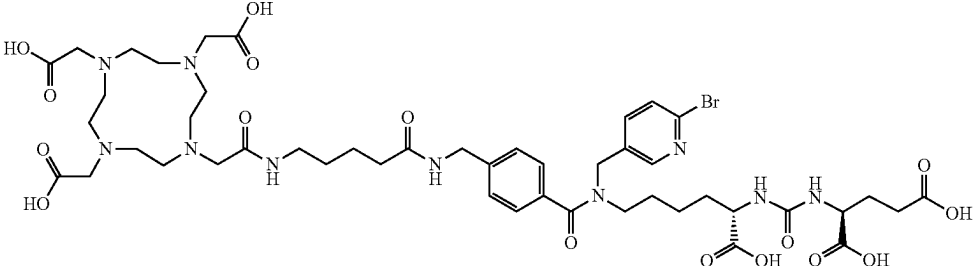
Cpd	Structure
62	
63	
64	
65	
66	

TABLE 1-continued

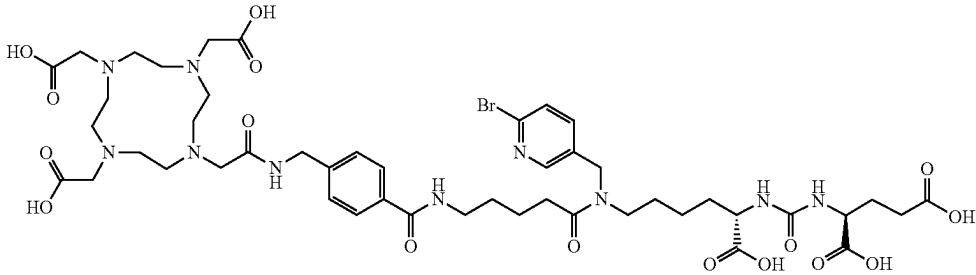
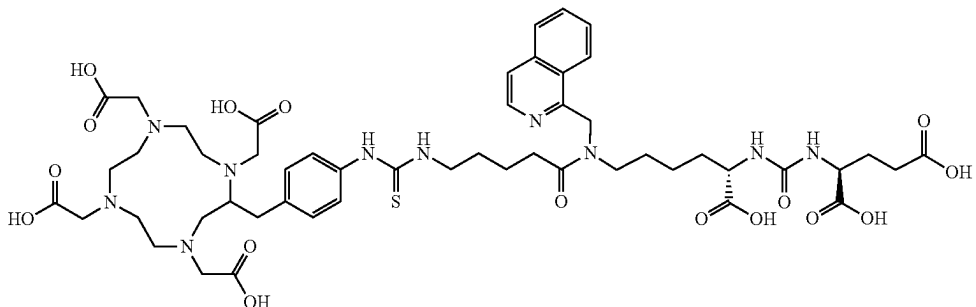
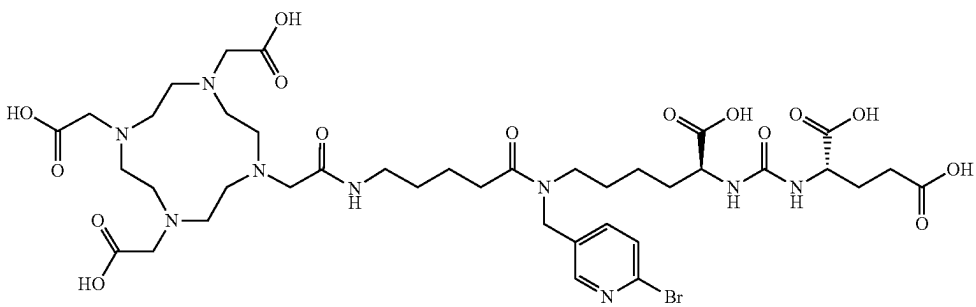
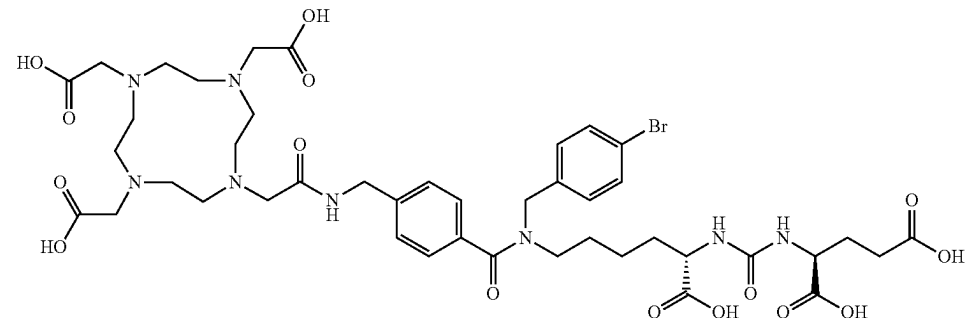
Cpd	Structure
67	
68	
69	
70	

TABLE 1-continued

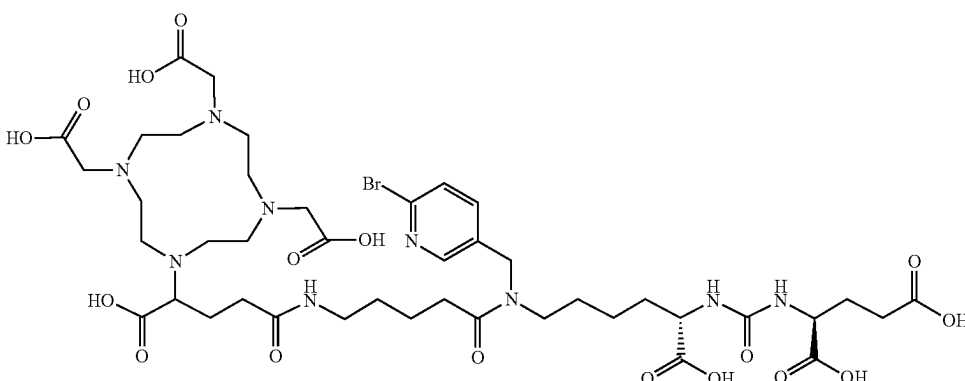
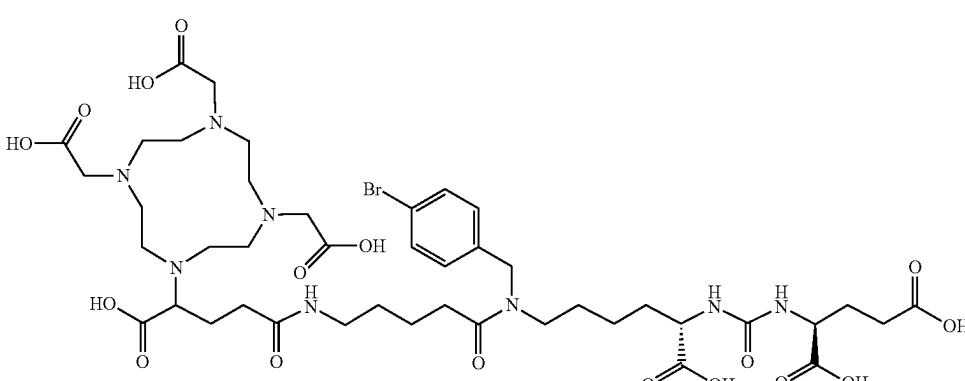
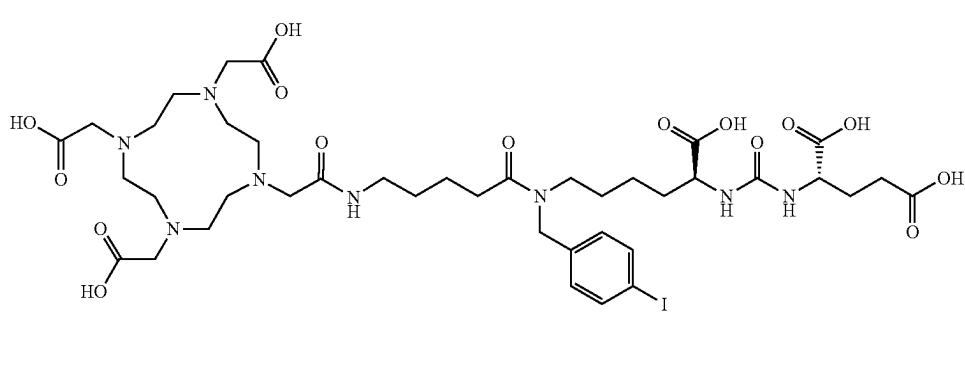
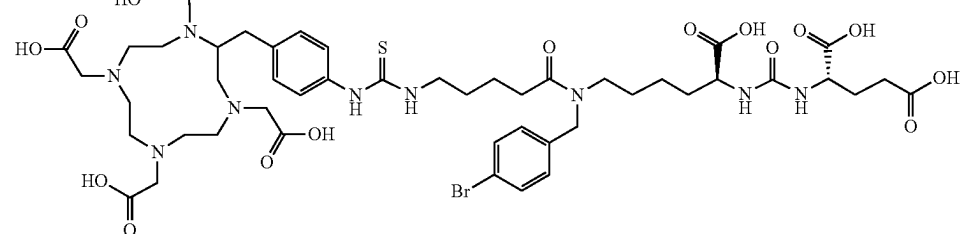
Cpd	Structure
71	
72	
73	
74	

TABLE 1-continued

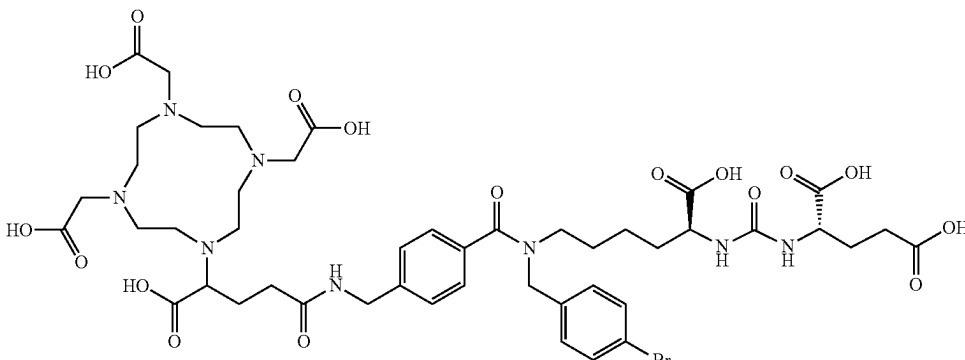
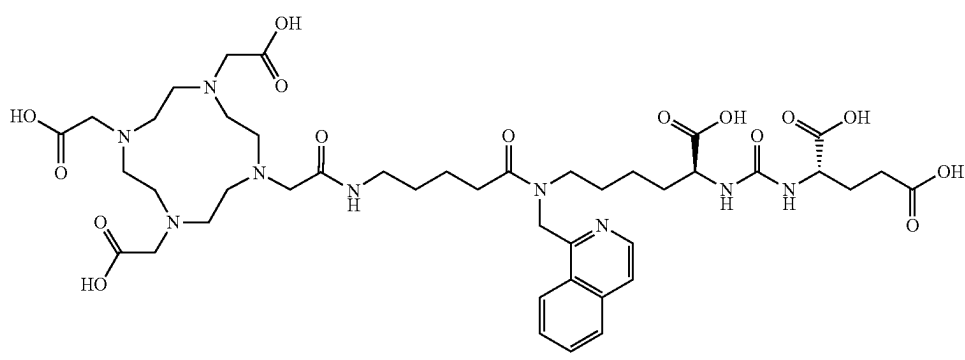
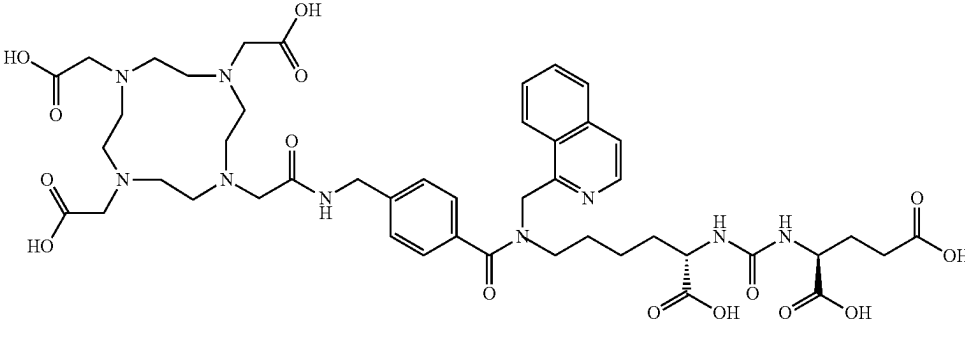
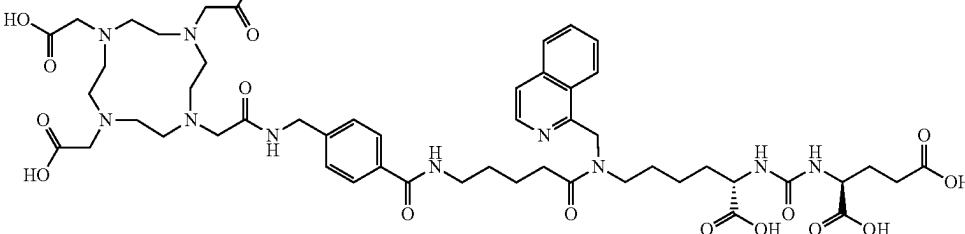
Cpd	Structure
75	
76	
77	
78	

TABLE 1-continued

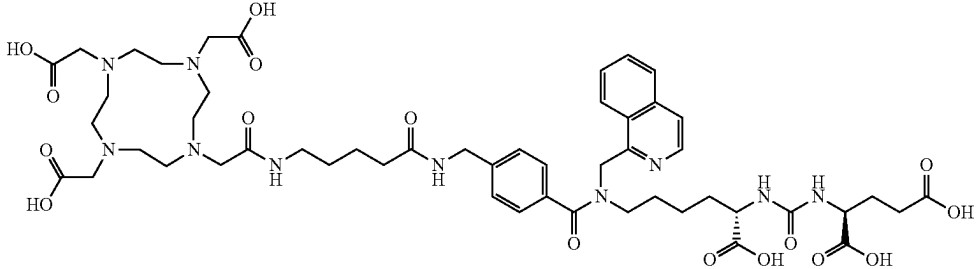
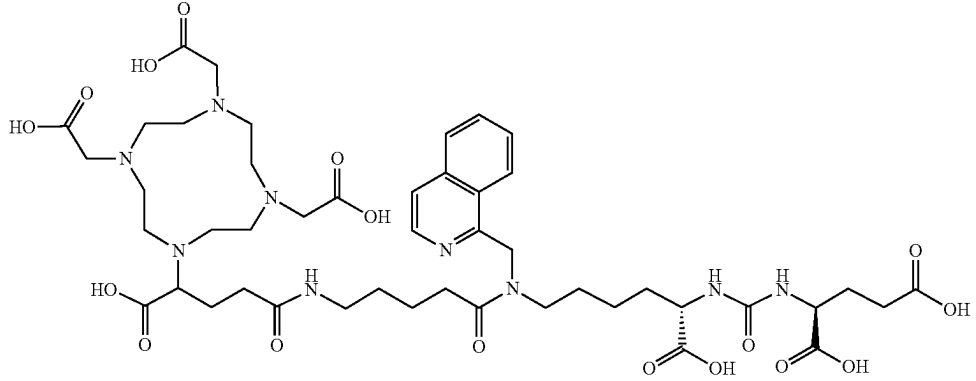
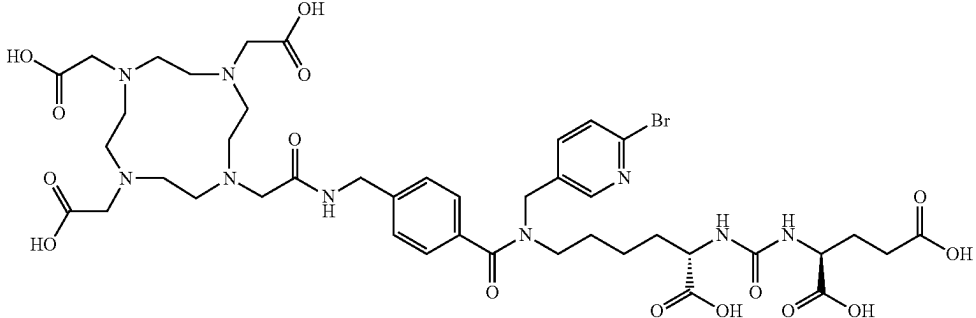
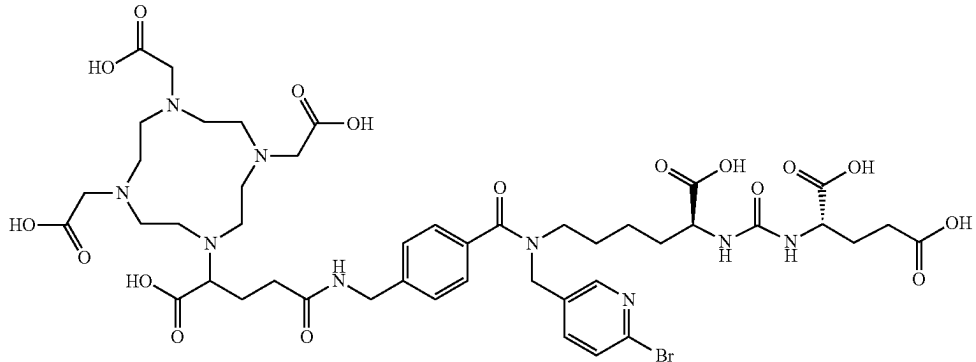
Cpd	Structure
79	
80	
81	
82	

TABLE 1-continued

Cpd	Structure
83	
84	
85	
86	
87	

TABLE 1-continued

Cpd	Structure
88	
89	
90	
91	

TABLE 1-continued

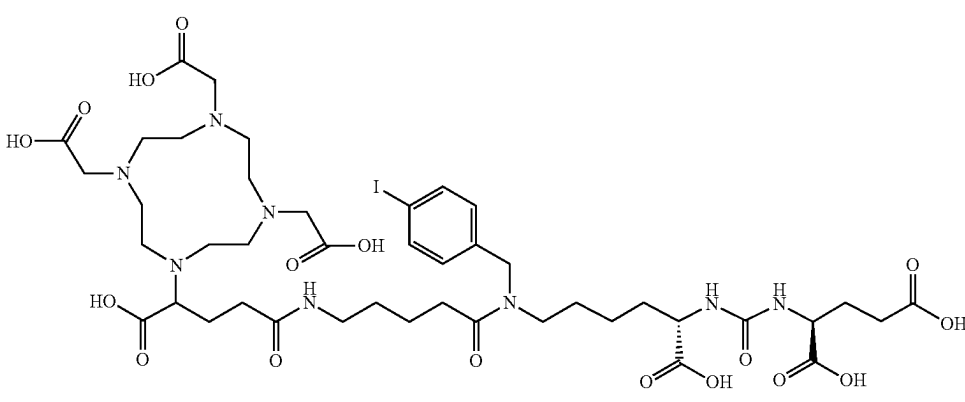
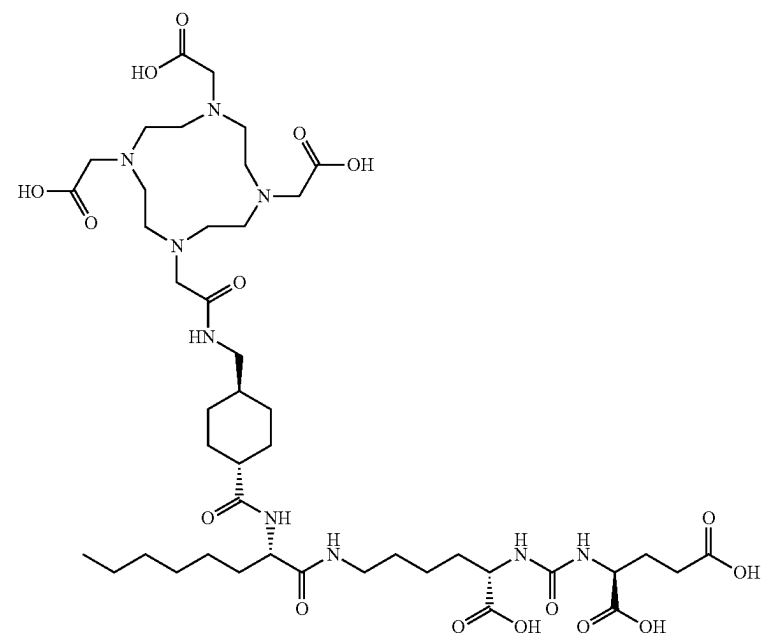
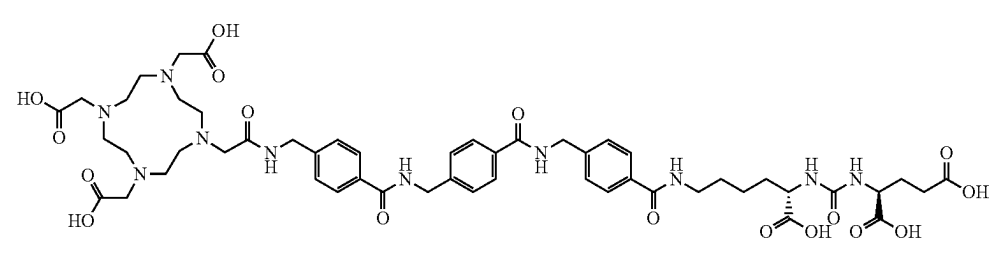
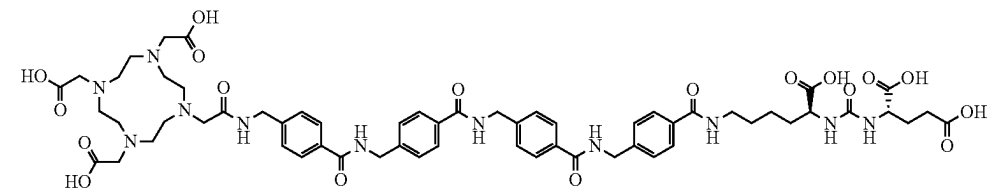
Cpd	Structure
92	
93	
94	
95	

TABLE 1-continued

Cpd	Structure
96	
97	
98	
99	
100	

TABLE 1-continued

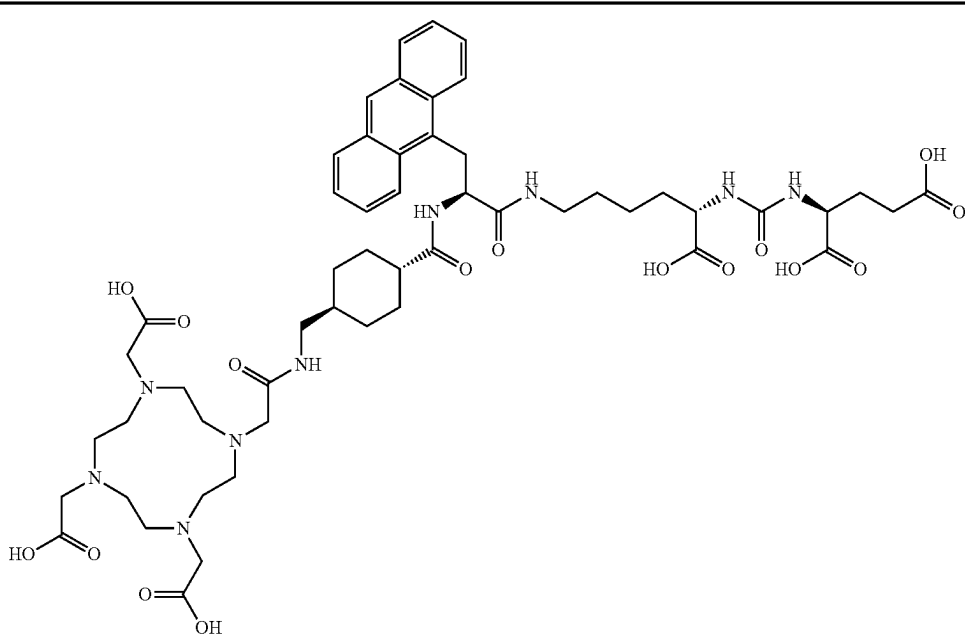
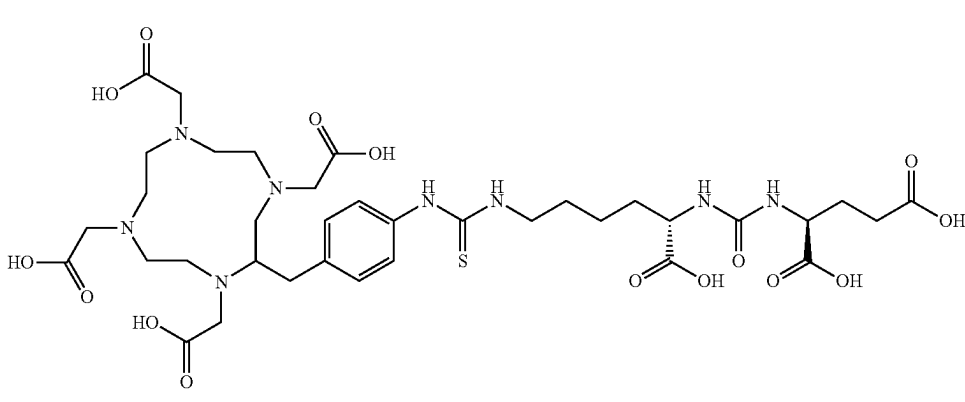
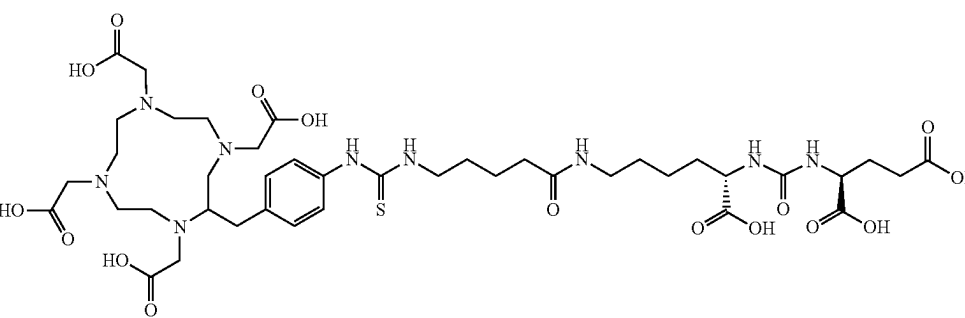
Cpd	Structure
104	 Chemical structure of compound 104: A complex molecule featuring a 1,3,5,7-tetraazabicyclo[3.3.1]nonane core. The core is substituted with four carboxylic acid groups. One of the nitrogen atoms is linked via a carbonyl group to a cyclohexyl ring, which is further connected to a long chain containing a fluorenyl group, a carbonyl, and a terminal carboxylic acid group. The chain also includes a thioamide group and a carboxylic acid group.
105	 Chemical structure of compound 105: A complex molecule featuring a 1,3,5,7-tetraazabicyclo[3.3.1]nonane core. The core is substituted with four carboxylic acid groups. One of the nitrogen atoms is linked via a carbonyl group to a phenyl ring, which is further connected to a long chain containing a thioamide group, a carboxylic acid group, and a terminal carboxylic acid group. The chain also includes a carboxylic acid group and a thioamide group.
106	 Chemical structure of compound 106: A complex molecule featuring a 1,3,5,7-tetraazabicyclo[3.3.1]nonane core. The core is substituted with four carboxylic acid groups. One of the nitrogen atoms is linked via a carbonyl group to a phenyl ring, which is further connected to a long chain containing a thioamide group, a carboxylic acid group, and a terminal carboxylic acid group. The chain also includes a carboxylic acid group and a thioamide group.

TABLE 1-continued

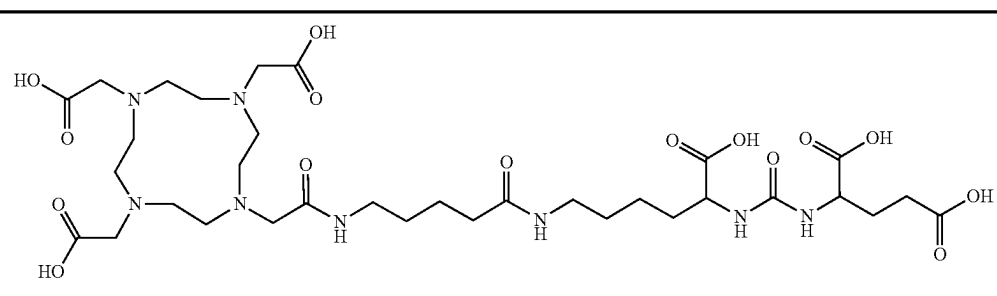
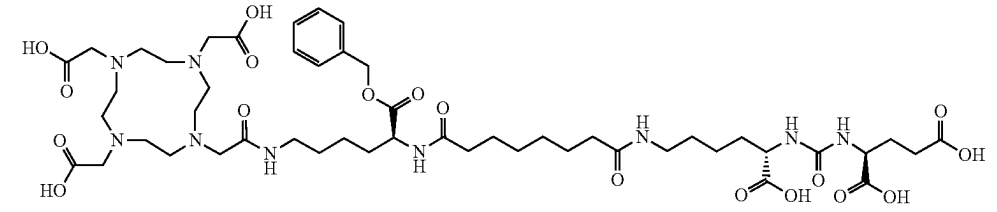
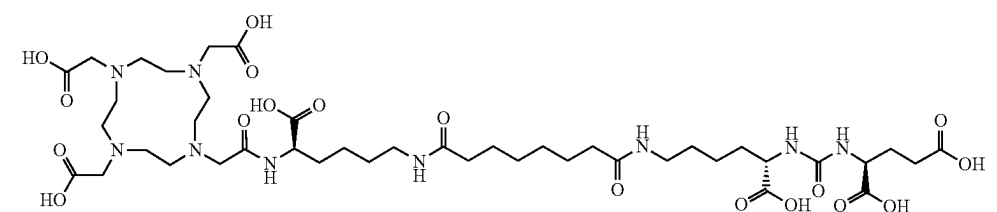
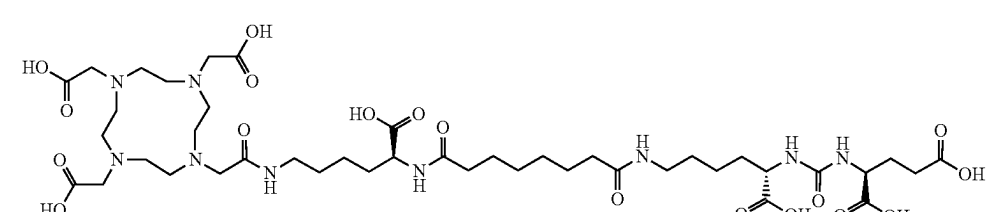
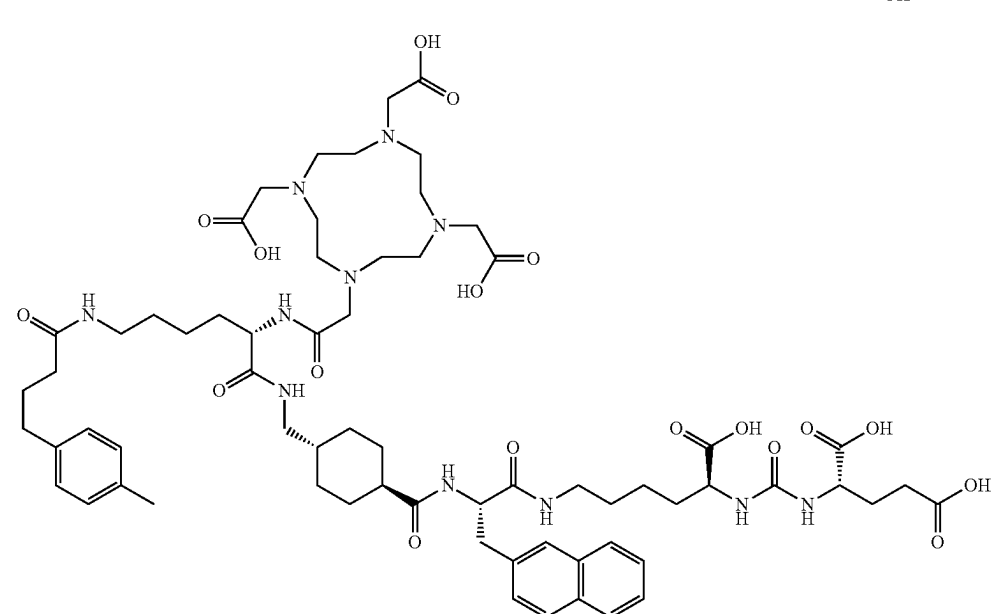
Cpd	Structure
107	
108	
109	
110	
111	

TABLE 1-continued

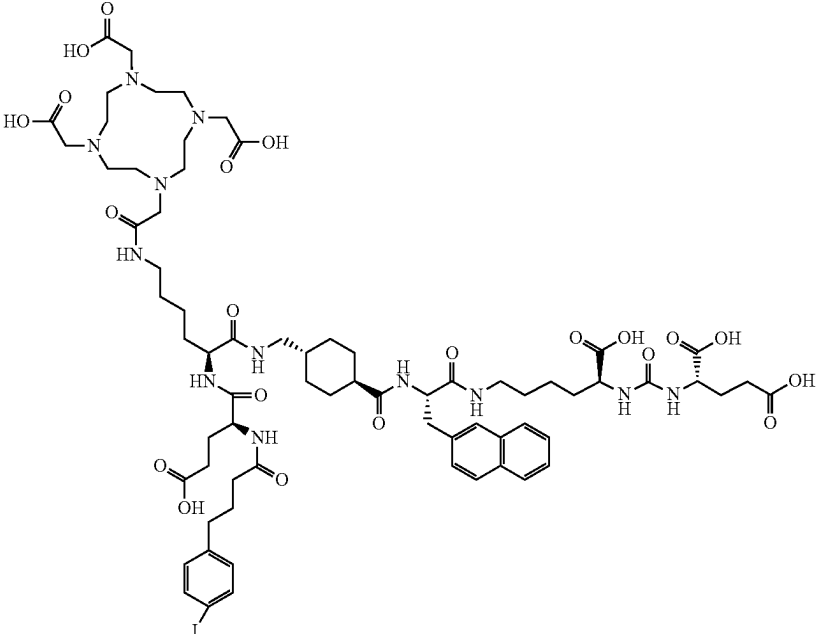
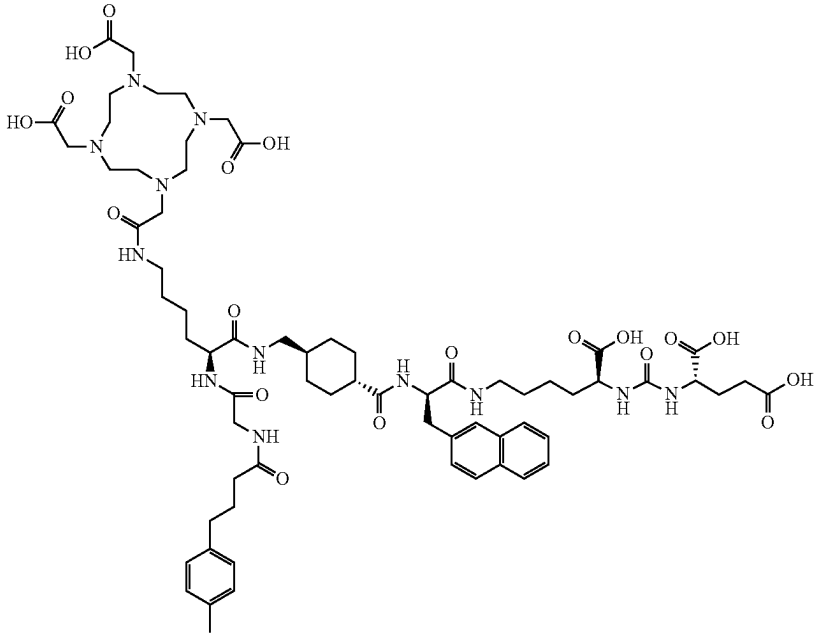
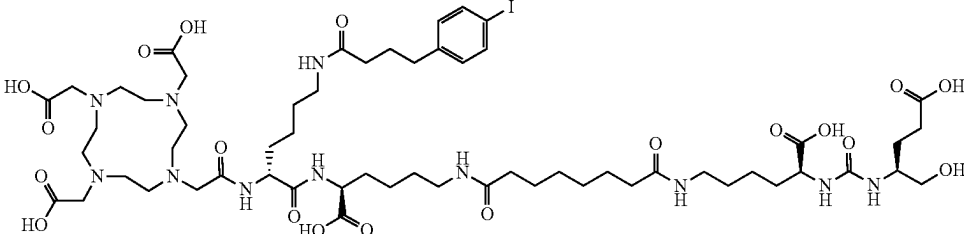
Cpd	Structure
112	
113	
114	

TABLE 1-continued

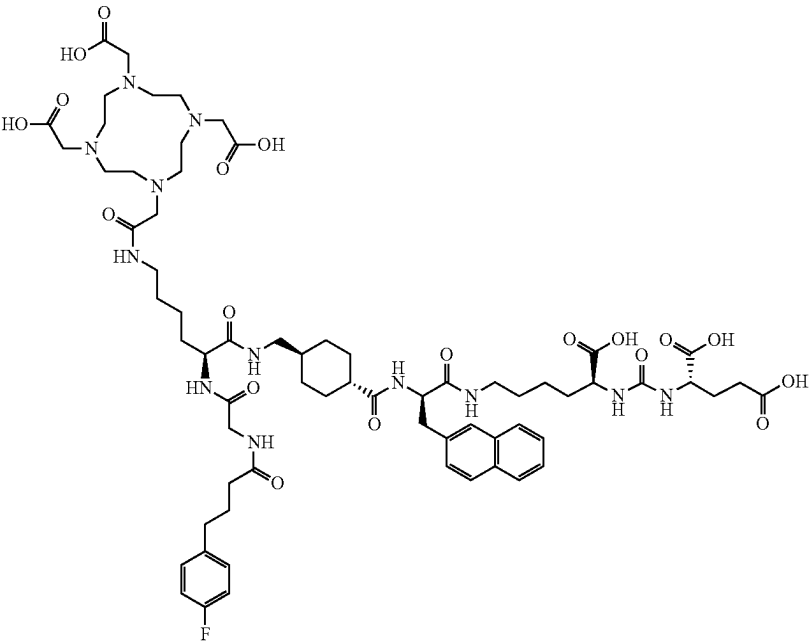
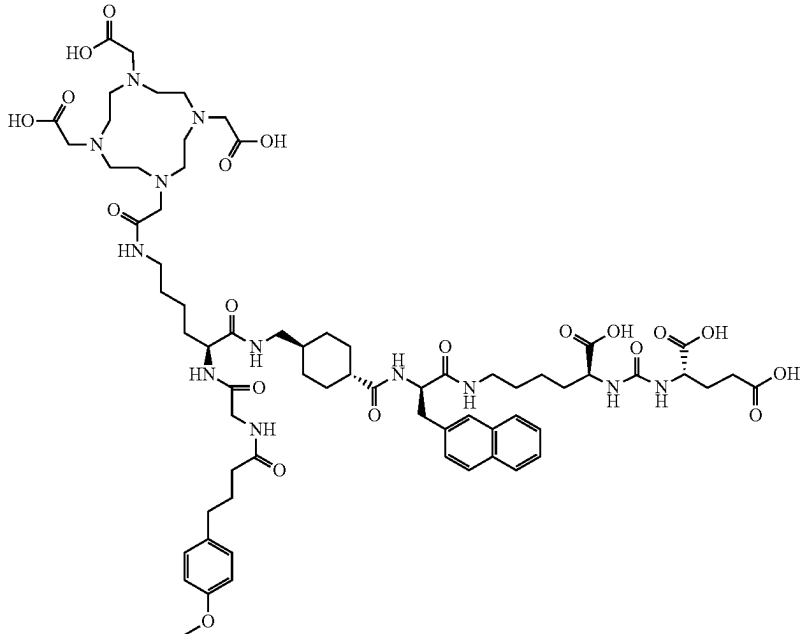
Cpd	Structure
115	 Chemical structure of compound 115. It features a 1,3,5,7-tetraazabicyclo[3.3.1]nonane core substituted with three carboxylic acid groups. One of the nitrogen atoms is linked via a carbonyl group to a long chain containing a cyclohexane ring, a naphthalene moiety, and a side chain with multiple hydroxyl and carboxylic acid groups. Another part of the chain includes a 4-fluorophenyl group.
116	 Chemical structure of compound 116. It is similar to compound 115, featuring the same 1,3,5,7-tetraazabicyclo[3.3.1]nonane core and side chains, but with a methoxy group (-OCH₃) on the phenyl ring instead of a fluorine atom.

TABLE 1-continued

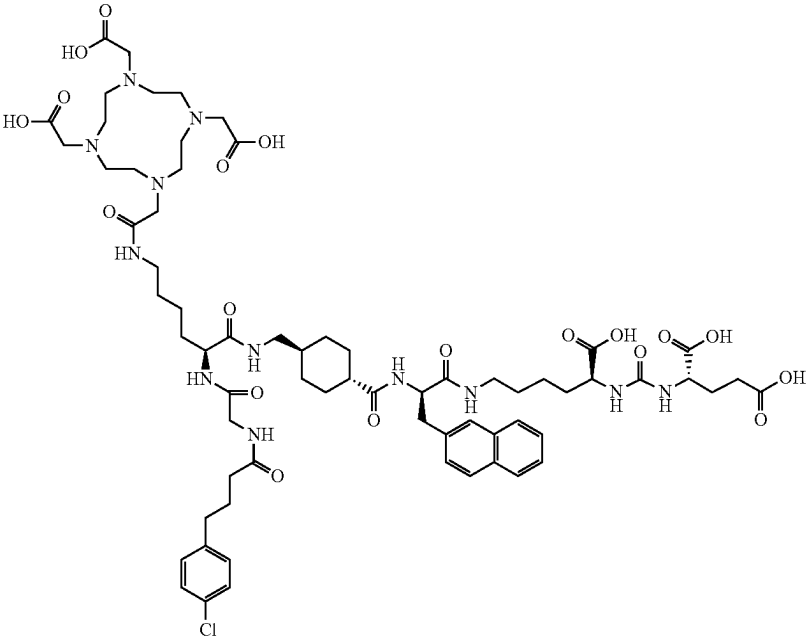
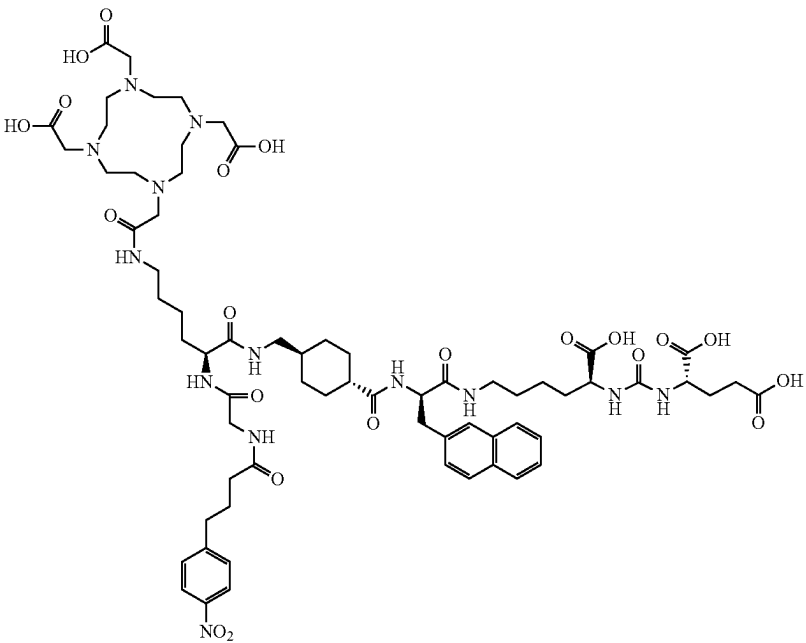
Cpd	Structure
117	 Chemical structure of compound 117. It features a 1,3,5,7-tetraazabicyclo[3.3.1]nonane core substituted with four carboxylic acid groups. One of the nitrogen atoms is linked via a carbonyl group to a long chain containing a cyclohexane ring, a naphthalene moiety, and a complex amide/urea structure with multiple hydroxyl and carboxylic acid groups. Another branch from the chain includes a 4-chlorophenyl group.
118	 Chemical structure of compound 118. It is similar to compound 117, featuring the same 1,3,5,7-tetraazabicyclo[3.3.1]nonane core and side chains, but with a nitro group (NO₂) instead of a chlorine atom on the phenyl ring.

TABLE 1-continued

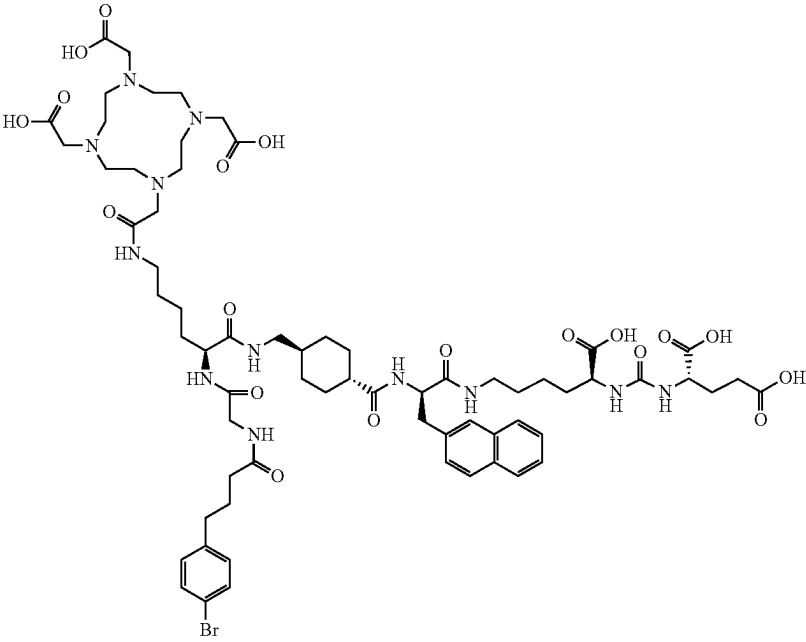
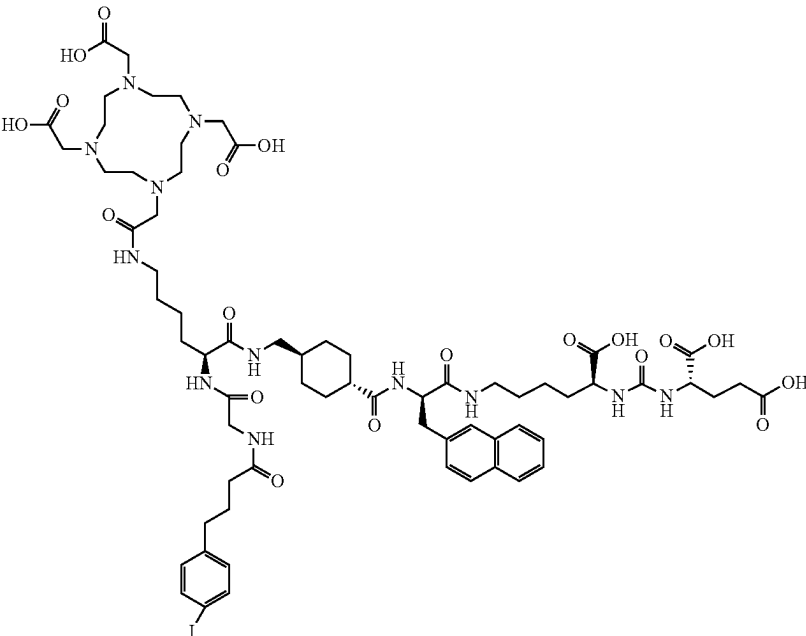
Cpd	Structure
119	 Chemical structure of compound 119. It features a 1,3,5,7-tetraazabicyclo[3.3.1]nonane core substituted with three carboxylic acid groups. One of the nitrogen atoms is linked via a carbonyl group to a long chain containing a cyclohexane ring, a naphthalene moiety, and a complex amide linkage. The chain also includes a 4-bromophenyl group and a carboxylic acid group.
120	 Chemical structure of compound 120. It is similar to compound 119, but the bromine atom on the phenyl ring is replaced by an iodine atom.

TABLE 1-continued

Cpd	Structure
121	
122	

TABLE 1-continued

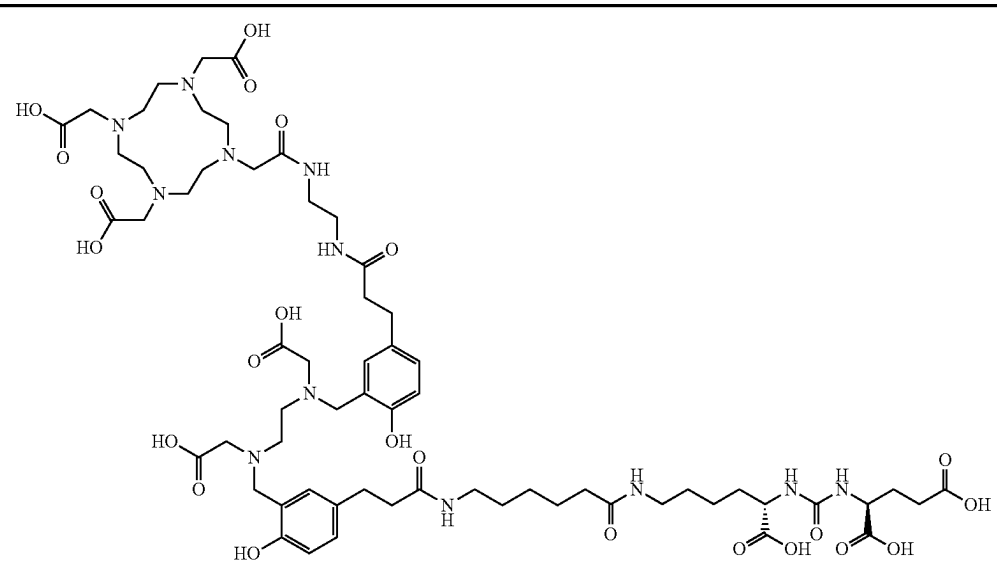
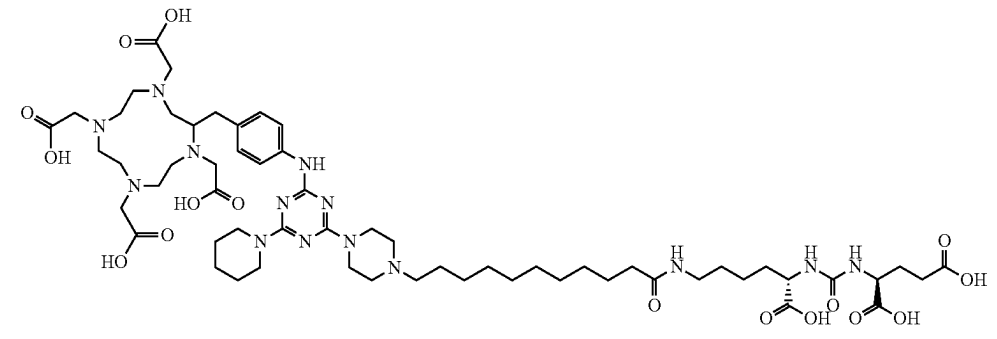
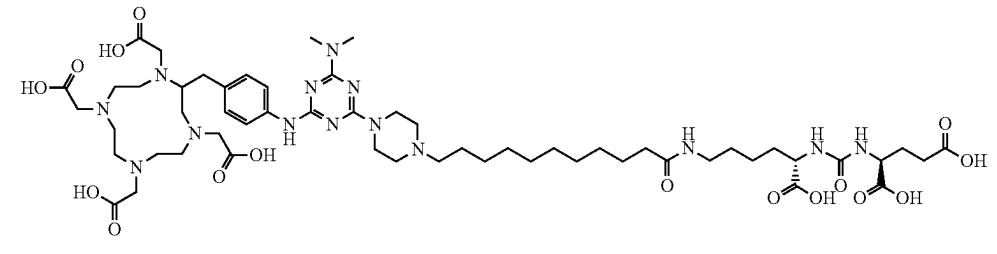
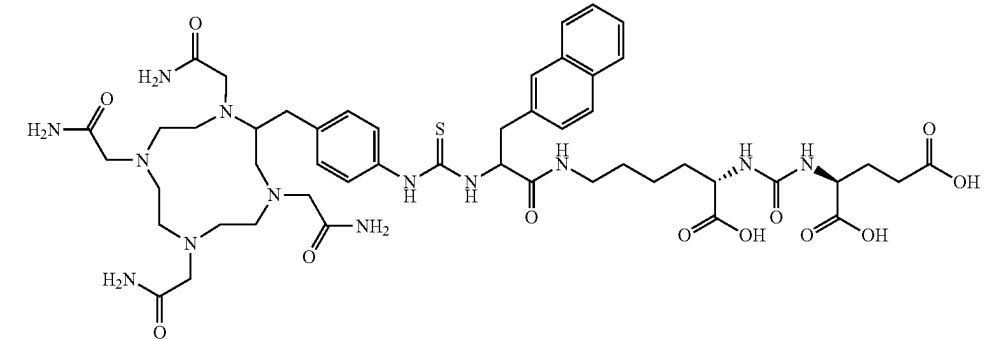
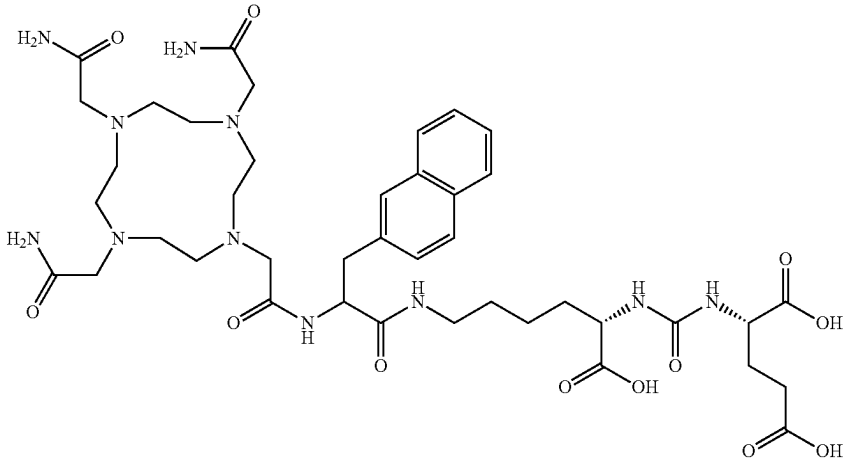
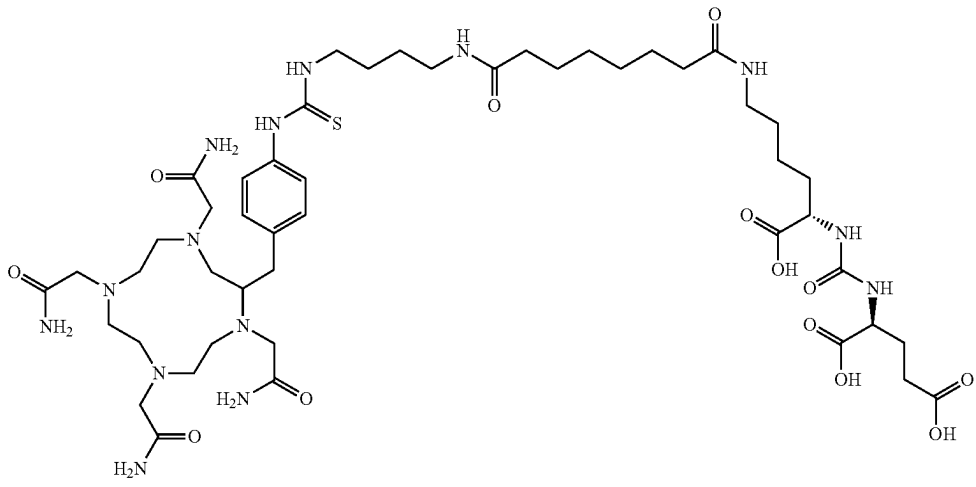
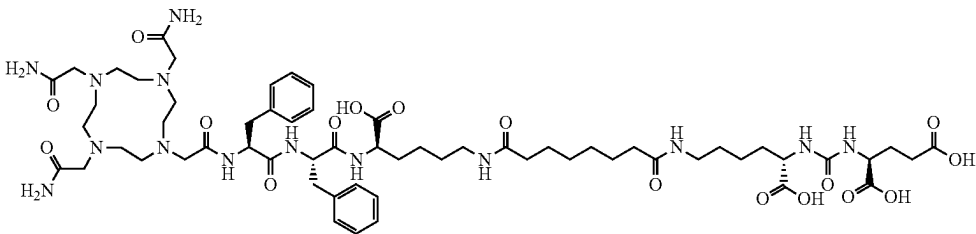
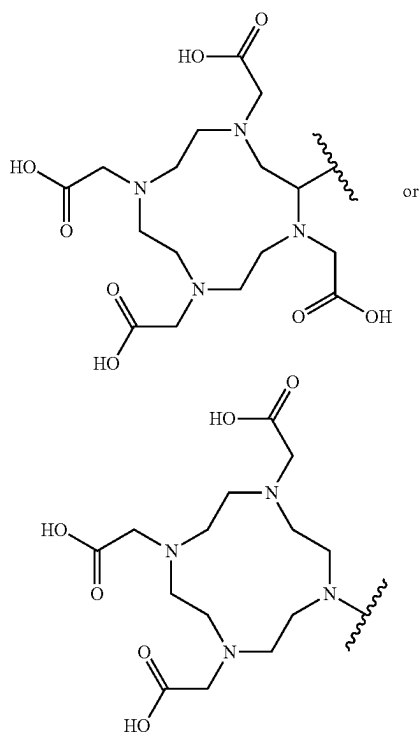
Cpd	Structure
123	
124	
125	
126	

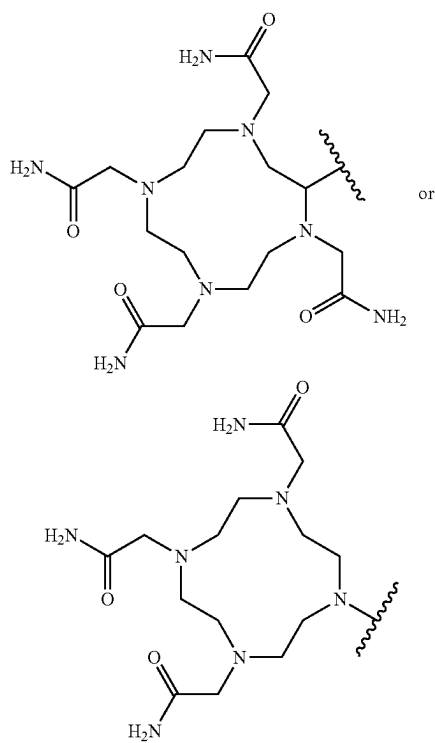
TABLE 1-continued

Cpd	Structure
127	
128	
129	

[0116] In some embodiments, a compound in Table 1 comprising the structure of

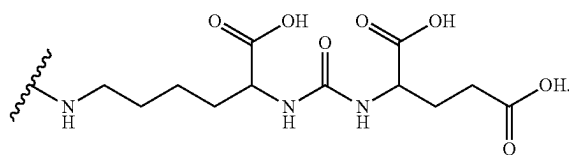


has the structure of

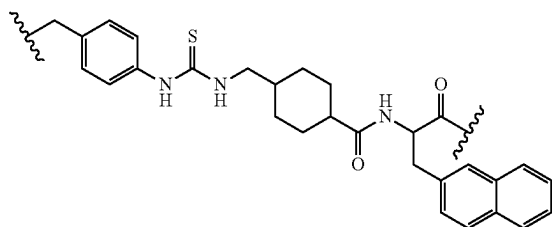


In some embodiments, a compound in Table 1 comprises a radionuclide bound to the metal chelator, wherein the radionuclide is ^{212}Pb .

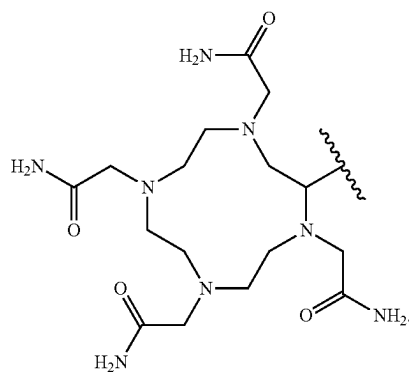
[0117] In one aspect, provided herein are radiopharmaceutical conjugates comprising a target binding moiety, a metal chelator, and a linking moiety covalently connecting the target binding moiety and the metal chelator. In some embodiments, the target binding moiety is a PSMA-binding moiety. In some embodiments, the radiopharmaceutical conjugate comprises a PSMA-binding moiety, a TCMC metal chelator, and a linking moiety covalently connecting the PSMA-binding moiety and the metal chelator. In some embodiments, the PSMA-binding moiety has the structure of



In some embodiments, the linking moiety has the structure of

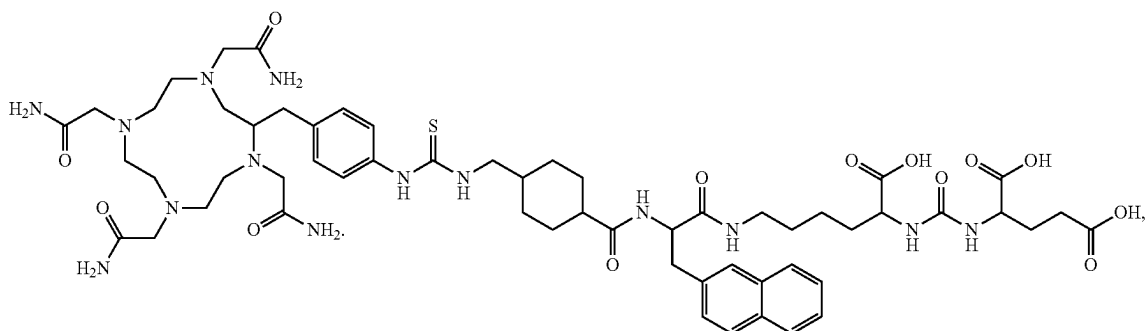


In some embodiments, the metal chelator has the structure of



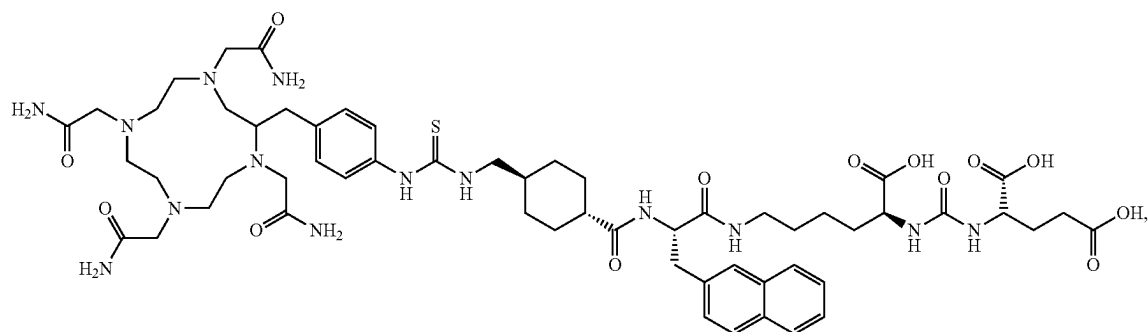
In some embodiments, the radiopharmaceutical conjugate further comprises a radionuclide bound to the metal chelator.

In some embodiments, the radiopharmaceutical conjugate comprises a compound having the structure

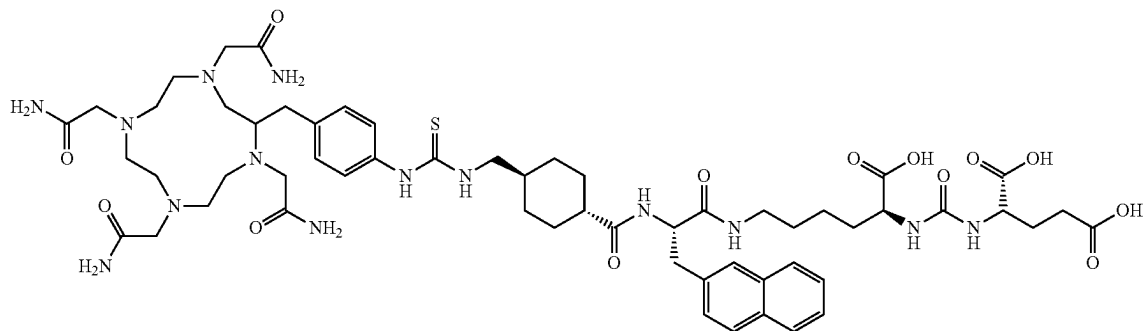


or a pharmaceutically acceptable salt thereof. In some embodiments, the compound has the structure

(Compound 1)

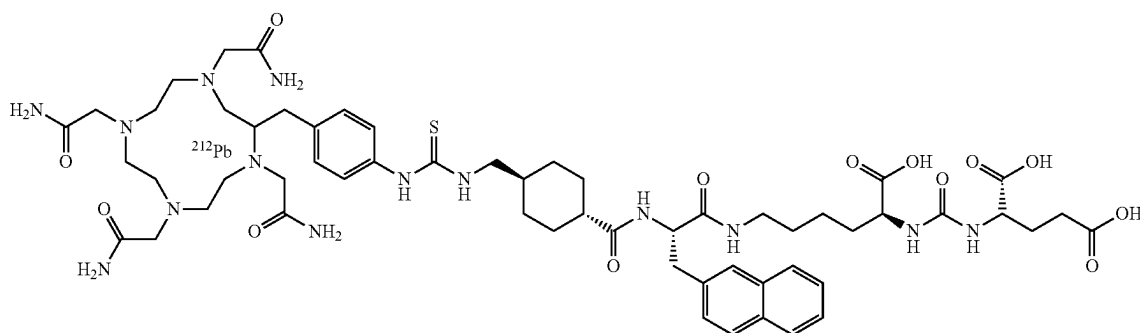


or a pharmaceutically acceptable salt thereof. In some embodiments, the radiopharmaceutical conjugate comprises a ^{212}Pb radionuclide bound to the metal chelator. In some embodiments, the radiopharmaceutical conjugate comprises a compound having the structure



of Compound 1, and a radionuclide bound to the metal chelator, wherein the radionuclide is ^{212}Pb (^{212}Pb -Compound 1). In some embodiments, provided herein is a radiopharmaceutical conjugate having a structure of

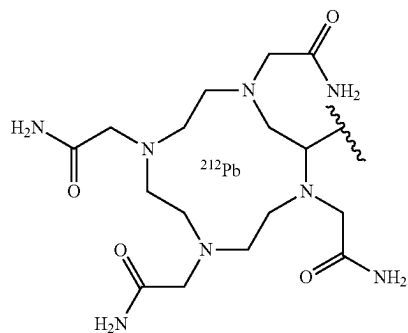
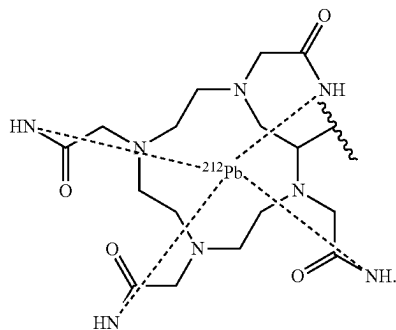
(²¹²Pb-Compound 1)



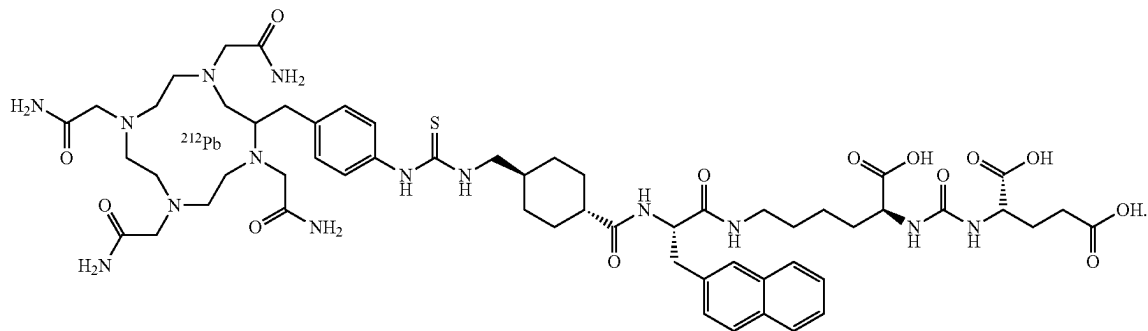
or a salt thereof.

[0118] The metal chelator, such as TCMC, can interact with the radionuclide (e.g., ^{212}Pb) via one or more functional groups and/or atoms. For example, the metal chelator can interact with the radionuclide via nitrogen and/or oxygen atoms. As another example, the metal chelator can interact with the radionuclide via amino and/or amide groups of the metal chelator. In some embodiments, the interaction of a metal chelator and a radionuclide of the conjugates disclosed herein can be illustrated as

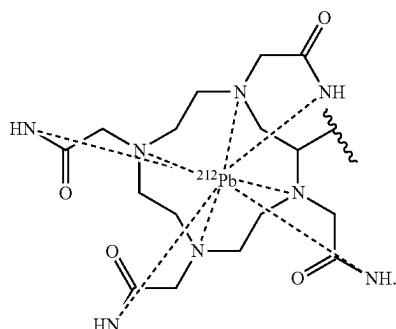
In some embodiments, the interaction of a metal chelator and a radionuclide of the conjugates disclosed herein can be illustrated as



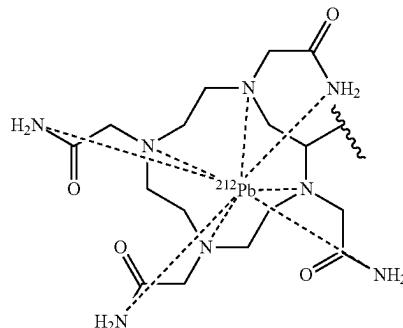
In some embodiments, ^{212}Pb -Compound 1 can be illustrated as



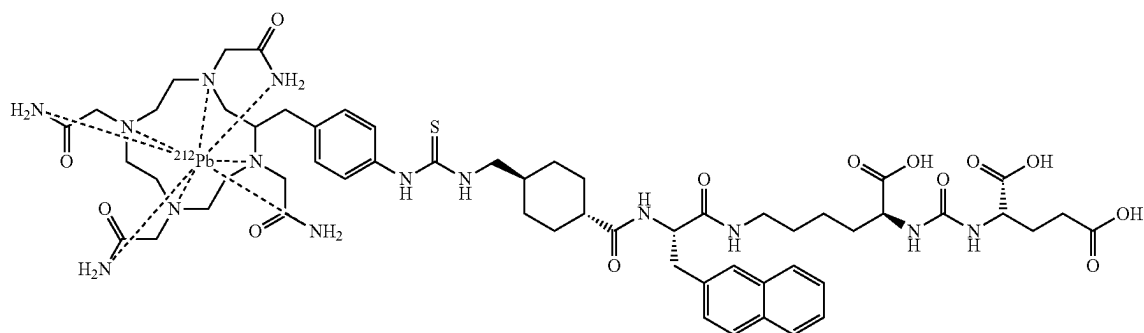
In some embodiments, the interaction of a metal chelator and a radionuclide of the conjugates disclosed herein can be illustrated as



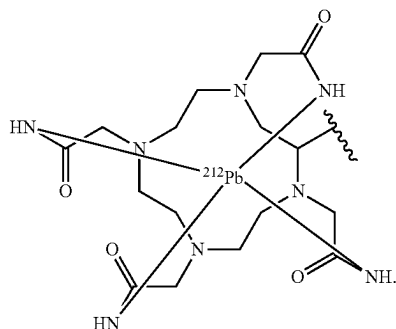
In some embodiments, the interaction of a metal chelator and a radionuclide of the conjugates disclosed herein can be illustrated as



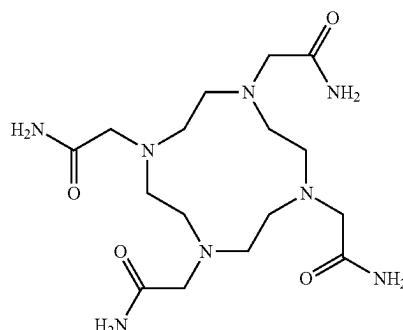
In some embodiments, ^{212}Pb -Compound 1 can be illustrated as



In some embodiments, the interaction of a metal chelator and a radionuclide of the conjugates disclosed herein can be illustrated as



[0119] TCMC (also known as DOTAM) has a structure of



when the attachment point is not shown.

[0120] In some embodiments, Compound 1 and ^{212}Pb -Compound 1 has a reduced uptake into salivary glands. In some embodiments, the reduced salivary gland uptake of ^{212}Pb -Compound 1 lowers a risk of a subject developing xerostomia. In some embodiments, the risk of a subject developing xerostomia is reduced with ^{212}Pb -Compound 1 compared to ^{177}Lu -PSMA-617 or ^{212}Ac -PSMA-617. In

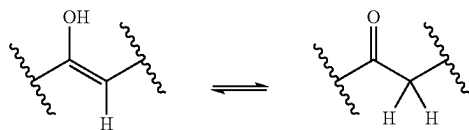
some embodiments, the subject does not develop xerostomia. In some embodiments, the ^{212}Pb -Compound 1 is administered without an agent for treating xerostomia. In some embodiments, the subject is not administered an agent for treating xerostomia. In some embodiments, a subject is selected for ^{212}Pb -Compound 1 administration due to a previous xerostomia diagnosis or is susceptible of developing xerostomia. In some embodiments, the subject was diagnosed with xerostomia prior to administering ^{212}Pb -Compound 1.

Isomers Stereoisomers

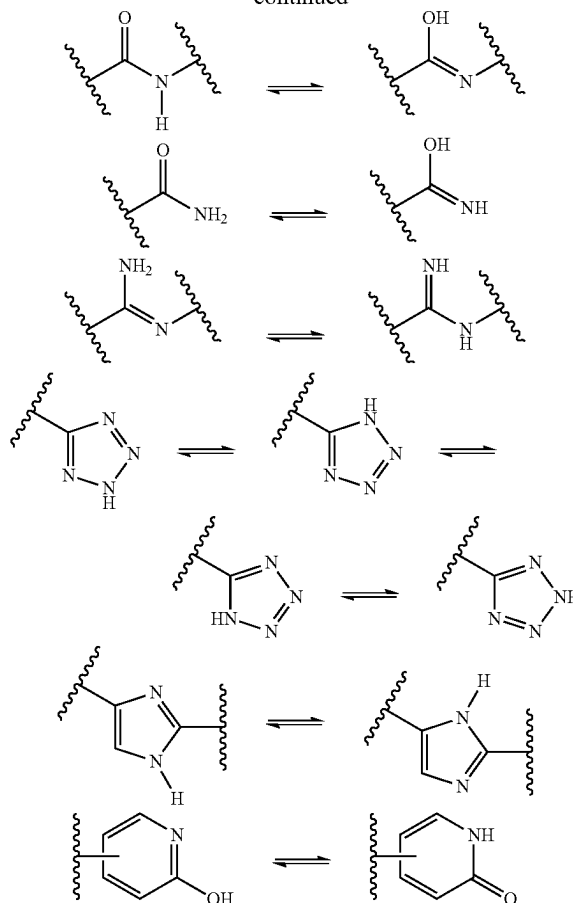
[0121] In some embodiments, the compounds described herein exist as geometric isomers. In some embodiments, the compounds described herein possess one or more double bonds. The compounds presented herein include cis, trans, syn, anti, entgegen (E), and zusammen (Z) isomers as well as the corresponding mixtures thereof. In some situations, the compounds described herein possess one or more chiral centers and each center exists in the R configuration or S configuration. The compounds described herein include diastereomeric, enantiomeric, and epimeric forms as well as the corresponding mixtures thereof. In additional embodiments of the compounds and methods provided herein, mixtures of enantiomers and/or diastereoisomers, resulting from a single preparative step, combination, or interconversion are useful for the applications described herein. In some embodiments, the compounds described herein are prepared as their individual stereoisomers by reacting a racemic mixture of the compound with an optically active resolving agent to form a pair of diastereoisomeric compounds, separating the diastereomers, and recovering the optically pure enantiomers. In some embodiments, dissociable complexes are preferred. In some embodiments, the diastereomers have distinct physical properties (e.g., melting points, boiling points, solubilities, reactivity, etc.) and are separated by taking advantage of these dissimilarities. In some embodiments, the diastereomers are separated by chiral chromatography, or preferably, by separation/resolution techniques based upon differences in solubility. In some embodiments, the optically pure enantiomer is then recovered, along with the resolving agent.

Tautomers

[0122] A “tautomer” refers to a molecule wherein a proton shift from one atom of a molecule to another atom of the same molecule is possible. The compounds presented herein, in certain embodiments, exist as tautomers. In circumstances where tautomerization is possible, a chemical equilibrium of the tautomers will exist. The exact ratio of the tautomers depends on several factors, including physical state, temperature, solvent, and pH. Some examples of tautomeric equilibrium include:



-continued



[0123] In some instances, the compounds disclosed herein exist in tautomeric forms. The structures of said compounds are illustrated in the one tautomeric form for clarity. The alternative tautomeric forms are expressly included in this disclosure.

Labeled Compounds.

[0124] In some embodiments, the compounds described herein exist in their isotopically-labeled forms. In some embodiments, the methods disclosed herein include methods of treating diseases by administering such isotopically-labeled compounds. In some embodiments, the methods disclosed herein include methods of treating diseases by administering such isotopically-labeled compounds as pharmaceutical compositions. Thus, in some embodiments, the compounds disclosed herein include isotopically-labeled compounds, which are identical to those recited herein, but for the fact that one or more atoms are replaced by an atom having an atomic mass or mass number different from the atomic mass or mass number usually found in nature. Examples of isotopes that can be incorporated into compounds described herein, or a solvate, or stereoisomer thereof, include isotopes of hydrogen, carbon, nitrogen, oxygen, phosphorous, sulfur, fluorine, and chloride, such as ^2H , ^3H , ^{13}C , ^{14}C , ^{15}N , ^{18}O , ^{17}O , ^{31}P , ^{32}P , ^{35}S , ^{18}F , and ^{36}Cl , respectively. Compounds described herein, and the pharmaceutically acceptable salts, solvates, or stereoisomers thereof

which contain the aforementioned isotopes and/or other isotopes of other atoms are within the scope of this disclosure. Certain isotopically-labeled compounds, for example those into which radioactive isotopes such as ^3H and ^{14}C are incorporated, are useful in drug and/or substrate tissue distribution assays. Tritiated, i.e., ^3H and carbon-14, i.e., ^{14}C , isotopes are notable for their ease of preparation and detectability. Further, substitution with heavy isotopes such as deuterium, i.e., ^2H , produces certain therapeutic advantages resulting from greater metabolic stability, for example increased in vivo half-life or reduced dosage requirements. In some embodiments, the isotopically labeled compound or a pharmaceutically acceptable salt, solvate, or stereoisomer thereof is prepared by any suitable method.

[0125] In some embodiments, the compounds described herein are labeled by other means, including, but not limited to, the use of chromophores or fluorescent moieties, bioluminescent labels, or chemiluminescent labels.

Pharmaceutically Acceptable Salts.

[0126] In some embodiments, the compounds described herein exist as their pharmaceutically acceptable salts. In some embodiments, the compounds disclosed herein include their pharmaceutically acceptable salts. As used herein, a “pharmaceutically acceptable salt” refers to any salt of a stabilizing agent that is useful for stabilizing the radiopharmaceutical compositions. As used herein, a “pharmaceutically acceptable salt” refers to any salt of a stabilizing agent that is useful for preventing or delaying the decomposition of the radiopharmaceutical within the compositions.

[0127] In some embodiments, the compounds described herein possess acidic or basic groups and therefore react with any of a number of inorganic or organic bases, and inorganic and organic acids, to form a pharmaceutically acceptable salt. In some embodiments, these salts are prepared in situ during the final isolation and purification of the compounds disclosed herein, or by separately reacting a purified compound in its free form with a suitable acid or base, and isolating the salt thus formed.

[0128] Examples of pharmaceutically acceptable salts include those salts prepared by reaction of the compounds described herein with a mineral acid, organic acid, or inorganic base, such salts including acetate, acrylate, adipate, alginate, aspartate, benzoate, benzenesulfonate, bisulfate, bisulfite, bromide, butyrate, butyn-1,4-dioate, camphorate, camphorsulfonate, caproate, caprylate, chlorobenzoate, chloride, citrate, cyclopentanepropionate, decanoate, digluconate, dihydrogenphosphate, dinitrobenzoate, dodecylsulfate, ethanesulfonate, formate, fumarate, glucoheptanoate, glycerophosphate, glycolate, hemisulfate, heptanoate, hexanoate, hexyne-1,6-dioate, hydroxybenzoate, γ -hydroxybutyrate, hydrochloride, hydrobromide, hydroiodide, 2-hydroxyethanesulfonate, iodide, isobutyrate, lactate, maleate, malonate, methanesulfonate, mandelate, metaphosphate, methanesulfonate, methoxybenzoate, methylbenzoate, monohydrogenphosphate, 1-naphthalenesulfonate, 2-naphthalenesulfonate, nicotinate, nitrate, palmoate, pectinate, persulfate, 3-phenylpropionate, phosphate, picrate, pivalate, propionate, pyrosulfate, pyrophosphate, propiolate, phthalate, phenylacetate, phenylbutyrate, propanesulfonate, salicylate, succinate, sulfate, sulfite, succinate, suberate, sebacate, sulfonate, tartrate, thiocyanate, tosylate, undecanoate, and xylenesulfonate.

[0129] Further, the compounds described herein can be prepared as pharmaceutically acceptable salts formed by reacting the free base form of the compound with a pharmaceutically acceptable inorganic or organic acid, including, but not limited to, inorganic acids such as hydrochloric acid, hydrobromic acid, sulfuric acid, nitric acid, phosphoric acid, metaphosphoric acid, and the like; and organic acids such as acetic acid, propionic acid, hexanoic acid, cyclopentanepropionic acid, glycolic acid, pyruvic acid, lactic acid, malonic acid, succinic acid, malic acid, maleic acid, fumaric acid, p-toluenesulfonic acid, tartaric acid, trifluoroacetic acid, citric acid, benzoic acid, 3-(4-hydroxybenzoyl) benzoic acid, cinnamic acid, mandelic acid, arylsulfonic acid, methanesulfonic acid, ethanesulfonic acid, 1,2-ethanedisulfonic acid, 2-hydroxyethanesulfonic acid, benzenesulfonic acid, 2-naphthalenesulfonic acid, 4-methylbicyclo-[2.2.2]oct-2-ene-1-carboxylic acid, glucoheptonic acid, 4,4'-methylenebis-(3-hydroxy-2-ene-1-carboxylic acid), 3-phenylpropionic acid, trimethylacetic acid, tertiary butylacetic acid, lauryl sulfuric acid, gluconic acid, glutamic acid, hydroxynaphthoic acid, salicylic acid, stearic acid, and muconic acid.

[0130] In some embodiments, the compounds described herein which comprise a free acid group react with a suitable base, such as the hydroxide, carbonate, bicarbonate, or sulfate of a pharmaceutically acceptable metal cation, with ammonia, or with a pharmaceutically acceptable organic primary, secondary, tertiary, or quaternary amine. Representative salts include the alkali or alkaline earth salts, like lithium, sodium, potassium, calcium, and magnesium, and aluminum salts, and the like. Illustrative examples of bases include sodium hydroxide, potassium hydroxide, choline hydroxide, sodium carbonate, $\text{N}^+(\text{C}_{1-4} \text{ alkyl})_4$, and the like.

[0131] Representative organic amines useful for the formation of base addition salts include ethylamine, diethylamine, ethylenediamine, ethanolamine, diethanolamine, piperazine, and the like. It should be understood that the compounds described herein also include the quaternization of any basic nitrogen-containing groups they contain. In some embodiments, water or oil-soluble or dispersible products are obtained by such quaternization.

Solvates.

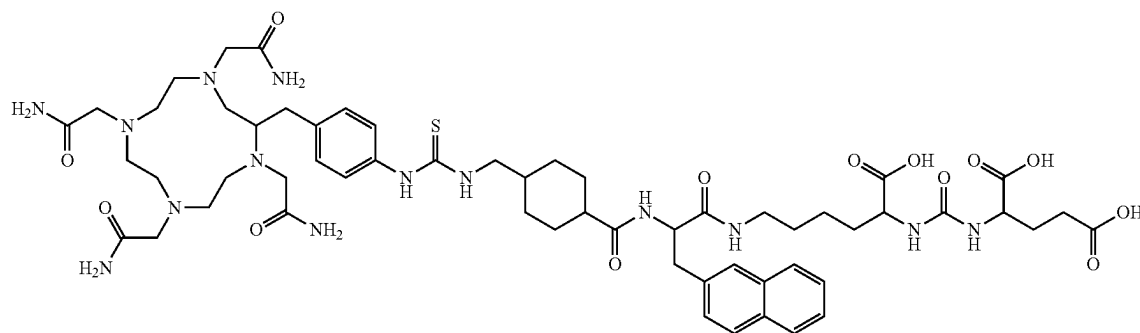
[0132] In some embodiments, the compounds described herein exist as solvates. Solvates contain either stoichiometric or non-stoichiometric amounts of a solvent, and, in some embodiments, are formed during the process of crystallization with pharmaceutically acceptable solvents such as water, ethanol, and the like. Hydrates are formed when the solvent is water, or alcoholates are formed when the solvent is alcohol. Solvates of the compounds described herein can be conveniently prepared or formed during the processes described herein. In addition, the compounds provided herein can exist in unsolvated as well as solvated forms. In general, the solvated forms are considered equivalent to the unsolvated forms for the purposes of the compounds and methods provided herein. Accordingly, one aspect of the present disclosure pertains to hydrates and solvates of compounds of the present disclosure and/or their pharmaceutical acceptable salts, as described herein, that can be isolated and characterized by methods known in the art, such as, thermogravimetric analysis (TGA), TGA-mass spectroscopy, TGA-Infrared spectroscopy, powder X-ray diffraction (PXRD), Karl Fisher titration, high resolution X-ray diffraction, and the like.

Pharmaceutical Compositions

[0133] The radiopharmaceutical conjugate described herein, including e.g., pharmaceutically acceptable salts or solvates thereof, can be administered as a component of a pharmaceutically acceptable formulation. In some embodiments, the radiopharmaceutical conjugate described herein is combined with a pharmaceutically suitable or acceptable carrier selected on the basis of a chosen route of administration and standard pharmaceutical practice as described, for example, in Remington: The Science and Practice of Pharmacy (Gennaro, 21st Ed. Mack Pub. Co., Easton, PA (2005)). Provided herein is a radiopharmaceutical pharmaceutical composition comprising a conjugate described herein, or a stereoisomer, pharmaceutically acceptable salt, amide, ester, solvate, or N-oxide thereof, together with one or more pharmaceutically acceptable carriers. The carrier(s) (or excipient(s)) is acceptable or suitable if the carrier is compatible with the other ingredients of the composition and not deleterious to the recipient (i.e., the subject or patient) of the composition.

[0134] In one aspect, the disclosure provides a liquid radiopharmaceutical composition comprising a herein described conjugate, or a pharmaceutically acceptable salt or solvate thereof, and a pharmaceutically acceptable excipient or carrier. In some embodiments, the liquid radiopharmaceutical composition comprises

[0135] (a) a compound having the structure:



or a pharmaceutically acceptable salt thereof; and

[0136] (b) a radionuclide bound to the metal chelator, wherein the radionuclide is ^{212}Pb , and wherein the liquid radiopharmaceutical composition has a radioactivity of about 75 MBq to about 300 MBq.

In some embodiments, the liquid radiopharmaceutical composition has a radioactivity of about 75 MBq to about 200 MBq. In some embodiments, the liquid radiopharmaceutical has a radioactivity of about 75 MBq. In some embodiments, the liquid radiopharmaceutical has a radioactivity of about 100 MBq. In some embodiments, wherein the liquid radiopharmaceutical has a radioactivity of about 150 MBq. In some embodiments, the liquid radiopharmaceutical has a radioactivity of about 200 MBq. In some embodiments, the liquid radiopharmaceutical has a radioactivity of about 300 MBq. In some embodiments, wherein the compound is present in an amount of about 1 μg to about 1,000 μg . In some embodiments, wherein the compound is present in an amount of about 50 μg to about 500 μg . In some embodiments, wherein the compound is present in an amount of

about 50 μg to about 300 μg . In some embodiments, wherein the compound is present in an amount of about 50 μg to about 150 μg . In some embodiments, wherein the compound is present in an amount of about 100 μg to about 200 μg . In some embodiments, wherein the compound is present in an amount of about 150 μg to about 250 μg . In some embodiments, wherein the compound is present in an amount of about 200 μg to about 300 μg . In some embodiments, wherein the compound is present in an amount of about 250 μg to about 350 μg . In some embodiments, wherein the compound is present in an amount of about 300 μg to about 400 μg . In some embodiments, wherein the compound is present in an amount of about 350 μg to about 450 μg . In some embodiments, wherein the compound is present in an amount of about 400 μg to about 500 μg . In some embodiments, wherein the compound is present in an amount of about 50 μg to about 100 μg . In some embodiments, wherein the compound is present in an amount of about 75 μg to about 125 μg . In some embodiments, wherein the compound is present in an amount of about 100 μg to about 150 μg . In some embodiments, wherein the compound is present in an amount of about 125 μg to about 175 μg . In some embodiments, wherein the compound is present in an amount of about 150 μg to about 200 μg . In some embodiments, wherein the compound is present in an amount of about 175 μg to about 225 μg . In some embodiments, wherein the compound is present in an amount of about 200 μg to about

250 μg . In some embodiments, wherein the compound is present in an amount of about 225 μg to about 275 μg . In some embodiments, wherein the compound is present in an amount of about 250 μg to about 300 μg .

[0137] The conjugates and pharmaceutical compositions of the current disclosure can be administered by any suitable means, e.g., by injection, including intravenous administration.

IV. Method of Treatment

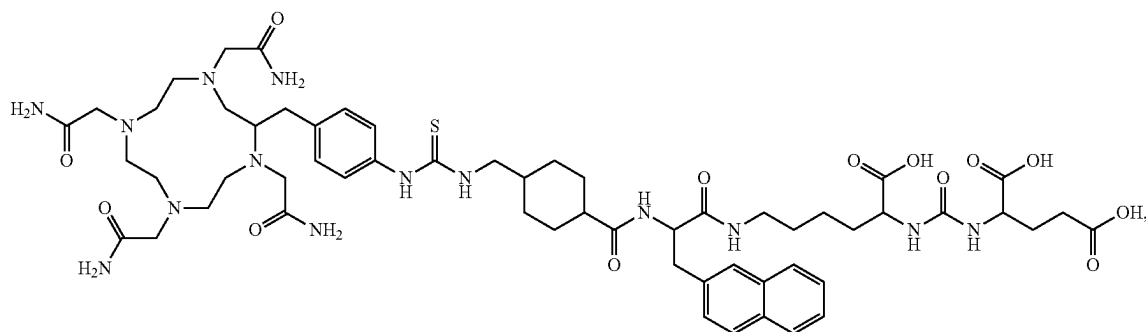
[0138] In one aspect, the disclosure provides methods of treating a disease or condition in a subject in need thereof. The methods can comprise administering a radiopharmaceutical conjugate, for example, a radiopharmaceutical of Formula (I), to the subject in need thereof. The methods can provide a therapeutic and/or prophylactic benefit to a subject in need thereof comprising administering a radiopharmaceutical conjugate described herein. In one aspect, the disclosure provides methods of preventing the formation of metastasis in a subject in need thereof.

[0139] The methods can comprise administering to a subject a therapeutically effective amount of a radiopharmaceutical conjugate or a pharmaceutically acceptable salt or solvate thereof. In some embodiments, the subject has cancer. In some embodiments, the cancer is a solid tumor. In some embodiments, the cancer is a PSMA-associated cancer. In some embodiments, the cancer is a PSMA positive cancer. In some embodiments, the cancer is prostate cancer. In some embodiments, the prostate cancer is metastatic castrate resistant prostate cancer (mCRPC). In some embodiments, the prostate cancer is progressive mCRPC. In some embodiments, provided herein is a method of treating a metastatic lesion in a subject in need thereof, comprising administering to the subject a radiopharmaceutical conjugate or a pharmaceutically acceptable salt thereof. In some embodiments, provided herein is a method of preventing or reducing the formation of a metastatic lesion in a subject in need thereof, comprising administering to the subject a radiopharmaceutical conjugate or a pharmaceutically acceptable salt thereof. In some embodiments, provided herein is a method of preventing the formation of a metastatic lesion in a subject in need thereof, comprising administering to the subject a radiopharmaceutical conjugate or a pharmaceutically acceptable salt thereof. In some embodiments, the subject has prostate cancer. In some embodiments, metastatic lesions are present in bone, viscera, lymph node(s), or a combination thereof. In some embodiments,

the metastatic lesion is present in bone. In some embodiments, the metastatic lesion is present in viscera. In some embodiments, the metastatic lesion is present in a lymph node(s). In some embodiments, the metastatic lesion is present in the lung, liver, heart, or kidney. In some embodiments, the metastatic lesions have an EANM PSMA-PET score of 1, 2, or 3. In some embodiments, the EANM PSMA-PET score of the metastatic lesion is 1. In some embodiments, the EANM PSMA-PET score of the metastatic lesion is 2. In some embodiments, the EANM PSMA-PET score of the metastatic lesion is 3. In some embodiments, the subject does not have a metastatic lesion. In some embodiments, the metastatic lesion is undetectable or not detected. In some embodiments, the subject does not have a PSMA-positive metastatic lesion.

[0140] In one aspect, provided herein is a method of treating a prostate cancer in a subject in need thereof, comprising administering to the subject a radiopharmaceutical conjugate of Formula (I), Formula (I-A), Formula (I-A-1), Formula (I-B-1), or Formula (I-B-2), wherein the radiopharmaceutical conjugate is administered without an agent for treating xerostomia. In some embodiments, provided herein is a method of treating a prostate cancer in a subject in need thereof, comprising administering to the subject a radiopharmaceutical conjugate, wherein the radiopharmaceutical conjugate comprises,

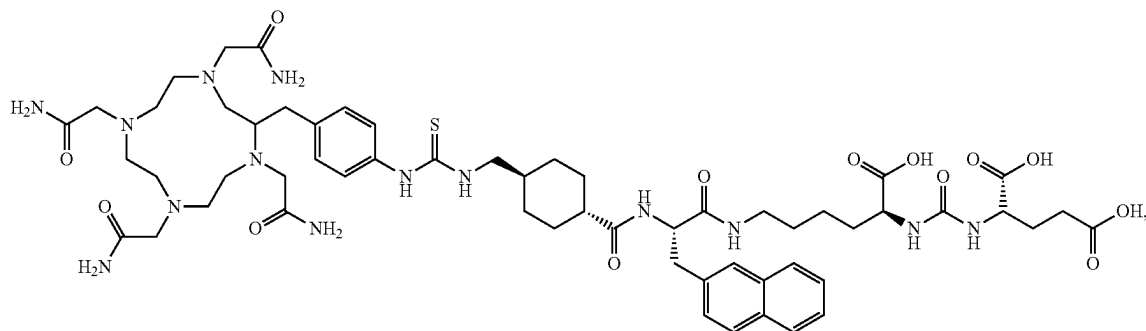
[0141] (a) a compound having the structure:



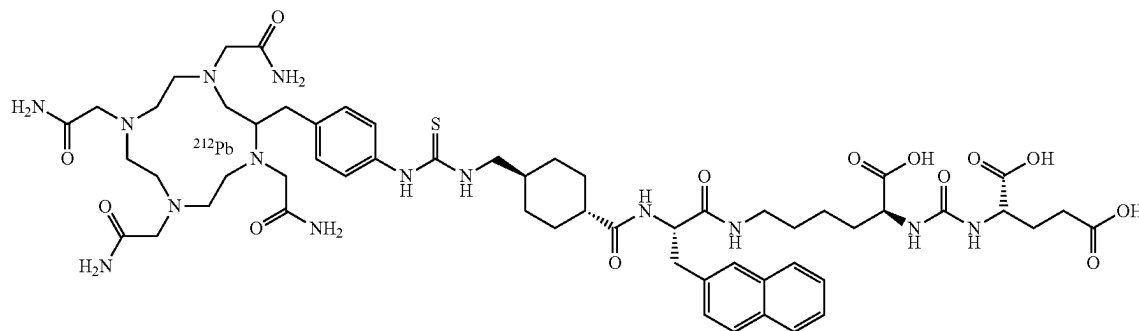
or a pharmaceutically acceptable salt thereof; and

[0142] (b) a radionuclide bound to the metal chelator, wherein the radionuclide is ^{212}Pb ;

wherein the radiopharmaceutical conjugate is administered without an agent for treating xerostomia. In some embodiments, the method comprises administering a therapeutically effective amount of the radiopharmaceutical conjugate to the subject. In some embodiments, the radiopharmaceutical conjugate comprises a compound having the structure of



or a pharmaceutically acceptable salt thereof. In some embodiments, the radiopharmaceutical conjugate has a structure of



or a salt thereof. In some embodiments, the subject has previously received an anti-cancer agent before the administering of the radiopharmaceutical conjugate, and wherein the anti-cancer agent is external beam radiation, chemotherapy, hormonal therapy, or a combination thereof. In some embodiments, the anti-cancer therapy is external beam radiation. In some embodiments, the anti-cancer therapy is chemotherapy. In some embodiments, the chemotherapy is a taxane (for example, docetaxel). In some embodiments, the anti-cancer therapy is hormonal therapy (for example, bicalutamide, flutamide, and goserelin). In some embodiments, the anti-cancer therapy is an androgen receptor axis targeted agent. In some embodiments, the radiopharmaceutical conjugate is administered in combination with a second anti-cancer agent, and wherein the second anti-cancer agent is administered concurrently or sequentially with the radiopharmaceutical conjugate. In some embodiments, the second anti-cancer agent is administered sequentially (e.g., after) the radiopharmaceutical conjugate. In some embodiments, the second anti-cancer agent is administered concurrently with the radiopharmaceutical conjugate. In some embodiments, the second anti-cancer agent is administered prior to administering the radiopharmaceutical conjugate. In some embodiments, the second anti-cancer agent is an androgen receptor-axis targeted agent. In some embodiments, the second anti-cancer agent is abiraterone, enzalutamide, nilutamide, flutamide, bicalutamide, ARN 509, galeterone, orteronel, or a salt thereof. In some embodiments, the second anti-cancer agent is enzalutamide, goserelin, leuporelin, or a salt thereof. In some embodiments, the radiopharmaceutical conjugate is administered an amount of 50 kBq to 500 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered an amount of 100 kBq to 500 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 1 MBq to 500 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 10 MBq to 500 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 50 MBq to 500 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 75 MBq to about 300 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered at a dose of about 10 MBq. In some embodiments, the radiopharmaceutical conjugate is administered at a dose of about 100 MBq. In some embodiments, the agent for treating xerostomia includes agents for treating dry mouth, apthyalism, or dry throat (e.g., spray substitution of saliva). In some embodiments, the subject is not administered the agent for treating xerostomia. In some

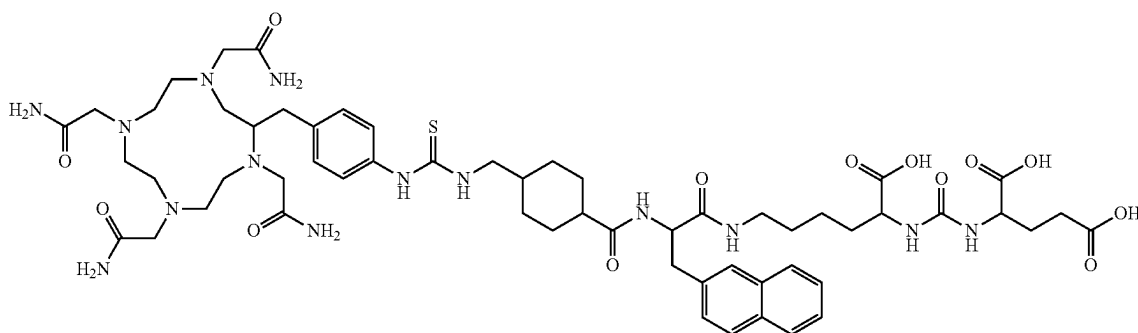
embodiments, the subject is not administered an agent for treating xerostomia for a period of 3 months, 6 months, 9 months, or 12 months since the administering of the radio-

pharmaceutical conjugate. In some embodiments, the subject is not administered an agent for treating xerostomia for a period of 3 months, 6 months, 9 months, or 12 months prior to the administering of the radiopharmaceutical conjugate. In some embodiments, the subject is not administered an agent for treating xerostomia for a period of at least 3 months, at least 6 months, at least 9 months, or at least 12 months before and after the administering of the radiopharmaceutical conjugate. In some embodiments, the subject is not administered an agent for treating xerostomia for a period of at least 4 weeks, at least 8 weeks, at least 12 weeks, or at least 20 weeks before and after the administering of the radiopharmaceutical conjugate. In some embodiments, administering the radiopharmaceutical conjugate reduces a risk of xerostomia in the subject, optionally wherein the risk is reduced relative to the administering of ^{177}Lu -PSMA-617 or ^{225}Ac -PSMA-617. In some embodiments, the risk is relative to the administering of ^{225}Ac -PSMA-617. In some embodiments, the subject has received a targeted radiopharmaceutical therapy prior to administering of the radiopharmaceutical conjugate. In some embodiments, the targeted radiopharmaceutical therapy comprises ^{177}Lu -PSMA-617 or ^{225}Ac -PSMA-617. In some embodiments, the targeted radiopharmaceutical therapy comprises ^{177}Lu -PSMA-617. In some embodiments, the subject has progressive prostate cancer after prior treatment with ^{177}Lu -PSMA-617 or ^{225}Ac -PSMA-617. In some embodiments, the subject is fasted prior to administering of the radiopharmaceutical conjugate. In some embodiments, the administering of the radiopharmaceutical conjugate does not cause or increase the risk of xerostomia in the subject. In some embodiments, the radiopharmaceutical conjugate is not significantly taken up in a salivary gland in the subject. In some embodiments, the subject does not develop xerostomia after the administering of the radiopharmaceutical conjugate. In some embodiments, the subject was previously diagnosed with xerostomia prior to administering the radiopharmaceutical conjugate. In some embodiments, the subject is susceptible of developing xerostomia. In some embodiments, the subject has an increased risk of developing xerostomia. In some embodiments, the subject is diagnosed as having an increased risk of developing xerostomia. In some embodiments, the administering does not cause any one or more of xerostomia, radiation nephropathy, or bone marrow toxicity, or hematological toxicity in the subject. In some embodiments, the prostate cancer is metastatic castrate resistant prostate cancer (mCRPC). In some embodiments, the prostate cancer is progressive mCRPC. In some embodiments, administering the radiopharmaceutical conjugate prevents or reduces the formation of a metastatic lesion in the subject.

In some embodiments, the metastatic lesion comprises a metastatic lesion in bone, viscera, lymph node(s), or a combination thereof. In some embodiments, the metastatic lesion is a metastatic bone lesion. In some embodiments, the metastatic lesion is present in the subject prior to administering the radiopharmaceutical conjugate, and administering the radiopharmaceutical conjugate treats, reduces, or eliminates the metastatic lesion. In some embodiments, the

subject has previously received an anti-cancer agent before the administering of the radiopharmaceutical conjugate. In some embodiments, provided herein is a method of treating a prostate cancer in a subject in need thereof, comprising administering to the subject a radiopharmaceutical conjugate, wherein the radiopharmaceutical conjugate comprises,

[0144] (a) a compound having the structure:

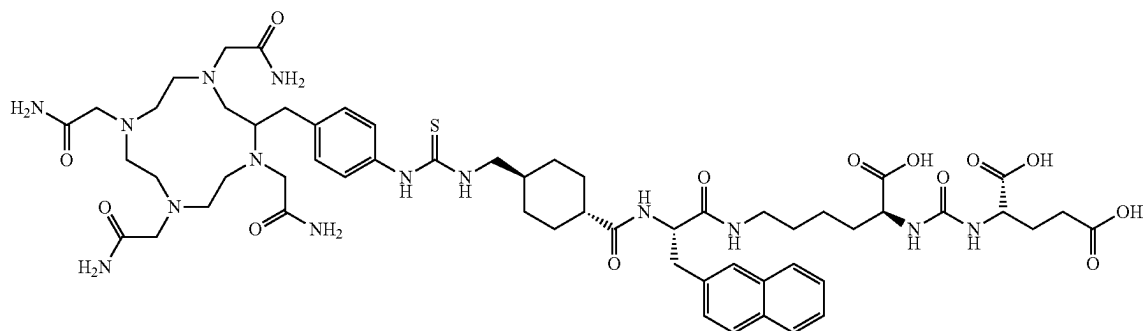


administering prevents or reduces metastasis formation. In some embodiments, the subject does not have a metastatic lesion (e.g., metastatic lesions in bone, viscera, lymph node(s), or a combination thereof) prior to administering the radiopharmaceutical conjugate, and the administering prevents or reduces formation of the metastatic lesions in the subject. In some embodiments, the prostate cancer is metastatic and comprises skeletal metastases prior to administering the radiopharmaceutical conjugate to the subject, and the administering prevents or reduces further skeletal metas-

or a pharmaceutically acceptable salt thereof; and

[0145] (b) a radionuclide bound to the metal chelator, wherein the radionuclide is ^{212}Pb ;

wherein the subject has previously received an anti-cancer agent before the administering of the radiopharmaceutical conjugate. In some embodiments, the method comprises administering a therapeutically effective amount of the radiopharmaceutical conjugate to the subject. In some embodiments, the radiopharmaceutical conjugate comprises a compound having the structure of



tases. In some embodiments, the prostate cancer is local and administering the radiopharmaceutical conjugate to the subject prevents or reduces skeletal metastases. In some embodiments, the prostate cancer is metastatic, but does not comprise bone lesions, and administering the radiopharmaceutical conjugate to the subject prevents or reduces further skeletal metastases. In some embodiments, the radiopharmaceutical conjugate is administered via intravenous injection. In some embodiments, the radiopharmaceutical conjugate is administered via bolus intravenous injection.

[0143] In another aspect, provided herein is a method of treating a prostate cancer in a subject in need thereof, comprising administering to the subject a radiopharmaceutical conjugate of Formula (I), Formula (I-A), Formula (I-A-1), Formula (I-B-1), or Formula (I-B-2), wherein the

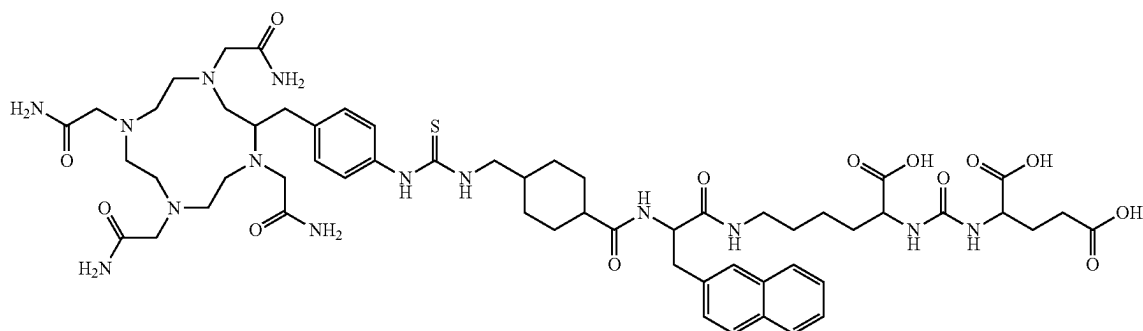
or a pharmaceutically acceptable salt thereof. In some embodiments, the anti-cancer agent is external beam radiation, chemotherapy, hormonal therapy, or a combination thereof. In some embodiments, the anti-cancer therapy is external beam radiation. In some embodiments, the anti-cancer therapy is chemotherapy. In some embodiments, the chemotherapy is a taxane (for example, docetaxel). In some embodiments, the anti-cancer therapy is hormonal therapy (for example, bicalutamide, flutamide, and goserelin). In some embodiments, the anti-cancer therapy is an androgen receptor axis targeted agent. In some embodiments, the radiopharmaceutical conjugate is administered in combination with a second anti-cancer agent, and wherein the second

anti-cancer agent is administered concurrently or sequentially with the radiopharmaceutical conjugate. In some embodiments, the second anti-cancer agent is administered sequentially (e.g., after) the radiopharmaceutical conjugate. In some embodiments, the second anti-cancer agent is administered concurrently with the radiopharmaceutical conjugate. In some embodiments, the second anti-cancer agent is administered prior to administering the radiopharmaceutical conjugate. In some embodiments, the second anti-cancer agent is an androgen receptor-axis targeted agent. In some embodiments, the second anti-cancer agent is abiraterone, enzalutamide, nilutamide, flutamide, bicalutamide, ARN 509, galeterone, orteronel, or a salt thereof. In some embodiments, the second anti-cancer agent is enzalutamide, goserelin, leuporelin, or a salt thereof. In some embodiments, the radiopharmaceutical conjugate is administered an amount of 50 kBq to 500 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered an amount of 100 kBq to 500 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 1 MBq to 500 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 10 MBq to 500 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 50 MBq to 500 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 75 MBq to about 300 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered at a dose of about 10 MBq. In some embodiments, the radiopharmaceutical conjugate is administered at a dose of about 100 MBq. In some embodiments, the radiopharmaceutical conjugate is administered without an agent for treating xerostomia. In some embodiments, the agent for treating xerostomia includes agents for treating dry mouth, aptyalism, or dry throat (e.g., spray substitution of saliva). In some embodiments, the subject is not administered the agent for treating xerostomia. In some embodiments, administering the radiopharmaceutical conjugate reduces a risk of xerostomia in the subject, optionally wherein the risk is reduced relative to the administering of ^{177}Lu -PSMA-617 or ^{225}Ac -PSMA-617. In some embodiments, the risk is relative to the administering of ^{225}Ac -PSMA-617. In some embodiments, the subject has received a targeted radiopharmaceutical therapy prior to administering of the radiopharmaceutical conjugate. In some embodiments, the targeted radiopharmaceutical therapy comprises ^{177}Lu -PSMA-617 or ^{225}Ac -PSMA-617. In some embodiments, the targeted radiopharmaceutical therapy comprises ^{177}Lu -PSMA-617. In some embodiments, the subject has progressive prostate cancer after prior treatment with ^{177}Lu -PSMA-617 or ^{225}Ac -PSMA-617. In some embodiments, the subject is fasted prior to administering of the radiopharmaceutical conjugate. In some embodiments, the administering of the radiopharmaceutical conjugate does

not cause or increase the risk of xerostomia in the subject. In some embodiments, the radiopharmaceutical conjugate is not significantly taken up in a salivary gland in the subject. In some embodiments, the subject does not develop xerostomia after the administering of the radiopharmaceutical conjugate. In some embodiments, the subject was previously diagnosed with xerostomia prior to administering the radiopharmaceutical conjugate. In some embodiments, the subject is susceptible of developing xerostomia. In some embodiments, the administering does not cause any one or more of xerostomia, radiation nephropathy, or bone marrow toxicity, or hematological toxicity in the subject. In some embodiments, the prostate cancer is metastatic castrate resistant prostate cancer (mCRPC). In some embodiments, the prostate cancer is progressive mCRPC. In some embodiments, administering the radiopharmaceutical conjugate prevents or reduces the formation of a metastatic lesion in the subject. In some embodiments, the metastatic lesion comprises a metastatic lesion in bone, viscera, lymph node(s), or a combination thereof. In some embodiments, the metastatic lesion is a metastatic bone lesion. In some embodiments, the metastatic lesion is present in the subject prior to administering the radiopharmaceutical conjugate, and administering the radiopharmaceutical conjugate treats, reduces, or eliminates the metastatic lesion. In some embodiments, the administering prevents or reduces metastasis formation. In some embodiments, the subject does not have a metastatic lesion (e.g., metastatic lesions in bone, viscera, lymph node(s), or a combination thereof) prior to administering the radiopharmaceutical conjugate, and the administering prevents or reduces formation of the metastatic lesions in the subject. In some embodiments, the prostate cancer is metastatic and comprises skeletal metastases prior to administering the radiopharmaceutical conjugate to the subject, and the administering prevents or reduces further skeletal metastases. In some embodiments, the prostate cancer is local and administering the radiopharmaceutical conjugate to the subject prevents or reduces skeletal metastases. In some embodiments, the prostate cancer is metastatic, but does not comprise bone lesions, and administering the radiopharmaceutical conjugate to the subject prevents or reduces further skeletal metastases. In some embodiments, the radiopharmaceutical conjugate is administered via intravenous injection. In some embodiments, the radiopharmaceutical conjugate is administered via bolus intravenous injection.

[0146] In another aspect, provided herein is a method of treating a metastatic prostate cancer in a subject in need thereof, comprising administering to the subject a radiopharmaceutical conjugate of Formula (I), Formula (I-A), Formula (I-A-1), Formula (I-B-1), or Formula (I-B-2). In some embodiments, the method of treating the metastatic prostate cancer in a subject in need thereof comprises administering to the subject a radiopharmaceutical conjugate comprising:

[0147] (a) a compound having the structure:



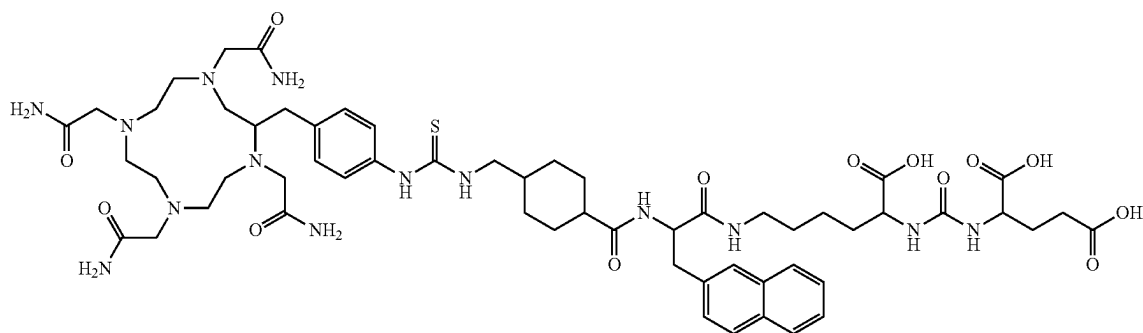
or a pharmaceutically acceptable salt thereof; and

[0148] (b) a radionuclide bound to the metal chelator, wherein the radionuclide is ^{212}Pb . In some embodiments, the metastatic prostate cancer comprises a metastatic lesion in bone, viscera, lymph node(s), or a combination thereof. In some embodiments, provided herein is a method of treating a metastatic prostate cancer comprising a metastatic lesion in bone, viscera, lymph node(s), or a combination thereof, the method comprising administering ^{212}Pb -Compound 1, or a pharmaceutically acceptable salt thereof, to a subject in need thereof. In some embodiments, the metastatic lesions (e.g., metastatic lesions in bone, viscera, lymph node(s), or a combination thereof) are present in the subject prior to administering ^{212}Pb -Compound 1 or a pharmaceutically acceptable salt thereof, and administering ^{212}Pb -Compound 1 or a pharmaceutically acceptable salt thereof treats, reduces, or eliminates the metastatic lesions. In some embodiments, the metastatic lesion is a bone lesion. In some embodiments, the metastatic lesion is a lesion in the viscera. In some embodiments, the metastatic lesion is a lesion in the lymph node. In some embodiments, the bone lesion is osteolytic bone lesion. In some embodiments, administering ^{212}Pb -Compound 1, or a pharmaceutically acceptable salt thereof, to a subject with prostate cancer prevents or reduces metastasis formation. In some embodiments, administering ^{212}Pb -Compound 1, or a pharmaceutically acceptable salt thereof, to a subject with prostate cancer prevents metastasis formation. In some embodiments, the subject does not have metastatic lesions (e.g., metastatic lesions in bone, viscera, lymph node(s), or a combination thereof) prior to the administering of ^{212}Pb -Compound 1 or a pharmaceutically acceptable salt thereof, and the administering of ^{212}Pb -Compound 1 or a pharmaceutically acceptable salt thereof prevents the formation of metastatic lesions. In some embodiments, the subject does not have metastatic lesions (e.g., metastatic lesions in bone, viscera, lymph node(s), or a combination thereof) prior to the administering of ^{212}Pb -Compound 1 or a pharmaceutically acceptable salt thereof, and the administering of ^{212}Pb -Compound 1 or a pharmaceutically acceptable salt thereof prevents the formation of metastatic lesions. In some embodiments, provided herein is a method of preventing or reducing prostate cancer metastases in a subject comprising administering ^{212}Pb -Compound 1, or a pharmaceutically acceptable salt thereof, to the subject. In some embodiments,

provided herein is a method of preventing prostate cancer metastases in a subject comprising administering ^{212}Pb -Compound 1, or a pharmaceutically acceptable salt thereof, to the subject. In some embodiments, provided herein is a method of treating, reducing, or eliminating prostate cancer metastases in a subject comprising administering ^{212}Pb -Compound 1, or a pharmaceutically acceptable salt thereof, to the subject. In some embodiments, the prostate cancer is metastatic and comprises skeletal metastases prior to administering ^{212}Pb -Compound 1, or a pharmaceutically acceptable salt thereof, to the subject. In some embodiments, the prostate cancer is metastatic and comprises skeletal metastases prior to administering ^{212}Pb -Compound 1, or a pharmaceutically acceptable salt thereof, to the subject. In some embodiments, the prostate cancer is metastatic and comprises skeletal metastases prior to administering ^{212}Pb -Compound 1, or a pharmaceutically acceptable salt thereof, to the subject, and administering ^{212}Pb -Compound 1, or a pharmaceutically acceptable salt thereof, to the subject prevents or reduces further skeletal metastases. In some embodiments, the prostate cancer is local and the administering of ^{212}Pb -Compound 1, or a pharmaceutically acceptable salt thereof, to the subject prevents or reduces skeletal metastases. In some embodiments, the prostate cancer is metastatic, but does not comprise bone lesions, and the administering of ^{212}Pb -Compound 1, or a pharmaceutically acceptable salt thereof, prevents further skeletal metastases. In some embodiments, the prostate cancer is metastatic, but does not comprise bone lesions, and the administering of ^{212}Pb -Compound 1, or a pharmaceutically acceptable salt thereof, prevents further skeletal metastases. In some embodiments, the administering of ^{212}Pb -Compound 1 or a salt thereof is more effective in preventing or reducing metastases than a control agent. In some embodiments, the control agent is saline solution. In some embodiments, the control agent is lutetium Lu 177 vipivotide tetraxetan, which is sold under the trade name PLUVICTO®.

[0149] In yet another aspect, provided herein is a method of treating a prostate cancer in a subject in need thereof, comprising administering to the subject a radiopharmaceutical conjugate of Formula (I), Formula (I-A), Formula (I-A-1), Formula (I-B-1), or Formula (I-B-2), wherein the radiopharmaceutical conjugate is administered in combination with a second anti-cancer agent. In some embodiments provided herein is a method of treating a prostate cancer in a subject in need thereof, comprising administering to the subject a radiopharmaceutical conjugate, wherein the radiopharmaceutical conjugate comprises,

[0150] (a) a compound having the structure:

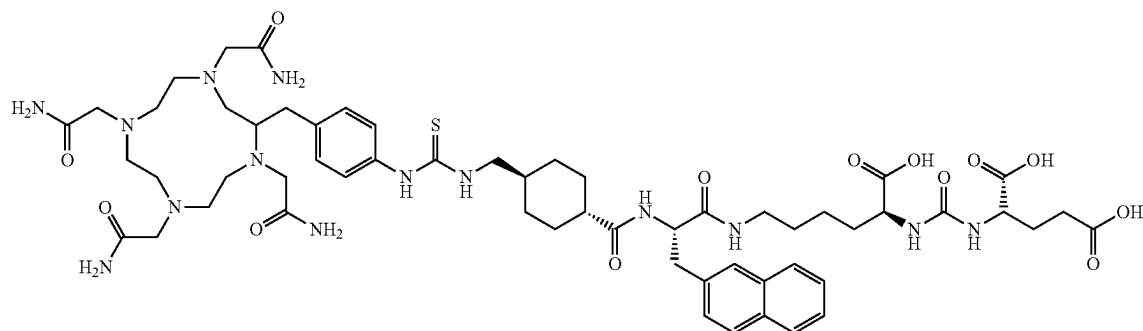


or a pharmaceutically acceptable salt thereof; and

[0151] (b) a radionuclide bound to the metal chelator, wherein the radionuclide is ^{212}Pb ;

wherein the radiopharmaceutical conjugate is administered in combination with a second anti-cancer agent. In some embodiments, the method comprises administering a therapeutically effective amount of the radiopharmaceutical conjugate to the subject. In some embodiments, the radiopharmaceutical conjugate comprises a compound having the structure of

a salt thereof. In some embodiments, the second anti-cancer agent is enzalutamide, goserelin, leuprorelin, or a salt thereof. In some embodiments, the radiopharmaceutical conjugate is administered an amount of 50 kBq to 500 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered an amount of 100 kBq to 500 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 1 MBq to 500 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 10 MBq to 500 MBq per dose. In some embodiments, the



or a pharmaceutically acceptable salt thereof. In some embodiments, the subject has previously received an anti-cancer agent before the administering of the radiopharmaceutical conjugate, and wherein the anti-cancer agent is external beam radiation, chemotherapy, hormonal therapy, or a combination thereof. In some embodiments, the anti-cancer therapy is external beam radiation. In some embodiments, the anti-cancer therapy is chemotherapy. In some embodiments, the chemotherapy is a taxane (for example, docetaxel). In some embodiments, the anti-cancer therapy is hormonal therapy (for example, bicalutamide, flutamide, and goserelin). In some embodiments, the anti-cancer therapy is an androgen receptor axis targeted agent. In some embodiments, the second anti-cancer agent is administered concurrently or sequentially with the radiopharmaceutical conjugate. In some embodiments, the second anti-cancer agent is administered sequentially (e.g., after) the radiopharmaceutical conjugate. In some embodiments, the second anti-cancer agent is administered concurrently with the radiopharmaceutical conjugate. In some embodiments, the second anti-cancer agent is administered prior to administering the radiopharmaceutical conjugate. In some embodiments, the second anti-cancer agent is an androgen receptor-axis targeted agent. In some embodiments, the second anti-cancer agent is abiraterone, enzalutamide, nilutamide, flutamide, bicalutamide, ARN 509, galeterone, orteronel, or

radiopharmaceutical conjugate is administered in an amount of about 50 MBq to 500 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 75 MBq to about 300 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered at a dose of about 10 MBq. In some embodiments, the radiopharmaceutical conjugate is administered at a dose of about 100 MBq. In some embodiments, the radiopharmaceutical conjugate is administered without an agent for treating xerostomia. In some embodiments, the agent for treating xerostomia includes agents for treating dry mouth, apyalism, or dry throat (e.g., spray substitution of saliva). In some embodiments, the subject is not administered the agent for treating xerostomia. In some embodiments, administering the radiopharmaceutical conjugate reduces a risk of xerostomia in the subject, optionally wherein the risk is reduced relative to the administering of ^{177}Lu -PSMA-617 or ^{225}Ac -PSMA-617. In some embodiments, the risk is relative to the administering of ^{225}Ac -PSMA-617. In some embodiments, the subject has received a targeted radiopharmaceutical therapy prior to administering of the radiopharmaceutical conjugate. In some embodiments, the targeted radiopharmaceutical therapy comprises ^{177}Lu -PSMA-617 or ^{225}Ac -PSMA-617. In some embodiments, the targeted radiopharmaceutical therapy comprises ^{177}Lu -PSMA-617. In some embodiments, the subject has

progressive prostate cancer after prior treatment with ^{177}Lu -PSMA-617 or ^{225}Ac -PSMA-617. In some embodiments, the subject is fasted prior to administering of the radiopharmaceutical conjugate. In some embodiments, the administering of the radiopharmaceutical conjugate does not cause or increase the risk of xerostomia in the subject. In some embodiments, the radiopharmaceutical conjugate is not significantly taken up in a salivary gland in the subject. In some embodiments, the subject does not develop xerostomia after the administering of the radiopharmaceutical conjugate. In some embodiments, the subject was previously diagnosed with xerostomia prior to administering the radiopharmaceutical conjugate. In some embodiments, the subject is susceptible of developing xerostomia. In some embodiments, the administering does not cause any one or more of xerostomia, radiation nephropathy, or bone marrow toxicity, or hematological toxicity in the subject. In some embodiments, the prostate cancer is metastatic castrate resistant prostate cancer (mCRPC). In some embodiments, the prostate cancer is progressive mCRPC. In some embodiments, administering the radiopharmaceutical conjugate prevents or reduces the formation of a metastatic lesion in the subject. In some embodiments, the metastatic lesion comprises a metastatic lesion in bone, viscera, lymph node(s), or a combination thereof. In some embodiments, the metastatic lesion is a metastatic bone lesion. In some embodiments, the metastatic lesion is present in the subject prior to administering the radiopharmaceutical conjugate, and administering the radiopharmaceutical conjugate treats, reduces, or eliminates the metastatic lesion. In some embodiments, the administering prevents or reduces metastasis formation. In some embodiments, the subject does not have a metastatic lesion (e.g., metastatic lesions in bone, viscera, lymph node(s), or a combination thereof) prior to administering the radiopharmaceutical conjugate, and the administering prevents or reduces formation of the metastatic lesions in the subject. In some embodiments, the prostate cancer is metastatic and comprises skeletal metastases prior to administering the radiopharmaceutical conjugate to the subject, and the administering prevents or reduces further skeletal metastases. In some embodiments, the prostate cancer is local and administering the radiopharmaceutical conjugate to the subject prevents or reduces skeletal metastases. In some embodiments, the prostate cancer is metastatic, but does not comprise bone lesions, and administering the radiopharmaceutical conjugate to the subject prevents or reduces further skeletal metastases. In some embodiments, the radiopharmaceutical conjugate is administered via intravenous injection. In some embodiments, the radiopharmaceutical conjugate is administered via bolus intravenous injection.

[0152] In another aspect, provided herein is a method of preventing or reducing the formation of a metastatic lesion in a subject in need thereof, comprising administering to the subject a radiopharmaceutical conjugate comprising

[0153] (a) a compound of Formula (I):



[0154] or a pharmaceutically acceptable salt thereof, wherein:

[0155] TL is a PSMA targeting ligand comprising a urea;

[0156] L is a bivalent linking moiety; and

[0157] R^M is a metal chelator; and

[0158] (b) a radionuclide bound to the metal chelator, wherein the radionuclide is ^{21}Pb .

In some embodiments, the subject has prostate cancer. In some embodiments, the metastatic lesion is a metastatic bone lesion. In some embodiments, the prostate cancer is metastatic and comprises a metastatic lesion in bone, vis-

cera, lymph node(s), or a combination thereof. In some embodiments, the metastatic lesion is present in the subject prior to administering the radiopharmaceutical conjugate, and administering the radiopharmaceutical conjugate treats, reduces, or eliminates the metastatic lesion. In some embodiments, the administering prevents or reduces metastasis formation. In some embodiments, the subject does not have a metastatic lesion (e.g., metastatic lesions in bone, viscera, lymph node(s), or a combination thereof) prior to administering the radiopharmaceutical conjugate, and the administering prevents or reduces formation of the metastatic lesions in the subject. In some embodiments, the prostate cancer is metastatic and comprises skeletal metastases prior to administering the radiopharmaceutical conjugate to the subject, and the administering prevents or reduces further skeletal metastases. In some embodiments, the prostate cancer is local and administering the radiopharmaceutical conjugate to the subject prevents or reduces skeletal metastases. In some embodiments, the prostate cancer is metastatic, but does not comprise bone lesions, and administering the radiopharmaceutical conjugate to the subject prevents or reduces further skeletal metastases.

[0159] In addition to the methods of treatment described above, the radiopharmaceutical conjugates described herein (e.g., a radiopharmaceutical conjugate of Formula (I), Formula (I-A), Formula (I-A-1), Formula (I-B-1), or Formula (I-B-2)) can be used to image, and/or as part of a treatment for diseases. Conjugates for imaging applications, e.g., single-photon emission computed tomography (SPECT) and positron emission tomography (PET), can comprise any radionuclide suitable for use as imaging isotopes. Accordingly, the conjugate can be administered to confirm target (e.g., PSMA) expression in a subject's tissues.

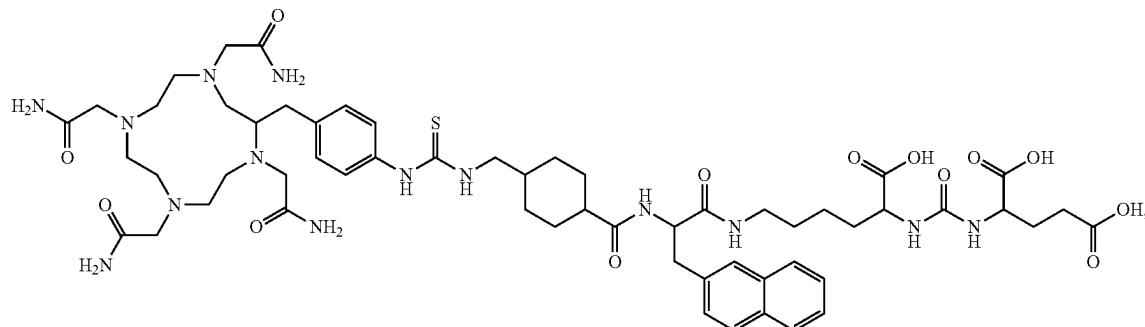
[0160] In one aspect, provided herein are methods for diagnosing or imaging a subject harboring a PSMA expressing cancer or tumor comprising administering to the subject a radiopharmaceutical described herein, or a pharmaceutically acceptable salt or solvate thereof. In one aspect, provided herein are methods for imaging a PSMA expressing cancer or tumor comprising administering to the subject a radiopharmaceutical described herein, or a pharmaceutically acceptable salt or solvate thereof. In some embodiments, the method further comprises selecting or confirming that a tumor in the subject expresses PSMA. In some embodiments, the method further comprises measuring the concentration of the radiopharmaceutical accumulated in the subject. In some embodiments, the method further comprises measuring the amount of radiation emitted from the radionuclide. In some embodiments, the method further comprises analyzing the elimination or clearance profile of the radiopharmaceutical in the subject. In some embodiments, the method further comprises measuring an elimination half-life of the radiopharmaceutical in the patient. In some embodiments, the method further comprises analyzing the clearance profile of the radiopharmaceutical in the subject. For example, radiopharmaceuticals of the present disclosure can be administered for patient selection purposes, such as to confirm the tumor has the appropriate expression of the PSMA target. As another example, radiopharmaceuticals of the present disclosure can be administered to a subject so that the subject's care team can make sure the radiopharmaceutical is cleared from the body in a suitable timeframe so that undesired irradiation of other tissues is minimized.

[0161] In one aspect, provided herein is a method of treating a prostate cancer in a subject in need thereof, comprising administering

[0162] (a) an imaging radiopharmaceutical agent comprising a first radionuclide; and

[0163] (b) a radiopharmaceutical conjugate comprising:

[0164] (i) a compound having the structure:

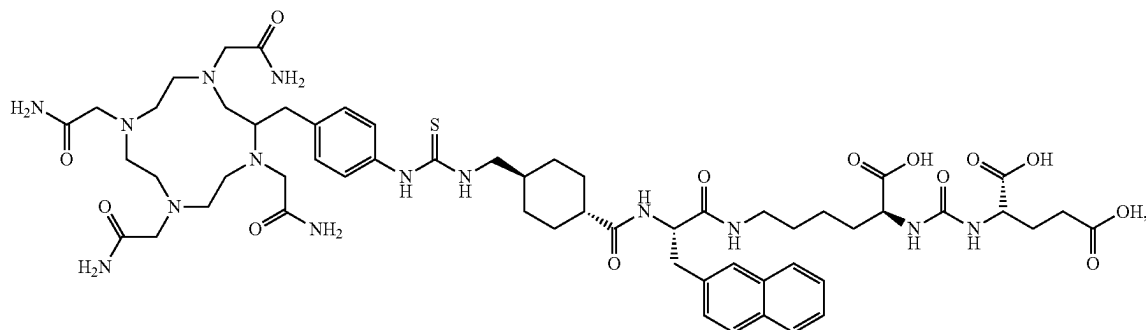


or a pharmaceutically acceptable salt thereof; and

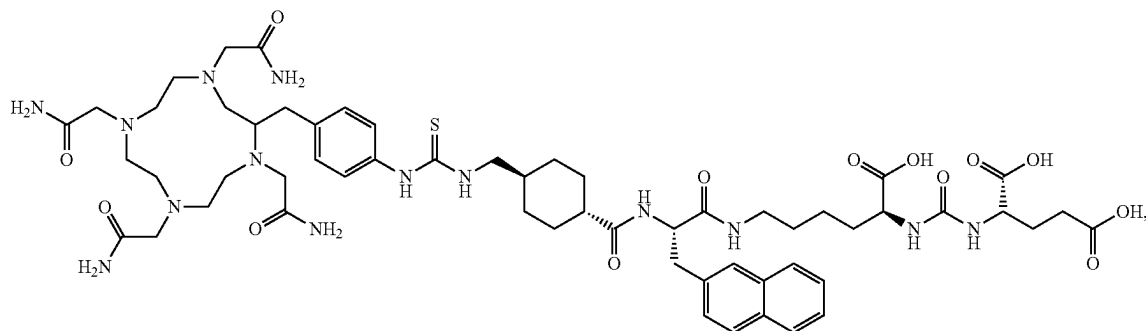
[0165] (ii) a second radionuclide bound to the metal chelator, wherein the second radionuclide is ^{212}Pb ;

wherein the imaging radiopharmaceutical agent is administered to the patient prior to the administering of the radiopharmaceutical conjugate. In some embodiments, the imaging radiopharmaceutical agent is ^{68}Ga -PSMA-11, capromab

pendetide, ^{89}Zr -J591, ^{111}In -J591, IAB2M, ^{18}F -DCFPyI, ^{59}mTc -MIP-1404, ^{18}F -flotufolostat, ^{18}F -piflufolostat, or ^{18}F -PSMA-1007, or a pharmaceutically acceptable salt thereof. In some embodiments, the imaging radiopharmaceutical agent is ^{68}Ga -PSMA-11. In some embodiments, the imaging radiopharmaceutical agent is ^{18}F -PSMA-1007. In some embodiments, the imaging radiopharmaceutical agent comprises a compound having the structure of



or a pharmaceutically acceptable salt thereof; and wherein the first radionuclide is ^{212}Pb or ^{203}Pb . In some embodiments, the first radionuclide is complexed with the TCMC. In some embodiments, the first radionuclide is ^{212}Pb . In some embodiments, the first radionuclide is ^{203}Pb . In some embodiments, the method comprises administering a therapeutically effective amount of the radiopharmaceutical conjugate to the subject. In some embodiments, the radiopharmaceutical conjugate comprises a compound having the structure of



or a pharmaceutically acceptable salt thereof. In some embodiments, the subject has previously received an anti-cancer agent before the administering of the radiopharmaceutical conjugate, and wherein the anti-cancer agent is external beam radiation, chemotherapy, hormonal therapy, or a combination thereof. In some embodiments, the anti-cancer therapy is external beam radiation. In some embodiments, the anti-cancer therapy is chemotherapy. In some embodiments, the chemotherapy is a taxane (for example, docetaxel). In some embodiments, the anti-cancer therapy is hormonal therapy (for example, bicalutamide, flutamide, and goserelin). In some embodiments, the anti-cancer therapy is an androgen receptor axis targeted agent. In some embodiments, the radiopharmaceutical conjugate is administered in combination with a second anti-cancer agent, and wherein the second anti-cancer agent is administered concurrently or sequentially with the radiopharmaceutical conjugate. In some embodiments, the second anti-cancer agent is administered sequentially (e.g., after) the radiopharmaceutical conjugate. In some embodiments, the second anti-cancer agent is administered concurrently with the radiopharmaceutical conjugate. In some embodiments, the second anti-cancer agent is administered prior to administering the radiopharmaceutical conjugate. In some embodiments, the second anti-cancer agent is an androgen receptor-axis targeted agent. In some embodiments, the second anti-cancer agent is abiraterone, enzalutamide, nilutamide, flutamide, bicalutamide, ARN 509, galeterone, orteronel, or a salt thereof. In some embodiments, the second anti-cancer agent is enzalutamide, goserelin, leuprorelin, or a salt thereof. In some embodiments, the radiopharmaceutical conjugate is administered an amount of 50 kBq to 500 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered an amount of 100 kBq to 500 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 1 MBq to 500 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 10 MBq to 500 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 50 MBq to 500 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 75 MBq to about 300 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered at a dose of about 10 MBq. In some embodiments, the radiopharmaceutical conjugate is administered at a dose of about 100 MBq. In some embodiments, the radiopharmaceutical conjugate is administered without an agent for treating xerostomia. In some embodiments, the agent for treating xerostomia includes agents for treating dry mouth, aptyalism, or dry throat (e.g., spray substitution of saliva). In some embodiments, the subject is not administered the agent for treating xerostomia. In some embodiments, administering the radiopharmaceutical conjugate reduces a risk of xerostomia in the subject, optionally wherein the risk is reduced relative to the administering of ^{177}Lu -PSMA-617 or ^{225}Ac -PSMA-617. In some embodiments, the risk is relative to the administering of ^{225}Ac -PSMA-617. In some embodiments, the subject has received a targeted radiopharmaceutical therapy prior to administering of the radiopharmaceutical conjugate. In some embodiments, the targeted radiopharmaceutical therapy comprises ^{177}Lu -PSMA-617 or ^{225}Ac -PSMA-617. In some embodiments, the targeted radiopharmaceutical therapy comprises ^{177}Lu -PSMA-617. In some embodiments, the subject has progressive prostate cancer after prior treatment with ^{177}Lu -PSMA-617 or ^{225}Ac -PSMA-617. In some embodiments, the

subject is fasted prior to administering of the radiopharmaceutical conjugate. In some embodiments, the administering of the radiopharmaceutical conjugate does not cause or increase the risk of xerostomia in the subject. In some embodiments, the radiopharmaceutical conjugate is not significantly taken up in a salivary gland in the subject. In some embodiments, the subject does not develop xerostomia after the administering of the radiopharmaceutical conjugate. In some embodiments, the subject was previously diagnosed with xerostomia prior to administering the radiopharmaceutical conjugate. In some embodiments, the subject is susceptible of developing xerostomia. In some embodiments, the administering does not cause any one or more of xerostomia, radiation nephropathy, or bone marrow toxicity, or hematological toxicity in the subject. In some embodiments, the prostate cancer is metastatic castrate resistant prostate cancer (mCRPC). In some embodiments, the prostate cancer is progressive mCRPC. In some embodiments, administering the radiopharmaceutical conjugate prevents or reduces the formation of a metastatic lesion in the subject. In some embodiments, the metastatic lesion comprises a metastatic lesion in bone, viscera, lymph node(s), or a combination thereof. In some embodiments, the metastatic lesion is a metastatic bone lesion. In some embodiments, the metastatic lesion is present in the subject prior to administering the radiopharmaceutical conjugate, and administering the radiopharmaceutical conjugate treats, reduces, or eliminates the metastatic lesion. In some embodiments, the administering prevents or reduces metastasis formation. In some embodiments, the subject does not have a metastatic lesion (e.g., metastatic lesions in bone, viscera, lymph node(s), or a combination thereof) prior to administering the radiopharmaceutical conjugate, and the administering prevents or reduces formation of the metastatic lesions in the subject. In some embodiments, the prostate cancer is metastatic and comprises skeletal metastases prior to administering the radiopharmaceutical conjugate to the subject, and the administering prevents or reduces further skeletal metastases. In some embodiments, the prostate cancer is local and administering the radiopharmaceutical conjugate to the subject prevents or reduces skeletal metastases. In some embodiments, the prostate cancer is metastatic, but does not comprise bone lesions, and administering the radiopharmaceutical conjugate to the subject prevents or reduces further skeletal metastases. In some embodiments, the radiopharmaceutical conjugate is administered via intravenous injection. In some embodiments, the radiopharmaceutical conjugate is administered via bolus intravenous injection.

[0166] In some embodiments, the radiopharmaceutical conjugate described herein (e.g., a radiopharmaceutical conjugate of Formula (I), Formula (I-A), Formula (I-A-1), Formula (I-B-1), or Formula (I-B-2)) can be administered alone or in combination with one or more additional therapeutic agents. For example, the combination therapy can include a radiopharmaceutical conjugate described herein co-formulated with, and/or co-administered with, one or more additional therapeutic agents, e.g., a second anti-cancer agent. In some embodiments, the second anti-cancer agent is an androgen receptor-axis targeted agent. In some embodiments, the second anti-cancer agent is abiraterone, enzalutamide, nilutamide, flutamide, bicalutamide, ARN 509, galeterone, orteronel, or a salt thereof. In some embodiments, the second anti-cancer agent is enzalutamide, goserelin, leuprorelin, or a salt thereof. Such combination therapies may advantageously utilize lower dosages of the

administered therapeutic agents, thus avoiding possible toxicities or complications associated with the various monotherapies.

[0167] When administered in combination, two (or more) different treatments can be delivered to the subject during the course of the subject's affliction with the disorder, e.g., the two or more treatments are delivered after the subject has been diagnosed with the disorder and before the disorder has been cured or eliminated. In some embodiments, the delivery of one treatment is still occurring when the delivery of the second begins, so that there is overlap. This is sometimes referred to herein as "simultaneous" or "concurrent delivery." In some embodiments, the delivery of one treatment ends before the delivery of the other treatment begins. In some embodiments of either case, the treatment is more effective because of combined administration. For example, the second treatment is more effective, e.g., an equivalent effect is seen with less of the second treatment, or the second treatment reduces symptoms to a greater extent, than would be seen if the second treatment were administered in the absence of the first treatment, or the analogous situation is seen with the first treatment. In some embodiments, delivery is such that the reduction in a symptom, or other parameter related to the disorder is greater than what would be observed with one treatment delivered in the absence of the other. The effect of the two treatments can be partially additive, wholly additive, or greater than additive. The delivery can be such that an effect of the first treatment delivered is still detectable when the second is delivered.

[0168] In some embodiments, provided herein is a method of treating a cancer in a subject in need thereof, wherein the subject has previously received an anti-cancer agent. In some embodiments, the cancer is prostate cancer. In some embodiments, the method comprises administering to the subject a radiopharmaceutical conjugate described herein (e.g., ^{212}Pb -Compound 1). In some embodiments, the subject has previously received an anti-cancer agent selected from external beam radiation, chemotherapy, and hormonal therapy, or a combination thereof. In some embodiments, the previous anti-cancer therapy is external beam radiation. In some embodiments, the previous anti-cancer therapy is chemotherapy. In some embodiments, the previous chemotherapy is a taxane (for example, docetaxel). In some embodiments, the previous anti-cancer therapy is hormonal therapy (for example, bicalutamide, flutamide, and goserelin). In some embodiments, the anti-cancer therapy is an androgen receptor axis targeted agent.

[0169] In some embodiments, the subject has a metastatic lesion visualized by PSMA PET imaging. In some embodiments, the PSMA-PET imaging co-localizes with ^{212}Pb -Compound 1 imaging.

[0170] In some embodiments, the subject has prostate cancer. In some embodiments, the subject has metastatic prostate cancer. In some embodiments, the subject has metastatic castrate resistant prostate cancer. In some embodiments, the cancer comprises PSMA-avid lesions demonstrated by PSMA PET/CT. In some embodiments, the cancer is stage IV metastatic castrate resistant prostate cancer. In some embodiments, the prostate cancer is locally extensive. In some embodiments, the prostate cancer comprises bone metastases. In some embodiments, the tumor is classified via the TNM classification system as T3N0M1b or T3NxM1b.

[0171] In some embodiments, the subject has an ECOG performance status of 0-2. In some embodiments, the subject has an ECOG performance status of 1. In some embodiments, the subject has a TEAE after administration of

^{212}Pb -Compound 1. In some embodiments, the TEAE is grade 1 or grade 2. In some embodiments, the TEAE are unrelated to ^{212}Pb -Compound 1 administration. The TEAE is selected from an injury, poisoning, or procedural complication (e.g., tooth fracture), a gastrointestinal disorder (e.g., abdominal pain, constipation, vomiting, etc.), a generalized disorder and administration site condition (e.g., localized oedema, etc.), a skin and subcutaneous tissue disorder (e.g., erythema, etc.), or a combination thereof. In some embodiments, the subject is administered a therapeutic agent for the treatment of the injury, poisoning, or procedural complication, the gastrointestinal disorder, the generalized disorder and administration site condition, or the skin and subcutaneous tissue disorder. In some embodiments, the TEAE does not include xerostomia. In some embodiments, the TEAE does not include xerostomia of any grade severity. In some embodiments, the TEAE does not include grade 1, 2, 3, or 4 xerostomia. In some embodiments, the TEAE does not include grade 2, 3, or 4 xerostomia. In some embodiments, the TEAE does not include grade 3 or 4 xerostomia. In some embodiments, the subject does not experience an adverse reaction, a serious adverse event, a suspected unexpected serious adverse event, or a death caused by administering ^{212}Pb -Compound 1. In some embodiments, the subject has a life expectancy of greater than 6 months. In some embodiments, the subject has adequate hematopoietic, kidney, and liver function.

[0172] In some embodiments, the subject has a prior orchiectomy. In some embodiments, the subject has a serum testosterone level of <50 ng/DL or <1.7 nmol/L. In some embodiments, the subject has previously received 1 or 2 regimens of taxane therapy before the administering of the radiopharmaceutical conjugate. In some embodiments, the subject is ineligible for taxane therapy. In some embodiments, the subject has at least one PSMA-avid distant metastatic lesion. In some embodiments, the subject has previously been treated with at least one androgen receptor axis targeted agent prior to the administering of the radiopharmaceutical conjugate. In some embodiments, the androgen receptor axis targeted agent is a novel androgen receptor axis targeted agent.

[0173] In some embodiments, the subject does not have a urinary obstruction. In some embodiments, the subject does not have an untreated CNS metastases. In some embodiments, the subject does not have a symptomatic medullary cord compression. In some embodiments, the subject does not have a diffuse bone or bone marrow involvement. In some embodiments, the subject does not have a radiation hypersensitivity. In some embodiments, the subject does not have a history of myelodysplastic syndrome. In some embodiments, the subject does not have a treatment-related acute myeloid leukemia. In some embodiments, the subject does not have a non-prostate cancer requiring treatment within 2 years prior to administering the radiopharmaceutical conjugate (excepting for treated basal cell carcinoma, squamous cell carcinoma, or carcinoma in situ). In some embodiments, the subject does not have an infection of CTCAE v5.0 Grade 2 not responding to therapy. In some embodiments, the subject does not have an infection of CTCAE Grade >2 . In some embodiments, the subject does not have a human immunodeficiency virus. In some embodiments, the subject does not have active hepatitis B virus. In some embodiments, the subject does not have active hepatitis C virus. In some embodiments, the subject does not have a non-healing wound or bone fracture. In some embodiments, the subject does not have a major surgery within 4 weeks prior to administering the radiopharmaceutical

tical conjugate. In some embodiments, the subject does not have impaired cardiac function or clinically significant cardiac disease. In some embodiments, the subject has not received prior systemic anti-cancer therapy or investigational therapy within 4 weeks prior to administering the radiopharmaceutical conjugate, with the exception of luteinizing hormone-releasing hormone or gonadotropin-releasing hormone. In some embodiments, the subject has not received prior PSMA-targeted radiopharmaceutical therapy, with the exception of ^{177}Lu -PSMA-617 or ^{225}Ac -PSMA-617. In some embodiments, the subject has not received prior treatment with systemic radiopharmaceutical therapy within 6 months prior to administering the radiopharmaceutical conjugate. In some embodiments, the subject has not received prior definitive radiotherapy within 6 weeks prior to administering the radiopharmaceutical conjugate. In some embodiments, the subject has not received external beam radiotherapy to a critical organ which has exceeded the organ tolerance dose limit. In some embodiments, the subject has not received hemi-body irradiation within 6 months prior to administering the radiopharmaceutical conjugate. In some embodiments, the subject has not received high-dose chemotherapy requiring hematopoietic-stem-cell-rescue. In some embodiments, the subject has not received an autologous or allogenic stem-cell transplant. In some embodiments, the subject does not have ongoing toxicity CTCAE Grade 2 or higher due to prior anti-cancer therapy that is not stabilized. In some embodiments, the subject has not been administered a live vaccine within 4 weeks prior to administration of the radiopharmaceutical conjugate. In some embodiments, the subject has not been administered a biological response modifier, including granulocyte colony-stimulating factor, granulocyte macrophage colony-stimulating factor, and erythropoietin within 3 weeks prior to administering the radiopharmaceutical conjugate. In some embodiments, the subject has not received systemic corticosteroids (e.g., >10 mg of prednisone/prednisolone per day) or herbal products that decrease PSA levels (e.g., saw palmetto) within 4 weeks prior to administering the radiopharmaceutical conjugate.

[0174] In some embodiments, ^{212}Pb -Compound 1 is administered to the subject via intravenous injection. In some embodiments, ^{212}Pb -Compound 1 is administered via slow bolus injection. In some embodiments, ^{212}Pb -Compound 1 is administered via slow bolus intravenous injection via a 3-way adapter coupled to a conventional isotonic saline solution.

[0175] In some embodiments, the subject is imaged by gamma camera imaging. In some embodiments, the subject is imaged by gamma planar imaging. In some embodiments, the subject is imaged by SPECT/CT imaging.

[0176] In some embodiments, in vivo half-life of ^{212}Pb -Compound 1 is determined via blood sampling and whole-body probe measurements. In some embodiments, in clearance of ^{212}Pb -Compound 1 is determined via blood sampling and whole-body probe measurements. In some embodiments, blood sampling and whole-body probe measurements are taken 0-30 minutes, 1-2 hours, 4-6 hours, and 16-24 hours after administration of ^{212}Pb -Compound 1.

[0177] In some embodiments, a subject is imaged via ^{18}F -PSMA-PET/CT prior to ^{212}Pb -Compound 1 administration. In some embodiments, the subject is imaged via ^{18}F -PSMA-1007. In some embodiments, the ^{18}F -PSMA-1007 is administered at a dose of 2.5 MBq/kg. In some embodiments, the ^{18}F -PSMA-1007 is administered at a maximum dose of 300 MBq. In some embodiments, the ^{18}F -PSMA-1007 is administered at a minimum dose of 200

MBq. In some embodiments, the ^{18}F -PSMA-1007 is administered at a dose between 200 MBq and 300 MBq.

[0178] In some embodiments, a subject has been previously treated with a systemic anti-cancer therapy prior to ^{212}Pb -Compound 1 administration. In some embodiments, the systemic anti-cancer therapy is hormonal therapy or chemotherapy. In some embodiments, the hormonal therapy is bicalutamide (e.g., 150 mg once daily), exulexin (e.g., 250 mg three times a day), zoladex (e.g., 10 mg every 12 weeks). In some embodiments, the chemotherapy is docetaxel (e.g., 150 mg every third week) or docetaxel-21 (e.g., 170 mg).

[0179] In some embodiments, a subject administered ^{212}Pb -Compound 1 has a concomitant disease. In some embodiments, the subject has a cardiac disorder (e.g., atrial fibrillation, myocardial ischemia, etc.), a gastrointestinal disorder (e.g., hiatus hernia, etc.), an injury (e.g., spinal compression fracture, etc.), a metabolic or nutrition disorder (e.g., folate deficiency, hypercholesterolemia, etc.), a musculoskeletal or connective tissue disorder (e.g., bone pain, osteoporosis, spinal stenosis, etc.), a nervous system disorder (e.g., polyneuropathy, etc.), a renal or urinary disorder (e.g., lower urinary tract symptoms, etc.), a prior surgical procedure (e.g., artificial urinary sphincter implant, etc.), or a vascular disorder (e.g., hypertension).

[0180] In some embodiments, a subject administered ^{212}Pb -Compound 1 is receiving a concomitant medication. In some embodiments, a subject is administered ^{212}Pb -Compound 1 in combination with a concomitant medication. In some embodiments, the concomitant medication is not a chemotherapy, a biologic therapy, an immune-oncology agent, or a radiation therapy. In some embodiments, the concomitant therapy comprises a drug for acid related disorders (e.g., esomeprazole, etc.), a drug for constipation (e.g., macrogol, sodium picosulfate, etc.), a vitamin (e.g., vitamin D, etc.), a mineral supplement (e.g., calcium, including combinations with vitamin D and/or other drugs, etc.), an antithrombotic agent (e.g., acetylsalicylic acid, warfarin, etc.), an antianemic preparation (e.g., cyanocobalamin, folic acid, etc.), an agent acting on the renin-angiotensin system (e.g., candesartan, losartan, diuretics, etc.), a lipid modifying agent (e.g., atorvastatin, ezetimibe, etc.), an endocrine therapy (e.g., enzalutamide, goserelin, leuporelin, etc.), drugs for the treatment of bone diseases (e.g., denosumab, zoledronic acid, etc.), analgesics (e.g., morphine, paracetamol, etc.), or an antihistamine (e.g., alimezine, etc.).

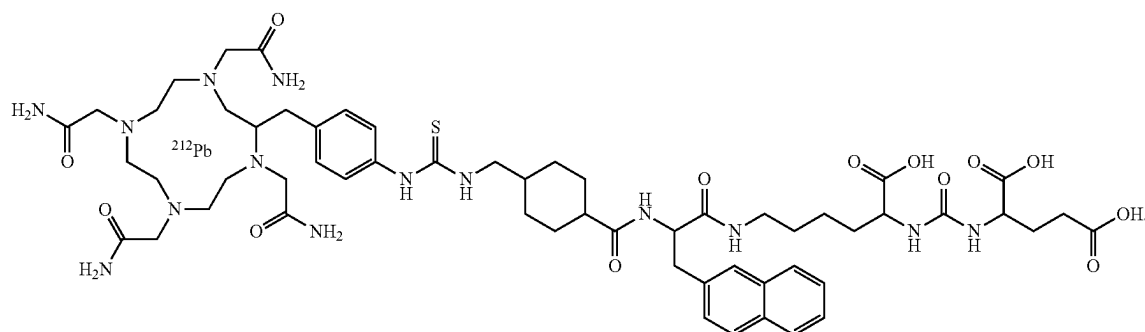
[0181] In some embodiments, ^{212}Pb -Compound 1 is not taken up in the salivary glands of a subject as determined by gamma imaging. In some embodiments, ^{212}Pb -Compound 1 is taken up by the kidneys in the subject as determined by gamma imaging. In some embodiments, the uptake of ^{212}Pb -Compound 1 in the kidneys of a subject is greater than or equal to the uptake of ^{212}Pb -Compound 1 in the liver of the subject. In some embodiments, ^{212}Pb -Compound 1 is not taken up by the spleen of the subject as determined by gamma imaging. In some embodiments, ^{212}Pb -Compound 1 is not taken up by the small intestines of the subject as determined by gamma imaging. In some embodiments, ^{212}Pb -Compound 1 is not taken up by the blood pool of the subject as determined by gamma imaging. In some embodiments, ^{212}Pb -Compound 1 is not taken up by the bladder of the subject as determined by gamma imaging. In some embodiments, ^{212}Pb -Compound 1 is taken up by the bladder of the subject as determined by gamma imaging. In some embodiments, the uptake of ^{212}Pb -Compound 1 in the bladder of a subject is equal to the uptake of ^{212}Pb -Compound 1 in the liver of the subject. In some embodiments,

²¹²Pb-Compound 1 is not taken up by the bone marrow of the subject as determined by gamma imaging. In some embodiments, uptake determined by gamma imaging is determined by visual assessment of a gamma planar imaging or SPECT/CT scan.

Dose and Frequency of Administration

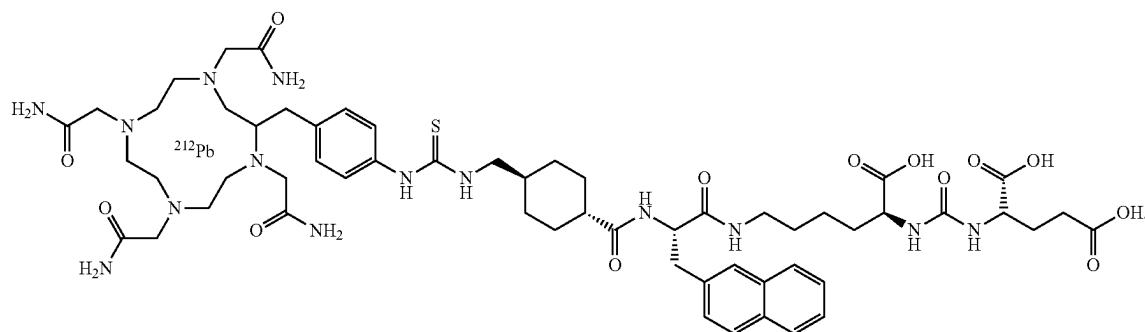
[0182] In one aspect, disclosed herein is a method of treating cancer in a subject in need thereof, wherein the method comprises administering to the subject a radiopharmaceutical conjugate, wherein the radiopharmaceutical conjugate comprises (a) a metal chelator; (b) a target-binding moiety, and (c) a radionuclide bound to the metal chelator, wherein the radionuclide is ^{212}Pb . In some embodiments, the radiopharmaceutical conjugate has the structure of Formula (I), Formula (I-A), Formula (I-A-1), Formula (I-B-1), or Formula (I-B-2). In some embodiments, the radiopharmaceutical conjugate is administered to the subject about every 1 to 7 weeks. In some embodiments, the radiopharmaceutical conjugate is administered to the subject about every 2 to 9 weeks. In some embodiments, the target-binding moiety binds to a PSMA. In some embodiments, the metal chelator is TCMC. In some embodiments, the metal chelator is DOTA. In some embodiments, the radiopharmaceutical conjugate is ^{212}Pb -Compound 1. In some embodiments, the radiopharmaceutical conjugate has a structure of

every 6 weeks. In some embodiments, the radiopharmaceutical conjugate is administered to the subject once a week. In some embodiments, the radiopharmaceutical conjugate is administered to the subject once every 2 weeks. In some embodiments, the radiopharmaceutical conjugate is administered to the subject once every 3 weeks. In some embodiments, the radiopharmaceutical conjugate is administered to the subject once every 4 weeks. In some embodiments, the radiopharmaceutical conjugate is administered to the subject once every 5 weeks. In some embodiments, the radiopharmaceutical conjugate is administered to the subject once every 6 weeks. In some embodiments, the radiopharmaceutical conjugate is administered to the subject once about every week. In some embodiments, the radiopharmaceutical conjugate is administered to the subject once about every 2 weeks. In some embodiments, the radiopharmaceutical conjugate is administered to the subject once about every 3 weeks. In some embodiments, the radiopharmaceutical conjugate is administered to the subject once about every 4 weeks. In some embodiments, the radiopharmaceutical conjugate is administered to the subject once about every 5 weeks. In some embodiments, the radiopharmaceutical conjugate is administered to the subject once about every 6 weeks. In some embodiments, the radiopharmaceutical conjugate is administered to the subject once about every 7 weeks. In some embodiments, the radiopharmaceutical con-



In some embodiments, the radiopharmaceutical conjugate has a structure of

jugate is administered to the subject at a 2-week interval. In some embodiments, the radiopharmaceutical conjugate is



In some embodiments, the radiopharmaceutical conjugate is administered to the subject about once every week, about once every 2 weeks, about once every 3 weeks, about once every 4 weeks, about once every 5 weeks, or about once

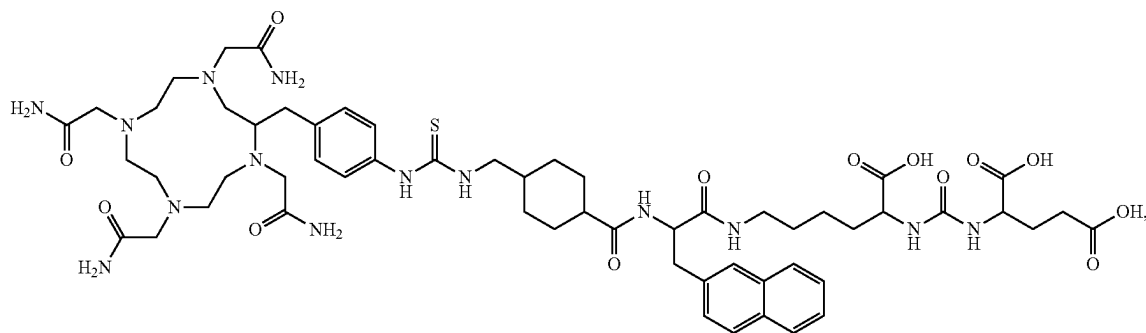
administered to the subject at a 3-week interval. In some embodiments, the radiopharmaceutical conjugate is administered to the subject at a 4-week interval. In some embodiments, the radiopharmaceutical conjugate is administered to

the subject at a 5-week interval. In some embodiments, the radiopharmaceutical conjugate is administered to the subject at a 6-week interval. In some embodiments, the radiopharmaceutical conjugate is administered to the subject at a 7-week interval. In some embodiments, the radiopharmaceutical conjugate is administered 2 times (i.e., for 2 cycles), 3 times, (i.e., for 3 cycles), 4 times, (i.e., 4 cycles), 5 times (i.e., 5 cycles), 6 times (i.e., 6 cycles), 7 times (i.e., 7 cycles), 8 times (i.e., 8 cycles), 9 times (i.e., 9 cycles), 10 times (i.e., 10 cycles), 11 times (i.e., 11 cycles), or 12 times (i.e., 12 cycles). In some embodiments, the radiopharmaceutical conjugate is administered 2 times (i.e., for 2 cycles), 3 times, (i.e., for 3 cycles), 4 times, (i.e., 4 cycles), 5 times (i.e., 5 cycles), 6 times (i.e., 6 cycles), 7 times (i.e., 7 cycles), or 8 times (i.e., 8 cycles). In some embodiments, the radiopharmaceutical conjugate is administered for at least one cycle. In some embodiments, the radiopharmaceutical conjugate is administered for 2 cycles. In some embodiments, the radiopharmaceutical conjugate is administered for 2 or more cycles. In some embodiments, the radiopharmaceutical conjugate is administered for 3 cycles. In some embodiments, the radiopharmaceutical conjugate is administered for 4 cycles. In some embodiments, the radiopharmaceutical conjugate is administered for 5 cycles. In some embodiments, the radiopharmaceutical conjugate is administered for 6

cycles. In some embodiments, the radiopharmaceutical conjugate is administered for 7 cycles. In some embodiments, the radiopharmaceutical conjugate is administered for 8 cycles. In some embodiments, the radiopharmaceutical conjugate is administered for 9 cycles. In some embodiments, the radiopharmaceutical conjugate is administered for 10 cycles. In some embodiments, the radiopharmaceutical conjugate is administered for 11 cycles. In some embodiments, the radiopharmaceutical conjugate is administered for 12 cycles. In some embodiments, the radiopharmaceutical conjugate is administered for 13 to 20 cycles.

[0183] In one aspect, provided herein is a method of treating a prostate cancer in a subject in need thereof, comprising administering to the subject a radiopharmaceutical conjugate. In some embodiments, the radiopharmaceutical conjugate has the structure of Formula (I), Formula (I-A), Formula (I-A-1), Formula (I-B-1), or Formula (I-B-2), wherein the radiopharmaceutical conjugate is administered to the subject in an amount between 50 kBq to 500 MBq per dose. In some embodiments, provided herein is a method of treating a prostate cancer in a subject in need thereof, comprising administering to the subject a radiopharmaceutical conjugate, wherein the radiopharmaceutical conjugate comprises,

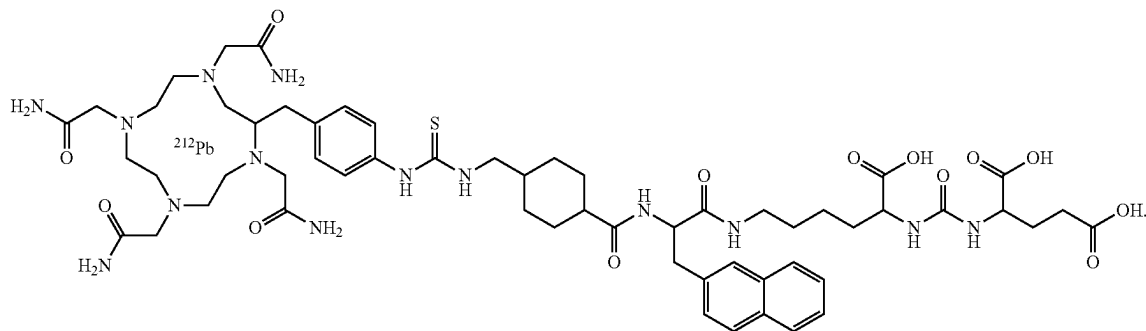
[0184] (a) a compound having the structure:



or a pharmaceutically acceptable salt thereof; and

[0185] (b) a radionuclide bound to the metal chelator, wherein the radionuclide is ^{212}Pb ;

wherein the radiopharmaceutical conjugate is administered to the subject in an amount between 50 kBq to 500 MBq per dose. In some embodiments, the radiopharmaceutical conjugate has a structure of



dose, about 95 MBq per dose, about 100 MBq per dose, about 105 MBq per dose, about 110 MBq per dose, about 115 MBq per dose, about 120 MBq per dose, about 125 MBq per dose, about 130 MBq per dose, about 140 MBq per dose, about 145 MBq per dose, about 150 MBq per dose, about 155 MBq per dose, about 160 MBq per dose, about 165 MBq per dose, about 170 MBq per dose, about 175 MBq per dose, about 180 MBq per dose, about 185 MBq per dose, about 190 MBq per dose, about 195 MBq per dose, about 200 MBq per dose, about 205 MBq per dose, about 210 MBq per dose, about 215 MBq per dose, about 220 MBq per dose, about 225 MBq per dose, about 230 MBq per dose, about 235 MBq per dose, about 240 MBq per dose, about 245 MBq per dose, about 250 MBq per dose, about 255 MBq per dose, about 260 MBq per dose, about 265 MBq per dose, about 270 MBq per dose, about 275 MBq per dose, about 280 MBq per dose, about 285 MBq per dose, about 290 MBq per dose, about 295 MBq per dose, or about 300 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 75 MBq per dose, about 100 MBq per dose, about 125 MBq per dose, about 150 MBq per dose, about 175 MBq per dose, about 200 MBq per dose, about 225 MBq per dose, about 250 MBq per dose, about 275 MBq per dose, or about 300 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 75 MBq per dose, about 100 MBq per dose, about 150 MBq per dose, about 200 MBq per dose, about 250 MBq per dose, or about 300 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 75 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 100 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 125 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 150 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 175 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 200 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 225 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 250 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 275 MBq per dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 300 MBq per dose. In some embodiments, a radiopharmaceutical conjugate described herein is administered to a subject in need thereof as a weight-based dose. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 0.02 kBq/kg to about 225 kBq/kg. In some embodiments, a radiopharmaceutical conjugate described herein is administered to a subject in need thereof as a weight-based dose. In some embodiments, the radiopharma-

[illegible]

administered in an amount of about 0.25 kBq/kg. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 0.5 kBq/kg. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 0.75 kBq/kg. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 1 kBq/kg. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 1.25 kBq/kg. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 1.5 kBq/kg. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 1.75 kBq/kg. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 2 kBq/kg. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 2.25 kBq/kg. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 2.5 kBq/kg. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 2.75 kBq/kg. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 3 kBq/kg. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 3.25 kBq/kg. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 3.5 kBq/kg. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 3.75 kBq/kg. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 4 kBq/kg. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 4.25 kBq/kg. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 4.5 kBq/kg. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 4.75 kBq/kg. In some embodiments, the radiopharmaceutical conjugate is administered in an amount of about 5 kBq/kg.

[0186] In some embodiments, the subject is administered folate in combination with a radiopharmaceutical described herein. In some embodiments, the subject is administered folate prior to, concurrently with, or after administering a radiopharmaceutical conjugate described herein. In some embodiments, the folate is administered concurrently (i.e., at the same time) as the radiopharmaceutical conjugate. In some embodiments, the subject is administered folate after administering a radiopharmaceutical conjugate described herein. In some embodiments, the folate is administered 0.25, 0.5, 1, 1.5, 2, 2.5, 3, 4, 5, 6, 7, 8, 9, 10, 11, or 12 hours after administering the radiopharmaceutical conjugate. In some embodiments, the radiopharmaceutical conjugate has the structure of Formula (I), Formula (I-A), Formula (I-A-1), Formula (I-B-1), or Formula (I-B-2). In some embodiments, the folate is administered up to 2 days prior to administering the radiopharmaceutical conjugate. In some embodiments, the folate is administered up to 24 hours prior to administering the radiopharmaceutical conjugate. In some embodiments, the folate is administered up to 12 hours prior to administering the radiopharmaceutical conjugate. In some embodiments, the folate is administered up to 8 hours prior to administering the radiopharmaceutical conjugate. In some embodiments, the folate is administered up to 4 hours prior

[illegible]

5 mg. In some embodiments, the folate is administered in an amount between 600 μ g and 6 mg. In some embodiments, the folate is administered in an amount between 700 μ g and 7 mg. In some embodiments, the folate is administered in an amount between 800 μ g and 8 mg. In some embodiments, the folate is administered in an amount between 900 μ g and 9 mg. In some embodiments, the folate is administered in an amount between 1 mg and 10 mg. In some embodiments, the folate is administered in an amount between 2 mg and 20 mg. In some embodiments, the folate is administered in an amount between 3 mg and 30 mg. In some embodiments, the folate is administered in an amount between 4 mg and 40 mg. In some embodiments, the folate is administered in an amount between 5 mg and 50 mg. In some embodiments, the folate is administered in an amount of 400 μ g, 800 μ g, 1 mg, 5 mg, 7.5 mg, 10 mg, or 15 mg. In some embodiments, the folate is folic acid, dihydrofolate (DHF), tetrahydrofolate (THF), 5, 10-methylenetetrahydrofolate (5, 10-MTHF), or 5-methyltetrahydrofolate (5-MTHF). In some embodiments, the folate is folic acid or 5-MTHF. In some embodiments, the folate is folic acid.

[0187] Further embodiments of the methods described herein are enumerated in the following clauses:

[0188] Clause 1: A method of preventing or reducing the formation of a metastatic bone lesion in a human subject in need thereof, comprising administering to the subject a radiopharmaceutical composition comprising

[0189] (a) a compound of Formula (I):



[0190] or a pharmaceutically acceptable salt thereof, wherein:

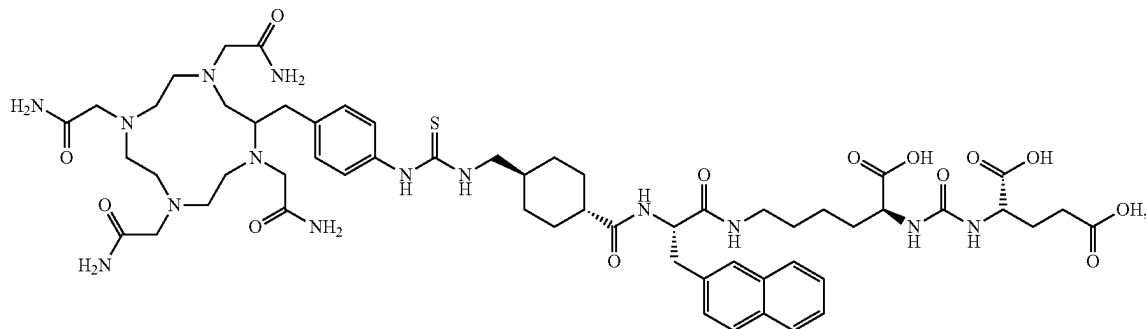
[0191] TL is a PSMA targeting ligand comprising a urea;

[0192] L is a bivalent linking moiety; and

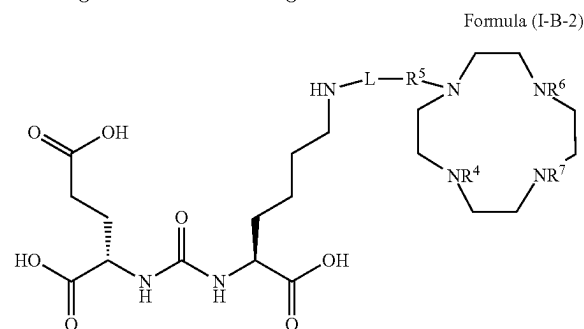
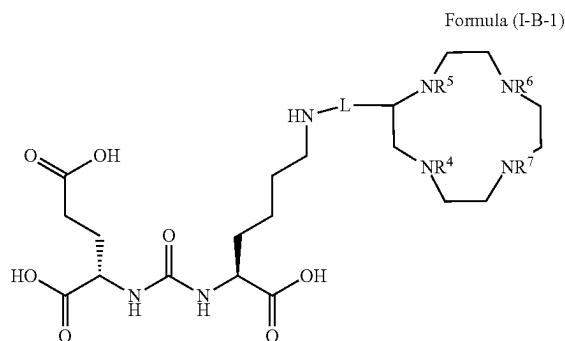
[0193] R^M is a metal chelator; and

[0194] (b) a radionuclide bound to the metal chelator, wherein the radionuclide is ^{212}Pb ;

wherein the human subject has prostate cancer; and wherein the radiopharmaceutical conjugate is administered to the human subject in an amount between 50 kBq to 500 MBq per dose.



[0195] Clause 2: The method of clause 1, wherein the radiopharmaceutical conjugate comprises a structure of Formula (I-B-1) or Formula (I-B-2):



[0196] or a pharmaceutically acceptable salt thereof, wherein:

[0197] R^4 , R^5 , R^6 , and R^7 are each independently selected from the group consisting of $-\text{C}_{1-3}\text{alkylene}-\text{C}(=\text{O})\text{OR}^3$, and $-\text{C}_{1-3}\text{alkylene}-\text{C}(=\text{O})\text{N}(\text{R}^3)_2$, wherein each $\text{C}_{1-3}\text{alkylene}$ is optionally substituted with $-\text{C}(=\text{O})\text{OH}$; and

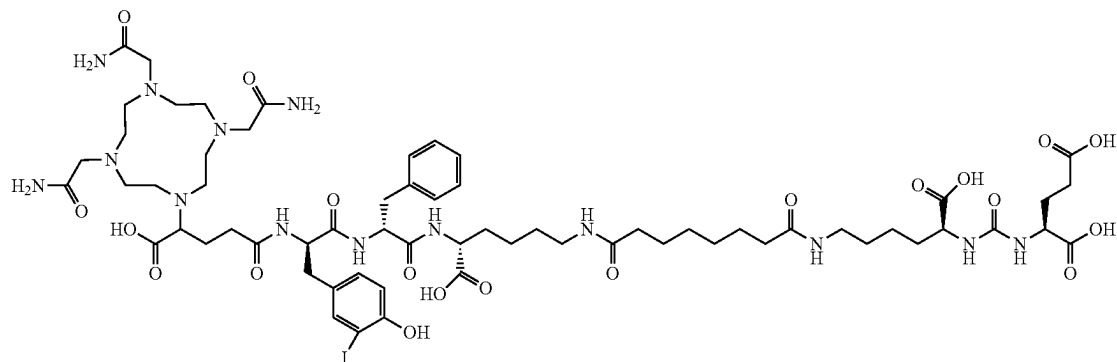
[0198] each R^3 is independently selected from the group consisting of H and $\text{C}_{1-6}\text{alkyl}$;

[0199] provided that R^5 in Formula (I-B-2) is $-\text{C}_{1-3}\text{alkylene}-\text{C}(=\text{O})-$, wherein the $\text{C}_{1-3}\text{alkylene}$ is optionally substituted with $-\text{C}(=\text{O})\text{OH}$.

[0200] Clause 3: The method of clause 1, wherein the compound of Formula (I) is

or a pharmaceutically acceptable salt thereof.

[0201] Clause 4: The method of clause 1, wherein the compound of Formula (I) is



or a pharmaceutically acceptable salt thereof.

[0202] Clause 5: The method of clause 1, wherein the prostate cancer is local and administering the radiopharmaceutical conjugate to the subject prevents or reduces skeletal metastases.

[0203] Clause 6: The method of clause 1, wherein the prostate cancer is metastatic, but does not comprise bone lesions, and administering the radiopharmaceutical conjugate to the subject prevents or reduces further skeletal metastases.

[0204] Clause 7: The method of clause 1, wherein the radiopharmaceutical conjugate is administered in an amount of about 75 MBq per dose to about 300 MBq per dose.

[0205] Clause 8: The method of clause 1, wherein the radiopharmaceutical conjugate is administered in an amount of about 75 MBq per dose, about 100 MBq per dose, about 150 MBq per dose, about 200 MBq per dose, about 250 MBq per dose, or about 300 MBq per dose.

[0206] Clause 9: The method of clause 1, wherein the radiopharmaceutical conjugate is administered about once every week, about once every 2 weeks, about once every 3 weeks, about once every 4 weeks, about once every 5 weeks, or about once every 6 weeks.

[0207] Clause 10: The method of clause 1, wherein the radiopharmaceutical conjugate is administered 2 times (i.e.,

for 2 cycles), 3 times (i.e., for 3 cycles), 4 times (i.e., 4 cycles), 5 times (i.e., 5 cycles), 6 times (i.e., 6 cycles), 7 times (i.e., 7 cycles), 8 times (i.e., 8 cycles), 9 times (i.e., 9 cycles), 10 times (i.e., 10 cycles), 11 times (i.e., 11 cycles), or 12 times (i.e., 12 cycles).

[0208] Clause 11: A method of treating a prostate cancer in a human subject in need thereof, comprising: administering to the human subject a radiopharmaceutical conjugate, wherein the radiopharmaceutical conjugate comprises,

[0209] (a) a compound of Formula (I):

TL-L-R^M

Formula (I)

[0210] or a pharmaceutically acceptable salt thereof, wherein:

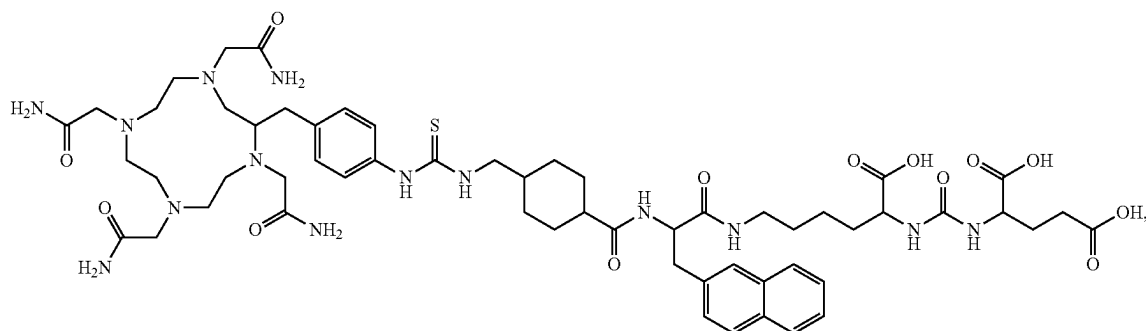
[0211] TL is a PSMA targeting ligand comprising a urea;

[0212] L is a bivalent linking moiety; and

[0213] R^M is a metal chelator; and

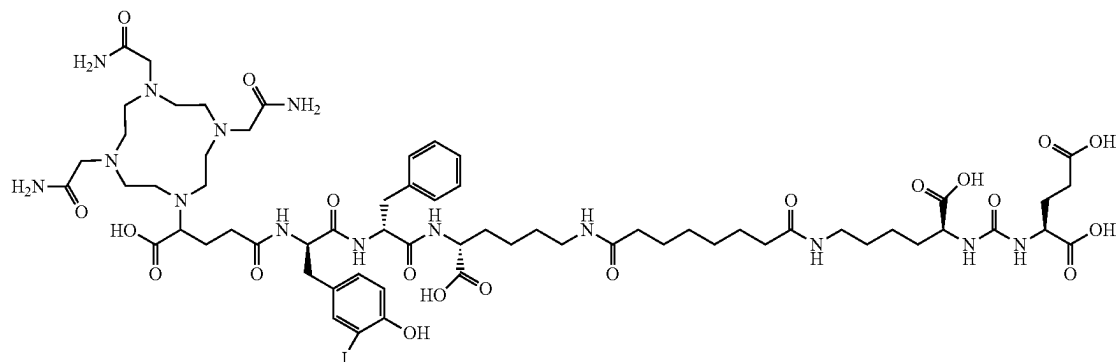
[0214] (b) a radionuclide bound to the metal chelator, wherein the radionuclide is ²¹²Pb; and wherein the radiopharmaceutical conjugate is administered in combination with folate.

[0215] Clause 12: The method of clause 11, wherein the compound has the structure of



or a pharmaceutically acceptable salt thereof.

[0216] Clause 13: The method of clause 11, wherein the compound has the structure of



or a pharmaceutically acceptable salt thereof.

[0217] Clause 14: The method of clause 11, wherein the folate is administered prior to, or concurrently with, the radiopharmaceutical conjugate.

[0218] Clause 15: The method of clause 11, wherein the folate is administered concurrently with the radiopharmaceutical conjugate.

[0219] Clause 16: The method of clause 11, wherein the folate is administered up to 12 hours prior to administering the radiopharmaceutical conjugate.

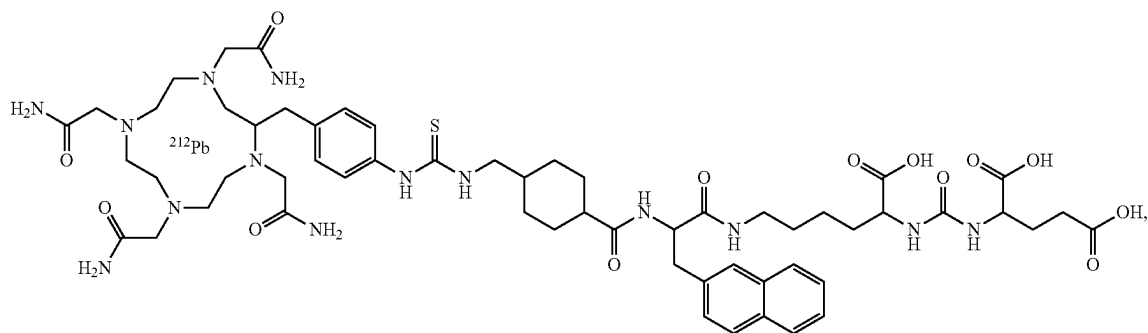
[0220] Clause 17: The method of clause 11, wherein the folate is administered in an amount between 100 µg and 25 mg.

[0221] Clause 18: The method of clause 11, wherein the folate is folic acid, dihydrofolate (DHF), tetrahydrofolate (THF), 5, 10-methylenetetrahydrofolate (5, 10-MTHF), or 5-methyltetrahydrofolate (5-MTHF).

[0224] Clause 21: The method of clause 11, wherein the radiopharmaceutical conjugate is administered about once every week, about once every 2 weeks, about once every 3 weeks, about once every 4 weeks, about once every 5 weeks, or about once every 6 weeks.

[0225] Clause 22: The method of clause 11, wherein the radiopharmaceutical conjugate is administered 2 times (i.e., for 2 cycles), 3 times (i.e., for 3 cycles), 4 times (i.e., 4 cycles), 5 times (i.e., 5 cycles), 6 times (i.e., 6 cycles), 7 times (i.e., 7 cycles), 8 times (i.e., 8 cycles), 9 times (i.e., 9 cycles), 10 times (i.e., 10 cycles), 11 times (i.e., 11 cycles), or 12 times (i.e., 12 cycles).

[0226] Clause 23: A method of treating a prostate cancer in a human subject in need thereof, comprising administering to the human subject a radiopharmaceutical conjugate, wherein the radiopharmaceutical conjugate comprises



[0222] Clause 19: The method of clause 11, wherein the radiopharmaceutical conjugate is administered in an amount between 50 kBq to 500 MBq per dose.

[0223] Clause 20: The method of clause 11, wherein the radiopharmaceutical conjugate is administered in an amount of about 75 MBq per dose to about 300 MBq per dose.

[0227] or a pharmaceutically acceptable salt thereof,

[0228] wherein the radiopharmaceutical conjugate is administered to the subject about once every week, about once every 2 weeks, about once every 3 weeks, about once every 4 weeks, about once every 5 weeks, or about once every 6 weeks.

[0229] Clause 24: The method of clause 23, wherein the radiopharmaceutical conjugate is administered to the subject about once every 3 weeks.

[0230] Clause 25: The method of clause 23, wherein the radiopharmaceutical conjugate is administered to the subject about once every 4 weeks.

[0231] Clause 26: The method of clause 23, wherein the radiopharmaceutical conjugate is administered to the subject about once every 5 weeks.

[0232] Clause 27: The method of clause 23, wherein the radiopharmaceutical conjugate is administered 2 times (i.e., for 2 cycles), 3 times (i.e., for 3 cycles), 4 times (i.e., 4 cycles), 5 times (i.e., 5 cycles), 6 times (i.e., 6 cycles), 7 times (i.e., 7 cycles), 8 times (i.e., 8 cycles), 9 times (i.e., 9 cycles), 10 times (i.e., 10 cycles), 11 times (i.e., 11 cycles), or 12 times (i.e., 12 cycles).

[0233] Clause 28: The method of clause 23, wherein the radiopharmaceutical conjugate is administered 4 times, 5 times, or 6 times.

[0234] Clause 29: The method of clause 23, wherein the radiopharmaceutical conjugate is administered in an amount between 50 kBq to 500 MBq per dose.

[0235] Clause 30: The method of clause 23, wherein the radiopharmaceutical conjugate is administered in an amount of about 75 MBq per dose to about 300 MBq per dose.

[0236] Although the present disclosure and its advantages have been described in detail, it should be understood that various changes, substitutions and alterations can be made herein without departing from the spirit and scope of the disclosure as defined in the appended claims.

[0237] The present disclosure is further illustrated in the following Examples which are given for illustration purposes only and are not intended to limit the disclosure in any way.

EXAMPLES

Example 1: Study Objectives and Overview

Study Objectives

Primary Objective

[0238] To explore whether the use of single-photon emission computed tomography (SPECT)/computed tomography (CT) gamma camera imaging is feasible for confirming uptake of ^{212}Pb -Compound 1 in various normal organs/tissue and in metastases.

Secondary Objectives

[0239] To determine safety and tolerability of first-in-man exposure of ^{212}Pb -Compound 1.

[0240] To study body clearance of ^{212}Pb -Compound 1 by gamma camera imaging and by sample and probe measurements.

[0241] Based on visual assessment, localize metastatic lesion(s) with ^{212}Pb -Compound 1 uptake, and compare to localization of lesions detected by PSMA PET imaging.

[0242] Based on visual assessment of uptake in normal tissue such as kidneys and salivary glands, to match overlap between the therapeutic and diagnostic images in order to make a go/no-go decision for the Phase 1 dose-escalation part of the study.

[0243] To explore any signals of therapeutic effects on disease-related biochemical markers prostate-specific antigen (PSA) and alkaline phosphatase (ALP).

Methods

[0244] The sample size was three participants.

[0245] Key inclusion criteria included progressive mCRPC, PSMA-avid lesions demonstrated by PSMA PET/CT, an ECOG performance status 0-2, life expectancy greater than 6 months, and adequate hematopoietic, kidney and liver function.

[0246] Study drug administration: a single injection of 10 MBq for ^{212}Pb -Compound 1 was administered for each participant. Administration was by slow bolus IV injection via a 3-way adapter coupled to a conventional isotonic saline solution.

[0247] The timepoint of injection of ^{212}Pb -Compound 1 was defined as Day 0, 0 hours.

Imaging, whole-body probe and blood assessments of clearance were performed at the following timepoints:

[0248] Gamma camera imaging (both planar and SPECT/CT) was scheduled 1-3 hours and 16-24 hours post injection.

[0249] Clearance was assessed with blood sampling and whole-body probe measurements at the following timepoints: 15 min after dose (+/-15 min window), 1 h after dose (+1 h window), 4 h after dose (+2 h window), 16 h after dose (+8 h window). Full blood and plasma were measured separately to investigate ligand in vivo stability.

Follow-up: Patients were followed-up for safety and tolerability for 28 days post-administration of ^{212}Pb -Compound 1, with visits on Day 14 and Day 28. Adverse events were reported per CTCAE v5 from time of informed consent until end of follow-up. Patients were followed-up for efficacy biomarkers (PSA, ALP) for 28 days post-administration of ^{212}Pb -Compound 1.

[0250] Table 2 depicts the study flow chart. A diagrammatic representation of the timepoints for planar and SPECT imaging, whole-body probe measurements, and blood sample draws is shown in FIG. 1.

TABLE 2

Study Flow-chart						
Clinical Assessments	Study period					
	Screening period	Treatment period				
		Visit (V) number				
		V1	V2	V3	V4	V5 (EOS)
		Weeks				
	-4 to 0	0	0	2	4	
	Study days					
		0				
	-28 to 0 days	Pre dose	Post dose	1	14 ± 2	28 ± 2
Written Informed consent	X					
Inclusion/Exclusion Criteria	X					
Demographic Data	X					
Medical History	X					
Prior Prostate Cancer Therapies	X					
Physical Examination	X	X		X	X	X
Vital Signs	X	X		X	X	X
ECOG	X	X			X	X
¹⁸ F-PSMA-PET	X ^a					
Diagnostic CT or MRI ^b	X					
12-lead ECG	X					
Haematology	X	X			X	X
Biochemistry	X	X			X	X
Urine analyses (dipstick)	X	X				
Whole blood (clearance analysis) ^e			X ^c	X ^c		
Administration of ²¹² Pb-Compound 1		X				
Planar gamma imaging and SPECT/CT			X ^d	X ^e		
Whole body probe measurements		X ^f	X ^f	X ^f		
Biomarkers (PSA, ALP)		X				X
Adverse events			X	X	X	X
Concomitant Medications			X	X	X	X

ALP, alkaline phosphatase; CT, computed tomography; ECG, electrocardiogram; ECOG, The Eastern Cooperative Oncology Group; EOS, end of study; IMP, investigational medicinal product; MRI, magnetic resonance imaging; PET, positron emission tomography; PI, principal investigator; PSA, prostate specific antigen; PSMA, prostate specific membrane antigen; SPECT, single-photon emission computed tomography.

^aDiagnostic PSMA-PET no older than six weeks prior to IMP administration. PSMA-PET/CT may be conducted using other approved PSMA-targeted PET radiotracers (including those based on ⁶⁸Ga).

^bNot mandatory, PI's discretion.

^cWhole blood samples for body clearance analysis should be taken at the following time-points: 15 min after dose (+/-15 min window), one hour after dose (+1 h window), four hours after dose (+2 h window), 16 hours after dose (+8 h window), to be collected if technically feasible at the site. An additional or replacement sample may be collected within the frame of the existing visit schedule. Such an additional sample should be collected only if it is deemed safe and needed for the technical assessment by the Investigator.

^dWithin 1-3 hours after IMP administration.

^eWithin 16-24 hours after IMP administration.

^fNon-invasive whole body probe measurements should be taken at the following time-points: predose, 15 min after dose (+/-15 min window), one hour after dose (+1 h window), four hours after dose (+2 h window), 16 hours after dose (+8 h window), to be collected if technically feasible at the site.

Example 2: Imaging

PSMA-PET/CT imaging

[0251] Participants underwent a standard of care diagnostic ¹⁸F-PSMA-PET/CT at baseline (i.e, within six weeks prior to administration of ²¹²Pb-Compound 1).

[0252] PSMA-PET/CT was conducted using the radiotracer ¹⁸F-PSMA-1007: dose 2.5 MBq/kg (minimum 200 MBq, maximum 300 MBq) with PET acquisition performed approximately 120 minutes post-injection.

Gamma Camera Imaging

[0253] Non-invasive gamma camera imaging with both planar gamma imaging (whole body scintigraphy) and SPECT/CT was performed at 2 timepoints following injection of ²¹²Pb-Compound 1:

[0254] 1-3 hours and 16-24 hours

[0255] The planar gamma camera imaging was acquired immediately before the SEPCT/CT imaging

[0256] Performed on a Siemens Symbia Intevo Bold hybrid scanner using a medium energy low penetration (MELP) or high energy (HE) collimator.

[0257] The same scanner was used for all acquisitions of all patients in the study.

[0258] Scanner room background radiation was measured 0.5 m from the SPECT/CT scanner table, over a time period of one minute, before the participant entered the scanner room. The measurement was repeated 0.5 m from the participant's parotid glands (from each side) and over the kidneys (posterior midline), respectively.

[0259] Planar gamma imaging whole body acquisition times were 30-40 minutes

[0260] SPECT/CT acquisition involved 2 or 3 bed positions with body contouring orbits starting from the vertex, 20-30 minutes per bed position.

[0261] Two energy windows, 40% at 79 keV and/or 20% at 239 keV.

[0262] Dual scatter windows of 20% for the 79 keV peak centered on 55 keV and 103 keV and 5% for the 239 keV peak centered on 209 keV and 268 keV for scatter correction.

[0263] Attenuation correction based on a low-dose non-enhanced CT.

[0264] Reconstruction parameters: 256x256 matrix, FLASH 3D or xSPECT and iterations x subsets: 15x1, 30x2, 30x4, 30x30.

[0265] Energy spectra indicating ²¹²Bi and ²¹²Pb windows for imaging is shown in FIG. 3.

Example 3: Patient Population: Demography, Cancer History and Medical History

[0266] Three male patients with progressive mCRPC on standard of care therapies were included in the study. The median age was 81 years at the time of informed consent (range 73-89 years), and mean height and weight were 181.5 m and 84.9 kg respectively. All patients had an ECOG status of 1 at study entry. Patient demographic data are shown in Table 3.

TABLE 3	
Demographics	
Characteristic	Phase 0 Participants (N = 3)
Ethnic Group, n (%)	
White	3 (100%)
Age at informed consent (years)	
Mean (SD)	81 (8.0)
Median (Min, Max)	81 (73,89)

TABLE 3-continued	
Demographics	
Characteristic	Phase 0 Participants (N = 3)
Height	
Mean (SD)	181.5 (4.9)
Median (Min, Max)	182 (178, 185)
Body weight (kg)	
Mean (SD)	84.9 (22.1)
Median (Min, Max)	86.5 (62.0, 106)

ID, identifier; SD, standard deviation.

[0267] All patients had stage IV metastatic castration resistant prostate cancer at the time of study entry. In each case this included the presence of locally extensive disease of the prostate and bone metastases. All patients had received prior external beam radiotherapy before enrolment to the clinical trial (see Table 4). No patient received external beam radiation therapy while on study. —P-GP-89-T

TABLE 4			
Previous anti-cancer radiotherapy			
Participant ID	Start Date	End Date	Reason for Ending Therapy
NO-01-001	22 Oct. 2009	25 Nov. 2009	Radiological progression
	NK May 2015	NK May 2015	
NO-01-002	23 Sep. 2015	NK November 2015	Completed
	8 Aug. 2019	26 Aug. 2019	Completed
	20 Aug. 2020	20 Aug. 2020	Completed
	20 Mar. 2023	31 Mar. 2023	Completed
NO-01-003	NK 2009	NK 2009	Completed

ID, identifier;

All patients had received prior systemic anticancer therapy before enrolment to the study. One patient had received prior taxane based chemotherapy (docetaxel), whereas two patients had received only prior hormonal based therapies (See Table 5).

TABLE 5							
Previous anti-cancer systemic therapy							
Participant ID	Type of Therapy	Description	Start Date	End Date	Dose (mg)	Frequency	TNM Stage of Disease
NO-01-001	Hormonal therapy	Bicalutamide	13 Mar. 2006	06 Oct. 2009	150 mg	Once daily	T3NoM1b
		Eulexin	06 Oct. 2009	27 Apr. 2015	250 mg	Three times a day	T3NoM1b
		Zoladex	06 Mar. 2015	18 Sep. 2015	10 mg	Every 12 weeks	T3NoM1b
NO-01-002	Chemotherapy	Docetaxel	1 Feb. 2016	25 Apr. 2016	150 mg	Every third week	T3NoM1b
		Docetaxel-21	11 Jan. 2016	11 Jan. 2016	170 mg	Once due to side effects	T3NoM1b
NO-01-003	Hormonal therapy	Bicalutamide	NK 2015	13 Feb. 2020	150 mg	Once daily	T3NxM1b

ID, identifier;

[0268] All patients enrolled had prior medical conditions reported at the time of informed consent (i.e. medical history) as described in Table 6.

TABLE 6

Medical history/concomitant diseases			
System Organ Class (SOC)	Preferred Term (PT)	Phase 0 patients	
		n	%
Cardiac disorders	Atrial fibrillation	1	33%
	Myocardial ischaemia	1	33%
	Any preferred term	2	67%
Gastrointestinal disorders	Hiatus hernia	1	33%
	Any preferred term	1	33%
Injury, poisoning and procedural complications	Spinal compression fracture	1	33%
	Any preferred term	1	33%
Metabolism and nutrition disorders	Folate deficiency	1	33%
	Hypercholesterolaemia	1	33%
	Any preferred term	2	67%
Musculoskeletal and connective tissue disorders	Bone pain	1	33%
	Osteoporosis	1	33%
	Spinal stenosis	1	33%
Nervous system disorders	Any preferred term	2	67%
	Polyneuropathy	1	33%
	Any preferred term	1	33%
Renal and urinary disorders	Lower urinary tract symptoms	1	33%
	Any preferred term	1	33%
Surgical and medical procedures	Artificial urinary sphincter implant	1	33%
	Any preferred term	1	33%
Vascular disorders	Hypertension	2	67%
	Any preferred term	2	67%
Any system organ class	Any preferred term	3	100%

[0269] All patients were reported to receive concomitant medication during the study as described in Table 7. All patients received hormonal therapy for mCRPC during the study, as permitted per protocol (e.g. enzalutamide, goserelin and leuporelin). During the study, no patients received concomitant chemotherapy, biological therapies, immunoncology agents or radiation therapies.

TABLE 7

Concomitant Medications	
ATC Level 2/ATC Level 5	Number (%) of Participants (N = 3)
A02 Drugs for acid related disorders	
Esomeprazole	1 (33)
A06 Drugs for constipation	
Macrogol, combinations	1 (33)
Sodium picosulfate	1 (33)
A11 Vitamins	
Vitamin D and analogues	1 (33)
A12 Mineral supplements	
Calcium, combinations with vitamin d and/or other drugs	2 (67)
B01 Antithrombotic agents	
Acetylsalicylic acid	1 (33)
Warfarin	1 (33)

TABLE 7-continued

Concomitant Medications	
ATC Level 2/ATC Level 5	Number (%) of Participants (N = 3)
B03 Antianemic preparations	
Cyanocobalamin	1 (33)
Folic acid	1 (33)
C09 Agents acting on the renin-angiotensin system	
Candesartan	1 (33)
Losartan and diuretics	1 (33)
Valsartan, amlodipine and hydrochlorothiazide	1 (33)
C10 Lipid modifying agents	
Atorvastatin	2 (67)
Ezetimibe	1 (33)
L02 Endocrine therapy	
Enzalutamide	3 (100)
Goserelin	2 (67)
Leuporelin	1 (33)
M05 Drugs for treatment of bone diseases	
Denosumab	1 (33)
Zoledronic acid	1 (33)
N02 Analgesics	
Morphine	1 (33)
Paracetamol	1 (33)
R06 Antihistamines for systemic use	
Alimemazine	1 (33)

Example 4: Administration of Investigational Medical Product (IMP) ²¹²Pb-Compound 1

[0270] All patients 3 consented for the clinical trial subsequently received a single administration of 9.4±0.3 MBq ²¹²Pb-Compound 1 by bolus intravenous injection, as described in Table 8.

TABLE 8

IMP Administration						
Participant ID	IMP dose prepared (MBq)	IMP preparation time/Date	Start time	Stop time	IMP infusion time (min)	IMP left in syringe
NO-01-001	9.99	15:22/30 Mar. 2023	15:49	15:50	1	0.07
NO-01-002	9.72	15:17/8 Jun. 2023	15:36	15:37	1	0.04
NO-01-003	9.25	14:16/11 May 2023	14:31	14:32	1	0.06

Example 5: Safety of Investigational Medical Product (IMP) ²¹²Pb-Compound 1

[0271] All 3 patients consented and administered ²¹²Pb-Compound 1 subsequently completed the planned safety follow up period (no withdrawals from the clinical trial).

[0272] The Eastern Cooperative Oncology Group (ECOG) performance status of all patients remained unchanged from baseline to end of study visit. Each patient had an ECOG performance score of 1 at all visits evaluated (see Table 9).

[0273] All 3 patients experienced at least 1 treatment emergent adverse event (TEAE) during the clinical trial, as shown in Tables 9-11. All TEAEs reported were CTCAE (v5) Grade 1 or 2; all TEAEs were considered by the Investigator as probably unrelated to ²¹²Pb-Compound 1. No adverse reactions, serious adverse events (SAEs), suspected unexpected serious adverse events (SUSARs) or deaths were reported.

[0274] In addition, there were no clinically significant differences or trends reported in hematology, biochemistry and urinalysis from baseline. This included the kidney and liver function parameters alanine aminotransferase (ALT), aspartate aminotransferase (AST), bilirubin, albumin, estimated glomerular filtration rate (egfr) and creatinine. See Table 12 for a summary of safety laboratory values.

[0275] No safety concerns arose from the clinical laboratory data, vital signs or any other observations related to safety.

[0276] Table 9 provides a listing of the ECOG status of each study participant for each study visit (screening to end of follow up).

TABLE 9			
ECOG Performance Status			
Participant ID	Visit	Date	Performance status (ECOG)
NO-01-001	Screening	2023 Mar. 6	1
	(Day -28 to -1)		
	Visit 2 (Day 0)	2023 Mar. 30	1
	Visit 4	2023 Apr. 13	1
NO-01-002	Visit 5	2023 Apr. 27	1
	Screening	2023 Apr. 27	1
	(Day -28 to -1)		
	Visit 2 (Day 0)	2023 Jun. 8	1
NO-01-003	Visit 4	2023 Jun. 22	1
	Visit 5	2023 Jul. 6	1
	Screening	2023 Apr. 28	1
	(Day -28 to -1)		
	Visit 2 (Day 0)	2023 May 11	1
	Visit 4	2023 May 25	1
	Visit 5	2023 Jun. 8	1

[0277] A summary of the reported TEAEs in the study participants is included in Table 10.

TABLE 10

Number and Proportion of Participants with TEAEs	
	Number (%) of Participants (N = 3)
Number (%) of patients with any adverse event	3 (100)
Number (%) of patients with 1 adverse event	3 (100)
Number (%) of patients with 2 adverse events	2 (67)
Number (%) of patients with ≥3 adverse events	1 (33)
Number (%) of patients with any possibly related adverse event (excluded unrelated)	0 (0)
Number (%) of patients with serious adverse events	0 (0)

[0278] TEAEs are summarized by system organ class (SOC) and preferred term (PT) in Table 11. None of the TEAE PTs were reported in more than one participant. Gastrointestinal disorder was the only SOC for which TEAEs were reported for more than one participant.

TABLE 11

Number and Proportion of Participants with TEAEs by body system and preferred term	
System Organ Class/Preferred Term	Number (%) of Participants (N = 3)
Injury, poisoning and procedural complications	1 (33)
Tooth fracture	1 (33)
Gastrointestinal disorders	3 (33)
Abdominal pain	1 (33)
Constipation	1 (33)
Vomiting	1 (33)
General disorders and administration site conditions	1 (33)
Localized oedema	1 (33)
Skin and subcutaneous tissue disorders	1 (33)
Erythema	1 (33)

[0279] All reported TEAEs were CTCAE Grade 1, with the exception of abdominal pain which was Common Terminology Criteria for Adverse Events (CTCAE) Grade 2. All reported TEAEs were considered by the Investigator as probably unrelated to study intervention (see Table 12). For laboratory parameter measurements over for each study visit, individual measurements of hematological and clinical chemistry parameters were within their normal reference ranges and variation between visits not considered clinically significant.

TABLE 12

Number of TEAEs by Severity and Relationship to Study Drug by body system and preferred term																	
										Relationship to treatment							
		CTCAE grade (v5)								Probably		Possibly		Probably		Not	
System Organ	Preferred	1		2		3-5		Unrelated		unrelated		related		related		assessable	
Class (SOC)	term (PT)	n	%	n	%	n	%	n	%	n	%	n	%	n	%	n	%
Injury, poisoning and procedural complications		1	(33)	0	0%	0	0%	0	0%	1	33%	0	0%	0	0%	0	0%
		1															
Gastrointestinal disorders	Tooth fracture	1	33%	0	0%	0	0%	0	0%	1	33%	0	0%	0	0%	0	0%
		2	67%	1	33%	0	0%	0	0%	3	100%	0	0%	0	0%	0	0%
	Abdominal pain	0	0%	1	33%	0	0%	0	0%	1	33%	0	0%	0	0%	0	0%
	Constipation	1	33%	0	0%	0	0%	0	0%	1	33%	0	0%	0	0%	0	0%
	Vomiting	1	33%	0	0%	0	0%	0	0%	1	33%	0	0%	0	0%	0	0%

TABLE 12-continued

Number of TEAEs by Severity and Relationship to Study Drug by body system and preferred term																	
System Organ Class (SOC)	Preferred term (PT)	CTCAE grade (v5)						Relationship to treatment									
		1		2		3-5		Unrelated	Probably unrelated	Possibly related	Probably related	Not assessable					
		n	%	n	%	n	%										
General disorders and administration site conditions		1	33%	0	0%	0	0%	0	0%	1	33%	0	0%	0	0%	0	0%
	Localised oedema	1	33%	0	0%	0	0%	0	0%	1	33%	0	0%	0	0%	0	0%
	Skin and subcutaneous tissue disorders	1	33%	0	0%	0	0%	0	0%	1	33%	0	0%	0	0%	0	0%
	Erythema	1	33%	0	0%	0	0%	0	0%	1	33%	0	0%	0	0%	0	0%

Example 6: Imaging Assessments

[0280] All 3 patients completed all planned study imaging procedures for assessment of biodistribution and clearance: PSMA-PET/CT scan at baseline, gamma camera imaging (SPECT/CT and planar gamma imaging) on Day 0 and Day 1 following ^{212}Pb -Compound 1 administration.

Baseline PSMA-PET CT

[0281] Each patient received a baseline PSMA-PET/CT evaluation utilizing an F18-PSMA-PET tracer during the screening period (i.e. prior to ^{212}Pb -Compound 1 administration).

[0282] Normal Tissue: PSMA-PET standard uptake values (SUVs) for normal tissues are summarized in Table 13. The pattern of PSMA-PET CT tracer uptake in normal organs is within expectations in each case; with notable uptake in the salivary glands (SUVmax range 20.06-30.7), kidneys (SUVmax range 18.7-35.2) and liver (SUVmax range 14.5-25.5).

[0283] Metastatic Lesions: Metastatic lesions were observed for all three participants at screening (<20 lesions each). PSMA-PET standard uptake values (SUVs) for metastatic lesions are summarized in Table 16. All 3 patients had at least three PSMA-avid metastatic lesions, with a wide range of F18-PSMA-PET uptake values with SUVmax of 10.1-77.4. For Participant NO-01-001 lesions were located in bone and viscera, for Participant NO 01 002 lesions were located in bone, and for Participant NO-01-003 lesions were located in bone and lymph node.

 ^{212}mPb -Compound 1 Gamma Imaging: Planar and SPECT CT

[0284] All patients received both planar imaging and SPECT/CT scans at the 1 hour and 16 hour timepoints post administration of ^{212}Pb -Compound 1. Note that as gamma imaging of ^{212}Pb is exploratory, the acquisition parameters (i.e., scan speeds, acquisition times etc.) were optimized during the clinical study between imaging the individual patients.

[0285] Normal Tissue: Visual assessment of uptake of ^{212}Pb -Compound 1 in normal tissues was defined as uptake clearly distinguishable from adjacent tissue.

Planar Gamma Imaging

[0286] Visual assessment of planar gamma imaging indicated ^{212}Pb -Compound 1 uptake by the kidneys for all three participants at both time points, by the liver in one participant at the 1 hr time point, and by the urinary bladder (including contents) for two participants at both time points.

[0287] There was no visualization of the salivary glands indicating ^{212}Pb -Compound 1 uptake on planar gamma imaging in any of the 3 patients at any timepoint.

[0288] Visual assessment of ^{212}Pb -Compound 1 by planar gamma imaging in normal tissues in comparison to the liver is presented in Table 14.

[0289] Visual assessment of ^{212}Pb -Compound 1 in participant NO-01-002 in normal tissue by planar gamma imaging is shown in FIG. 5A (anterior view 1 hour post injection), FIG. 5B (posterior view 1 hour post injection), FIG. 5C (anterior view 16 hours post injection), and FIG. 5D (posterior view 16 hours post injection).

SPECT/CT

[0290] Visual assessment of SPECT/CT images indicated ^{212}Pb -Compound 1 uptake by the kidneys for all three participants for at least one of the time points, by the liver in one participant at one time point, in the blood pool for two participants for at least one of the time points, and in the urinary bladder (including contents) for one participant at one time point.

[0291] There was no visualization of the salivary glands indicating ^{212}Pb -Compound 1 uptake on SPECT/CT in any of the 3 patients at any timepoint.

[0292] Visual assessment of ^{212}Pb -Compound 1 by SPECT/CT in normal tissues in comparison to the liver is presented in Table 15.

[0293] Visual assessment of ^{212}Pb -Compound 1 in participant NO-01-001 in normal tissue by SPECT/CT is shown in FIG. 4A (whole body MIP, anterior view) and FIG. 4B (axial section through the cranium at the level of the parotid glands). ^{212}Pb -Compound 1 uptake could not be visualized in the parotid glands.

[0294] Visual assessment of ^{212}Pb -Compound 1 in participant NO-01-003 in the abdominal retrocaval lymph node is shown in FIG. 6, including paired SPECT (summed) scans of the axial section (FIG. 6A), sagittal

section (FIG. 6C), and coronal section (FIG. 6E), and non-contrast enhanced CT scans of the axial section (FIG. 6B), sagittal section (FIG. 6D), and coronal section (FIG. 6F). The retrocaval lymph node metastasis is indicated with white arrows on the CT images, and the areas with increased ^{212}Pb -Compound 1 uptake (darker regions) are shown in the SPECT images.

[0295] Metastatic Lesions: Visual assessment of SPECT/CT images indicated ^{212}Pb -Compound 1 uptake at the 1 hour time point for one lesion in Participant NO-01-003, a

retrocaval lymph node; 11 mm in short axis and 16 mm in longest diameter. This lesion was the most PSMA-avid lesion for any participant, with the highest SUV mean and max demonstrated on PSMA-PET (77.4 Bq/mL and 45 Bq/mL, respectively).

[0296] ^{212}Pb -Compound 1 was not visualized for any other lesion, at any timepoint, by SPECT/CT or planar gamma imaging. Visual assessment of ^{212}Pb -Compound 1 by planar gamma imaging and SPECT/CT in metastatic lesions in comparison to the liver is presented in Table 17.

TABLE 13

F18-PSMA-PET: Normal Tissue Evaluation at Screening																		
Participant Id	How many lesions observed?	Regional locations	Salivary gland		Kidneys		Liver		Spleen		Small Intestine, including contents		Blood pool		Bladder, including contents		Bone Marrow	
			SUV max	SUV mean	SUV max	SUV mean	SUV max	SUV mean	SUV max	SUV mean	SUV max	SUV mean	SUV max	SUV mean	SUV max	SUV mean	SUV max	SUV mean
			Bq/mL	Bq/mL	Bq/mL	Bq/mL	Bq/mL	Bq/mL	Bq/mL	Bq/mL	Bq/mL	Bq/mL	Bq/mL	Bq/mL	Bq/mL	Bq/mL	Bq/mL	Bq/mL
NO-01-001	</=20	Bone, Viscera	30.7	18.5	35.2	19.2	14.5	11.3	16.2	12.6	7.1	5.4	1.2	0.7	8.3	6.2	1.8	1.3
NO-01-002	</=20	Bone	20.6	13	26.8	16.1	25.5	21.8	7.2	6.1	22.5	17.1	0.8	0.6	2.8	2.3	1.3	0.9
NO-01-003	</=20	Lymph Node, Bone	30.7	19.8	18.7	11.4	14.8	12.5	15.3	13.7	22	16.1	1.3	1.1	3	2.3	1	0.7

TABLE 14

Gamma Imaging (Planar): Normal Tissue Evaluation										
		Salivary gland			Kidneys			Spleen		
Participant Id	Time point	Relative time of scan to dose (H.M)	Scan speed (cm/min)	Did visual assessment indicate ²¹² Pb-Compound 1 uptake?	Visual assessment of ²¹² Pb-Compound 1 uptake in comparison to liver?	Did visual assessment indicate ²¹² Pb-Compound 1 uptake?	Visual assessment of ²¹² Pb-Compound 1 uptake in comparison to liver?	Liver Did visual assessment indicate ²¹² Pb-Compound 1 uptake?	Did visual assessment indicate ²¹² Pb-Compound 1 uptake?	Visual assessment of ²¹² Pb-Compound 1 uptake in comparison to liver?
NO-01-001	1 h	1.59	7	No		Yes	Greater than	No	No	
	16 h	17.44	5	No		Yes	Greater than	No	No	
NO-01-002	1 h	2.06	3	No		Yes	Greater than	No	No	
	16 h	17.54	3	No		Yes	Equal to	No	No	
NO-01-003	1 h	2.00	4	No		Yes	Equal to	Yes	No	
	16 h	19.32	3	No		Yes	Greater than	No	No	
Small Intestine, including contents				Blood pool		Bladder, including contents		Bone Marrow		
Did visual assessment indicate ²¹² Pb-Compound 1 uptake?	Visual assessment of ²¹² Pb-Compound 1 uptake in comparison to liver?		Did visual assessment indicate ²¹² Pb-Compound 1 uptake?	Visual assessment of ²¹² Pb-Compound 1 uptake in comparison to liver?		Did visual assessment indicate ²¹² Pb-Compound 1 uptake?	Visual assessment of ²¹² Pb-Compound 1 uptake in comparison to liver?	Did visual assessment indicate ²¹² Pb-Compound 1 uptake?	Visual assessment of ²¹² Pb-Compound 1 uptake in comparison to liver?	
No			No			No			No	
No			No			No			No	
No			No			Yes	Equal to		No	
No			No			Yes	Equal to		No	
No			No			Yes	Equal to		No	
No			No			Yes	Equal to		No	

TABLE 15

Gamma Imaging (SPECT/CT): Normal Tissue Evaluation									
Participant Id	Time point	dose (H.M)	Salivary gland		Kidneys		Liver		
			Did visual assessment of ^{212}Pb -indicate Compound 1 uptake?	Visual assessment of ^{212}Pb -Compound 1 uptake in comparison to liver?	Did visual assessment of ^{212}Pb -indicate Compound 1 uptake?	Visual assessment of ^{212}Pb -Compound 1 uptake in comparison to liver?	Did visual assessment of ^{212}Pb -indicate Compound 1 uptake?	Did visual assessment of ^{212}Pb -indicate Compound 1 uptake?	Visual assessment of ^{212}Pb -Compound 1 uptake in comparison to liver?
			Compound 1 uptake?	Compound 1 uptake?	Compound 1 uptake?	Compound 1 uptake?	Compound 1 uptake?	Compound 1 uptake?	Compound 1 uptake?
NO-01-001	1 h	2.45	No		No		No	No	
	16 h	18.34	Not Applicable		Yes	Greater than	No	No	
NO-01-002	1 h	3.03	No		Yes	Greater than	No	No	
	16 h	18.52	No		Yes	Greater than	No	No	
NO-01-003	1 h	2.52	No		Yes	Equal to	Yes	No	
	16 h	20.38	No		Yes	Greater than	No	No	
Small Intestine, including contents		Blood pool		Bladder, including contents		Bone Marrow			
Did visual assessment of ^{212}Pb -indicate Compound 1 uptake?	Visual assessment of ^{212}Pb -Compound 1 uptake in comparison to liver?	Did visual assessment of ^{212}Pb -indicate Compound 1 uptake?	Visual assessment of ^{212}Pb -Compound 1 uptake in comparison to liver?	Did visual assessment of ^{212}Pb -indicate Compound 1 uptake?	Visual assessment of ^{212}Pb -Compound 1 uptake in comparison to liver?	Did visual assessment of ^{212}Pb -indicate Compound 1 uptake?	Visual assessment of ^{212}Pb -Compound 1 uptake in comparison to liver?	Did visual assessment of ^{212}Pb -indicate Compound 1 uptake?	Visual assessment of ^{212}Pb -Compound 1 uptake in comparison to liver?
No		Yes	Equal to	No		No		No	
No		Not Applicable		No		No		No	
No		No		Yes	Equal to	No		No	
No		No		Not Applicable		No		No	
No		Yes	Equal to	Not Applicable		No		No	
No		No		No		No		No	

TABLE 16

F18-PSMA-PET: Metastatic Lesion Evaluation								
Participant ID		Lesion No.	Regional location	Anatomical location	Size (mm)	SUV max Bq/mL	SUV mean Bq/mL	EANM PSMA-PET score
NO-01-001	Screening	Lesion 1	Bone	Right pubic bone ramus superior	60	10.1	5.7	1 = Equal to or above blood pool and lower than liver
		Lesion 2	Visceral	Dorsal bladder wall	28 × 13 × 17	15.9	9	2 = Equal to or above liver and lower than parotid gland
		Lesion 3	Visceral	Left seminal vesicle	12 × 8 × 10	10.6	6.2	1 = Equal to or above blood pool and lower than liver
NO-01-002	Screening	Lesion 1	Bone	Sternum	150	28.6	14.7	2 = Equal to or above liver and lower than parotid gland
		Lesion 2	Bone	Left 3rd rib dorsal	25	30.9	20.6	3 = Equal to or above parotid gland
		Lesion 3	Bone	Cervical vertebra C7	25	18.9	11.3	2 = Equal to or above liver and lower than parotid gland
NO-01-003	Screening	Lesion 1	Lymph node	Retrocaval	11 × 14 × 16	77.4	45	3 = Equal to or above parotid gland
		Lesion 2	Bone	Right 4th rib anterolateral	40	17.4	11.1	2 = Equal to or above liver and lower than parotid gland
		Lesion 3	Bone	Lumbar vertebra L3	10	13.6	8	2 = Equal to or above liver and lower than parotid gland

TABLE 17

Gamma Imaging (planar and SPECT/CT): Metastatic Lesion Evaluation											
Participant ID	Time Point	Lesion No.	Regional location	Anatomical location	Size (mm)	Relative time of scan to dose (h:mm)	Planar		SPECT-CT		
							Compound 1 uptake?	Visual assessment of ²¹² Pb-Compound 1 uptake in comparison to liver?	Relative time of scan to dose (h:mm)	Compound 1 uptake?	Visual assessment of ²¹² Pb-Compound 1 uptake in comparison to liver?
NO-01-001	1 H	Lesion 1	Bone	Right pubic bone ramus superior	60	1.59	No		2.45	No	
		Lesion 2	Visceral	Dorsal bladder wall	28 × 13 × 17		No		2.45	No	
		Lesion 3	Visceral	Left seminal vesicle	12 × 8 × 10		No		2.45	No	
	16 H	Lesion 1	Bone	Right pubic bone ramus superior	60	17.44	No		18.34	No	
		Lesion 2	Visceral	Dorsal bladder wall	28 × 13 × 17		No		18.34	No	
		Lesion 3	Visceral	Left seminal vesicle	12 × 8 × 10		No		18.34	No	
NO-01-002	1 H	Lesion 1	Bone	Sternum	150	2.06	No		3.03	No	
		Lesion 2	Bone	Left 3rd rib dorsal	25		No		3.03	No	
		Lesion 3	Bone	Cervical vertebra C7	25		No		3.03	No	
	16 H	Lesion 1	Bone	Sternum	150	17.54	No		18.52	No	
		Lesion 2	Bone	Left 3rd rib dorsal	25		No		18.52	No	
		Lesion 3	Bone	Cervical vertebra C7	25		No		18.52	No	
NO-01-003	1 H	Lesion 1	Lymph node	Retrocaval	11 × 14 × 16	2.00	No		2.52	Yes	Greater than
		Lesion 2	Bone	Right 4th rib anterolateral	40		No		2.52	No	
		Lesion 3	Bone	Lumbar vertebra L3	10		No		2.52	No	
	16 H	Lesion 1	Lymph node	Retrocaval	11 × 14 × 16	19.32	No		20.38	No	
		Lesion 2	Bone	Right 4th rib anterolateral	40		No		20.38	No	
		Lesion 3	Bone	Lumbar vertebra L3	10		No		20.38	No	

Example 7: Summary of Findings

[0297] The primary objective of the study was to explore whether the use of gamma imaging (SPECT/CT and planar gamma imaging) was feasible for confirming uptake of ²¹²Pb-Compound 1 in various normal organs/tissue and in metastases. Further objectives included assessment of safety and tolerability, assessment of body clearance of ²¹²Pb-Compound 1, comparison to PSMA PET imaging, and assessment of the disease related biomarkers PSA and ALP.

[0298] Three adult male participants with progressive mCRPC were screened, met the inclusion criteria of the study, received 10 MBq ²¹²Pb-Compound 1 as planned, and completed the study imaging and follow up procedures. ²¹²Pb-Compound 1 was administered to each of the participants without safety concerns from the reported TEAEs, clinical laboratory data, vital signs or any other observations. The study findings support feasibility of gamma imaging for assessing uptake of ²¹²Pb-Compound 1 in various normal organs/tissues and in metastases by visual qualitative assessment.

[0299] The most PSMA-avid metastatic lesion, a retrocaval lymph node metastasis with short-axis diameter 11 mm, was visualized with ²¹²Pb-Compound 1 uptake on

post-therapy SPECT/CT; however, visualization on planar imaging was not possible. For the other metastatic lesions, visualization of ²¹²Pb-Compound 1 on either SPECT/CT or planar imaging was not clearly demonstrated. This may be due to lower PSMA-expression of these metastases indicated by the lower SUVs on PSMA PET, combined with the low dose of ²¹²Pb-Compound 1 and limitations of gamma imaging described above.

[0300] The distribution of ²¹²Pb-Compound 1 observed on planar gamma imaging and SPECT/CT is compatible with the observed distribution of PSMA/PET tracer at screening for most organs, including the kidneys, liver, and urinary bladder, except for the salivary glands which were not detectable.

[0301] The absence of observed salivary gland uptake of ²¹²Pb-Compound 1 is notable. Salivary glands express PSMA; both PSMA-targeted PET/CT tracer (as demonstrated at baseline PET/CT imaging) and other PSMA-targeted radiotherapeutic agents typically demonstrate substantial uptake in the salivary glands.

[0302] Of note, salivary gland toxicity (xerostomia) is a common for most PSMA-targeting RLTs (e.g. ¹⁷⁷Lu-PSMA-617), and may be the dose limiting toxicity for several of

these (e.g. ^{225}Ac -PSMA-617). Furthermore, xerostomia even at low severity has a strong impact on patient quality of life (e.g. CTCAE grade 1 or 2); whereas higher severity (e.g. CTCAE grade 3) may render patients unable to receive food by mouth.

[0303] The observation in the study that on both SPECT/CT and planar gamma imaging that no ^{212}Pb -Compound 1 uptake could be observed to the salivary glands, indicates a lower risk of xerostomia for ^{212}Pb -Compound 1.

Example 8: Phase 1/2 Study to Investigate the Safety, Tolerability, Pharmacokinetics, Biodistribution and Antitumour Activity of the Alpha-Therapeutic Radioligand ^{212}Pb -Compound 1

[0304] This is a Phase 1/2 dose escalation study to investigate if participants with advanced mCRPC can be treated with ^{212}Pb -Compound 1 with an acceptable safety profile and expected positive benefit/risk ratio, and to obtain preliminary data on antitumor activity. The dose escalation design will allow determination of the recommended dose and treatment schedule of ^{212}Pb -Compound 1, for further evaluation and refinement during the dose expansion part of the trial. Furthermore, evaluation will be undertaken of the biodistribution, pharmacokinetics (PK), and body clearance for ^{212}Pb -Compound 1 to support development of this Alpha Radioligand Therapy (ART).

[0305] Dose Escalation: A step-wise escalation of the ^{212}Pb dose (MBq) of ^{212}Pb -Compound 1 will be conducted in cohorts of participants with mCRPC who have not received prior treatment with ^{177}Lu -PSMA, to establish the recommended dose and regimen to be explored in Dose Expansion.

[0306] Dose Expansion: The Dose Expansion Groups define mCRPC patient populations with unmet medical need: Group A: participants who have not received prior treatment with ^{177}Lu -PSMA-617, and Group B: participants who have been treated with ^{177}Lu -PSMA-617. Participants in Group A will receive the recommended dose and schedule from Dose Escalation. For Group B, a safety run-in cohort will first be opened at one dose level lower to ensure no safety concerns before proceeding to the recommended dose and schedule.

Example 9: Phase 1/2 study objectives and endpoints

Objectives: Primary

[0307] To determine the safety and tolerability profile of ^{212}Pb -Compound 1 in participants with mCRPC

[0308] Dose Escalation: To determine the Recommended Dose and Schedule for Dose Expansion for ^{212}Pb -Compound 1

[0309] Dose Expansion: To determine the Recommended Dose and Schedule for clinical development for ^{212}Pb -Compound 1

Endpoints: Primary

[0310] Incidence and severity of TEAEs (including adverse event of special interest and treatment-emergent serious adverse event)

[0311] Incidence of dose limiting toxicity

[0312] Preliminary antitumor activity (ORR and PSA_{50})

[0313] Recommended dose and schedule based on safety and preliminary antitumour activity assessments (including incidence of dose limiting toxicities (DLTs) and PSA_{50} response)

[0314] Recommended dose and schedule based on safety and preliminary antitumour efficacy (including incidence and severity of [S]AEs, PSA_{50} response, and ORR)

Objectives: Secondary

[0315] To determine the preliminary antitumor activity of ^{212}Pb -Compound 1

[0316] To evaluate the antitumor efficacy of ^{212}Pb -Compound 1

[0317] To evaluate SPECT/CT gamma camera imaging for ^{212}Pb -Compound 1

[0318] To evaluate the pharmacokinetics of ^{212}Pb -Compound 1

Endpoints: Secondary

[0319] Objective response rate (ORR) (per PCWG3 guidelines)

[0320] PSA_{50} response at >12 weeks

[0321] Best overall PSA response

[0322] Radiologic progression-free survival (rPFS) (per Prostate Cancer Working Group 3 (PCWG3) guidelines)

[0323] Duration of response (DOR) (per PCWG3 guidelines)

[0324] Duration of PSA_{50} response

[0325] Overall survival (OS)

[0326] Assessment of uptake in regions of interest (ROIs) in normal tissues and tumor lesions

[0327] Blood activity of ^{212}Pb -Compound 1 after single and multiple doses (including C_{max} , C_{min} , AUC, etc.)

Objectives: Exploratory

[0328] To evaluate early antitumour activity of ^{212}Pb -Compound 1 using PSMA/PET-CT

[0329] To further evaluate the antitumour efficacy of ^{212}Pb -Compound 1

[0330] To evaluate body clearance of ^{212}Pb -Compound 1

[0331] To evaluate biodistribution of ^{212}Pb -Compound 1 by SPECT/CT imaging

[0332] To evaluate localisation of lesions detected by SPECT/CT of ^{212}Pb -Compound 1 and baseline PSMA-PET/CT

[0333] To investigate PSMA radiotracer uptake in tumour lesions and potential relationship to the antitumour activity of ^{212}Pb -Compound 1

[0334] To investigate uptake in tumour lesions of ^{212}Pb -Compound 1 by gamma camera imaging and potential relationship the antitumour activity of ^{212}Pb -Compound 1

[0335] To assess potential pharmacodynamic and safety biomarkers for ^{212}Pb -Compound 1

[0336] To investigate the safety profile of ^{212}Pb -Compound 1 relative to prior radiation exposure

[0337] To assess impact on quality of life for participants with mCRPC receiving ^{212}Pb -Compound 1

Endpoints: Exploratory

- [0338] ORR per RECIP 1.0 between baseline to Week 12
- [0339] Overall ORR using RECIP 1.0 criteria
- [0340] Overall survival (OS)
- [0341] Quantification of activity from SPECT/CT imaging
- [0342] Quantification of activity from whole-body measurements
- [0343] Urine clearance (up to 48 h)
- [0344] Assessment of uptake in ROIs in normal tissues
- [0345] Dosimetry estimations
- [0346] Compare ²¹²Pb-Compound 1 and PSMA-PET radiotracer uptake in corresponding tumour lesions
- [0347] Correlation of baseline mean standardised uptake values (SUVs) in tumour lesions from PSMA-PET/CT with antitumour activity (radiological response, PSA50, etc.)
- [0348] Correlation ROI measurements in tumour lesions from gamma camera imaging with antitumour activity (radiological response, PSA50, etc.)
- [0349] Various biomarkers (e.g., tumour genomic alterations, circulating safety biomarkers for critical organs, potentially predictive biomarkers, etc)
- [0350] Assessment of TEAEs with exposure from EBRT and RTIs
- [0351] Health-related Quality of Life (HRQoL) questionnaires FACT-P and SXI, optionally FACT-RNT

Example 10: Dose Escalation Cohort

- [0352] The Dose Escalation part of the study aims to assess the safety and tolerability of ²¹²Pb-Compound 1 and identify the recommended dose and schedule of ²¹²Pb-Compound 1 for the Dose Expansion groups. See the Inclusion and Exclusion Criteria for definition of the participant population for Dose Escalation, including requirements for prior anticancer therapies.
- [0353] Dose Escalation of ²¹²Pb-Compound 1 will be based on assessment of clinical safety data supported by available PK and tumor response profiles, supported by a Time-to-Event Bayesian Optimal Interval Design (TITE-BOIN) method which includes information over the periods of exposure to ²¹²Pb-Compound 1To enable early dosage optimisation of ²¹²Pb-Compound 1, the Dose Escalation part of the study will aim to evaluate the planned dose levels of ²¹²Pb-Compound 1 and the schedule of treatment (cycle duration).
- [0354] Dose Escalation will be initiated with a starting dose of 100 MBq of ²¹²Pb for the first cohort of participants. ²¹²Pb dose levels will be increased for subsequent cohorts in a planned stepwise fashion (see Table 18). Dose levels below 100 MBq or increments between the dose levels described in Table 18 may be explored if indicated by clinical efficacy and/or safety data.
- [0355] The ligand dose for ²¹²Pb-Compound 1 will be in the range of 50-200 µg with no planned escalation. For example, the ligand dose for ²¹²Pb-Compound 1 will be in the range of 50-100 µg (no planned escalation).

TABLE 18

Exemplary ²¹² Pb Dose Escalation Levels				
Dose	²¹² Pb	% Change from	No. Treatment cycles	
Level	Dose	previous Level	Planned	Additional ^a
1	100 MBq	NA	4	+2
2	150 MBq	50%	4	+2
3	200 MBq	33%	4	+2
-1 ^b	75 MBq	-25%	4	+2

²¹²Pb, lead-212; NA, not applicable; No., number.
^aIndividual participants may receive an additional two treatment cycles after the four planned if there is clinical benefit in the opinion of the Investigator, and safety criteria are met.
^bDose Level -1 may be explored if indicated by clinical data (efficacy and/or safety). Reference for % change for Dose Level -1 is relative to Dose Level 1.

- [0356] Dose Escalation will be initiated with ²¹²Pb-Compound 1 administered by slow bolus injection on Day 1 of a 6-week (42-day) cycle. Study treatment administration after Cycle 1 may be delayed by up to an additional six weeks if required to manage treatment-emergent adverse events (TEAEs). Participants may only receive study treatment after Cycle 1 if they meet all safety criteria required to continue, have not progressed according to the Prostate Cancer Working Group 3 (PCWG3) guidelines, and after individual positive benefit/risk assessment by the Investigator and Sponsor.
- [0357] Four cycles of study treatment are planned. Individual participants may continue up to a maximum of 6 treatment cycles provided they demonstrate all of the following:
- [0358] Clinical benefit in the opinion of the Investigator
 - [0359] A radiographic response, and/or a PSA50 response, and/or clinical improvement
 - [0360] No progressive disease per PCWG3 criteria
 - [0361] No acute toxicity greater than CTCAE Grade 1. Exception may be made for chronic, stabilized Grade 2 events, which may have contribution from past therapies or the underlying disease (e.g. alopecia, neuropathy, fatigue, anorexia, nail changes, etc.)
- [0362] Cycle Optimization: During Dose Escalation, dependent on the safety and efficacy profile of ²¹²Pb-Compound 1, additional cohorts may be opened to assess optimal cycle duration, including shorter cycle periods (e.g., three or four weeks). Cohorts to assess cycle optimization may be opened in parallel to assessing dose level escalation.
- Dose Escalation Cohorts and Process
- [0363] Participants will be assigned to cohorts at ²¹²Pb dose levels (per the Cohort Management Plan). In the first ²¹²Pb dose level cohort there will be an interval of at least 7 days between the first and second participants' first dose, and an interval of at least 1 day between the first dose for subsequent participants. In all subsequent cohorts, there will be an interval of at least 1 day between the first and second participants' first dose; however, if a DLT occurs in the first week of treatment for a participant in any cohort, all subsequent cohorts will apply an interval of at least seven days between the first and second participants' first dose as an additional precaution.
- [0364] The TITE-BOIN approach (Yuan Y, Lin R, Li D, Nie L, Warren K E, "Time-to-event Bayesian optimal interval design to accelerate phase I trials," Clin Cancer Res. 2018; 24(20):4921-30. doi:10.1158/1078-0432.CCR-18-0246) with a target DLT rate of 30% will be used to guide

the decision on the next dose level/schedule that can be taken. If the observed DLT rate at the current dose is <0.236 , escalate the dose to the next higher dose level. If the observed DLT rate at the current dose is >0.359 , de-escalate the dose to the next lower dose level. If the observed DLT rate at the current dose is between 0.236 and 0.359 then the current dose is maintained.

[0365] Initially, three participants will be treated in each Dose Escalation cohort, and a minimum of two participants will have to be DLT-evaluable prior to adjusting dose or dosing schedule. To be considered DLT-evaluable, a participant is required to receive at least 80% of the planned dose at Cycle 1 Day 1 and will have either completed the 'DLT Observation Period' of six weeks in the absence of DLT, or have experienced a DLT. Participants who discontinue during the first six weeks due to any reason other than a DLT, and participants who received less than 80% of the planned dose will not be considered evaluable, and will be replaced. 12271 The Phase 1 trial will commence at 100 MBq administered on Day 1 of a 6-week cycle to ensure participant safety. Initially, three participants will be treated. The decision to escalate, maintain, or de-escalate the dosage will be guided by the TITE-BOIN design. This approach allows for dynamic decision-making by not requiring all pending participants to be DLT evaluable before making dosage adjustments. Exemplary TITE-BOIN decisions are represented in Table 19.

TABLE 19

Dose Escalation/De-escalation Rules for the TITE-BOIN Design					
#Pts	#DLTs	#Pending	Escalate	Stay	De-escalate
3	0	≤ 1	Y		
3	0	≥ 2		Suspend	
3	1	0		Y	
3	1	1		>0.88	≤ 0.88
3	1	≥ 2		Suspend	
3	2	≤ 1			Y
3	3	0			Y & Elim
6	0	≤ 3	Y		
6	0	≥ 4		Suspend	
6	1	≤ 1	Y		
6	1	2	≥ 0.60	<0.60	
6	1	3	≥ 1.96	<1.96	
6	1	≥ 4		Suspend	
6	2	0	Y		
6	2	1		>0.73	≤ 0.73
6	2	2		>1.80	≤ 1.80
6	2	3		>2.87	≤ 2.87
6	2	≥ 4		Suspend	
6	3	≤ 3			Y
6	≥ 4	≤ 2			Y & Elim
9	0	≤ 4	Y		
9	0	≥ 5		Suspend	
9	1	≤ 4	Y		
9	1	≥ 5		Suspend	
9	2	0	Y		
9	2	1	≥ 0.59	<0.59	
9	2	2	≥ 1.65	<1.65	
9	2	3	≥ 2.71	<2.71	
9	2	4	≥ 3.77	<3.77	
9	2	≥ 5		Suspend	
9	3	0		Y	
9	3	1		>0.58	≤ 0.58
9	3	2		>1.65	≤ 1.65
9	3	3		>2.72	≤ 2.72
9	3	4		>3.79	≤ 3.79
9	3	≥ 5		Suspend	
9	4	≤ 5			Y
9	≥ 5	≤ 4			Y & Elim

In Table 19: #, number; DLT, Dose Limiting Toxicity; Elim, eliminate; No., number; Pts, participants; STFT, standardised total follow-up time; TITE-BOIN, Time-to-Event Bayesian Optimal Interval Design; Y, yes. Note: “#Pts” is the total number of evaluable participants treated at the current dose level, “#DLTs” is the number of participants who experienced DLT at the current dose level, “#Pending” denotes the number of participants whose DLT data are pending at the current dose level, “STFT” is the standardised total follow-up time for the participants with data pending, defined as the total follow-up time for the participants with data pending divided by the length of the DLT assessment window. “Y” represents “Yes”, and “Y&Elim” represents “Yes and Eliminate”. When a dose is eliminated, all higher doses should also be eliminated.

[0366] Exemplary DLTs are shown in Table 20.

TABLE 20

Dose Limiting Toxicities	
Hematologic	Non-Hematologic
Grade 4 neutropenia lasting >7 days.	ALT or AST $>3 \times$ upper limit of normal (ULN) and concomitant total bilirubin $>2 \times$ ULN (except for participants with liver metastases or hepatic comorbidities, e.g., Gilbert's syndrome)
Any febrile neutropenia (i.e., $\geq$ Grade 3)	AST or ALT $>8 \times$ ULN of any duration or AST or ALT $>5 \times$ ULN for ≥ 14 days (all participants: with or without liver metastases)
Grade 4 thrombocytopenia	Any Grade ≥ 3 non-haematologic toxicity, except: Isolated, asymptomatic elevations of biochemical laboratory values, including electrolyte abnormalities, that are suspected to be spurious or respond to medical intervention within 72 hours.
	Nausea, vomiting, or diarrhoea controlled by anti-emetic or antidiarrheal therapy within 72 hours.
	Transient fatigue that improves within 72 hours
Grade 3 thrombocytopenia with bleeding event	Any death not clearly due to the underlying disease or extraneous causes
Grade 4 anaemia	Any AE that requires a dose delay of >42 days
	Any other clinically significant and/or unacceptable toxicity judged to be a DLT by the Investigator and Sponsor

Parallel Cohort Evaluation to Optimise Cycle Duration.

[0367] Cohorts to assess cycle optimization may be opened in parallel to cohorts assessing higher ^{212}Pb dose levels. However, it is not permitted to increase the ^{212}Pb Dose level and decrease cycle duration in the same Dose Escalation Cohort. Example nomenclature for Cohorts is provided in Table 21.

TABLE 21

Dose Escalation Cohorts		
Cohort	Dose Level (DL)	Schedule
1	DL 1: 100 MBq	6-week cycle
1S*	DL 1: 100 MBq	Cycle length optimization: e.g. 4-week cycle (+/-1 week)
2	DL 2: 150 MBq	6-week cycle
2S*	DL 2: 150 MBq	Cycle length optimization: e.g. 4-week cycle (+/-1 week)
3	DL 3: 200 MBq	6-week cycle
3S*	DL 3: 200 MBq	Cycle length optimization: e.g. 4-week cycle (+/-1 week)
X	Other dose level X (e.g. 75 MBq)	6-week cycle
XS*	Other dose level X (e.g. 75 MBq)	Cycle length optimization: e.g. 4-week cycle (+/-1 week)

[0368] Cohorts with shorter cycle durations may follow the same Dose Escalation procedure guided by the TITE-BOIN design (assumptions as for the 6-week cycle schedule), but with the additional precaution that Dose Escalation will only be permitted to a Dose Level already ‘cleared’ for a 6-week cycle. The Table 22 below outlines the decisions tree to open parallel cohorts for cycle optimisation based on prior Dose Level clearance.

TABLE 22

Decision Table for Permitting Opening Cohorts for Cycle Length Optimisation in Parallel During ²¹² Pb Dose Level Escalation			
Dose Level	Dose Escalation cleared for 6-week cycle at Dose Level	Dose Escalation cleared for shorter cycle at lower (-1) Dose Level	Initiate parallel Cohort with shorter cycle at Dose Level
100 MBq	Yes	NA	Yes
100 MBq	No	NA	No
150 MBq	Yes	Yes	Yes
150 MBq	No	Yes	No
150 MBq	Yes	No	No
150 MBq	No	No	No
200 MBq	Yes	Yes	Yes
200 MBq	No	Yes	No
200 MBq	Yes	No	No
200 MBq	No	No	No

[0369] In table 22: NA, not applicable.

Dose Decisions

[0370] Safety data will be reviewed on an ongoing basis during the study for all participants and will be regularly discussed with the Investigators. At the time of a potential cohort Dose Decision, indicated by the TITE-BOIN (e.g., Table 19), all available clinical safety, tolerability, PK, and tumour response data will be discussed and evaluated.

[0371] Dose Decisions will be made jointly by the Sponsor and Investigators based on the totality of the data from all preceding Dose Escalation cohorts, and may include: dose escalation to the next Dose Level, dose de-escalation to a lower Dose Level, further recruitment at the same Dose Level, adjustment of the dosing schedule (e.g., cycle duration based on toxicity incidence and rate of recovery), and/or termination of Dose Escalation.

[0372] Dose Decisions will be guided by the TITE-BOIN design according to prespecified rules based on actual obser-

vations of TEAEs that fulfil the DLT criteria, and observed preliminary antitumour activity (see e.g., Table 19).

[0373] Evaluation of Antitumour Activity: During Dose Escalation, once a cohort with a given Dose Level/Schedule is ‘cleared’ by DLT evaluation, the cohort may be backfilled to up to nine evaluable participants if a primary efficacy signal is observed. This approach allows for the gathering of antitumour activity data (e.g., radiological and PSA50 responses), which will be used, along with the clinical safety data and available PK and biodistribution data, to select the Recommended Dose and Schedule for Dose Expansion. The safety data of these additional participants will be discussed by the Investigators, the Sponsor, and the CRO. Any DLTs or Regimen Impacting Toxicities (RITs) will be taken into consideration for subsequent Dose Escalation Decisions.

End of Dose Escalation:

[0374] Dose Escalation will end if one of the stopping criteria is met:

[0375] The recommended dose for Expansion has been determined.

[0376] The Maximum Tolerated Dose (MTD) is determined (or the maximum planned dose is tolerable).

[0377] The minimum dose tested is toxic.

[0378] At any time where there are already nine or more evaluable participants treated at the ²¹²Pb dose level recommended for the next Dose Escalation cohort.

[0379] The Dose Escalation part of the study will progress into the Dose Expansion part. Based on available data, participants in the Dose Escalation Cohort at the Recommended Dose and Schedule for Expansion may be included in Dose Expansion Group A..

Example 11: Dose Expansion

[0380] The Dose Expansion part of the study will progress seamlessly from Dose Escalation. The Dose Expansion part of the study aims to determine the preliminary antitumor activity and safety of ²¹²Pb-Compound 1 and confirm the recommended dose and schedule for further development. Participants in Dose Expansion will be monitored for adverse events (for example, DLTs) and antitumor activity to allow continual monitoring of the recommended dose and schedule throughout the clinical study.

[0381] There will be 2 separate Groups in Dose Expansion of 30 participants, with indication-based populations of mCRPC specified based on participant’s prior anticancer therapies, namely:

[0382] Group A: patients who have not received prior treatment with ¹⁷⁷Lu-PSMA

[0383] Group B: patients who have been treated with ¹⁷⁷Lu-PSMA.

[0384] Participants in Group A will receive the recommended dose and schedule of ²¹²Pb-Compound 1 from Dose Escalation. For Group B, a safety run-in cohort of at least 3 evaluable participants will first be opened at one dose level lower than the recommended dose and Schedule from Escalation. Clearance of the safety run-in cohort by Dose Decision Meeting (using the same process as described for Dose Escalation) must occur before Group B may proceed to open at the recommended dose and schedule.

[0385] More than one dose level or schedule of ²¹²Pb-Compound 1 may be selected to be tested in any Dose Expansion Group, either sequentially or concurrently, based

on the benefit-risk profile observed during Dose Escalation. Randomization will be performed if more than one dose level/schedule is tested concurrently in same Dose Expansion group.

[0386] The Dose Expansion part of the study will be conducted in an adaptive two-stage design with one interim assessment in each Group. The expected number of evaluable participants in each group is 30. An interim futility assessment will be performed after approximately 50% of the intended total number of evaluable participants in each group have completed the tumor assessments indicated before the end of Cycle 3.

Example 12: Phase 1/2 Study Population

Inclusion Criteria

[0387] Participants are eligible to be included in the study only if all the following criteria apply:

[0388] Male participants at least 18 years of age at the time of the signing informed consent.

[0389] Eastern Cooperative Oncology Group performance status (ECOG PS) of 0 or 1.

[0390] Life expectancy of at least 6 months.

[0391] Metastatic castration-resistant prostate cancer (mCRPC) with histological confirmation of prostate adenocarcinoma. The mCRPC may be progressive.

[0392] Availability of fresh or archival tumor samples obtained on or after the most recent anticancer therapy. Tumor samples obtained before the last anticancer treatment may be accepted on agreement between the Investigator and Sponsor. Availability of alternative profiling of tumour mutational status may be accepted instead on agreement between the Investigator and Sponsor.

[0393] Previous treatment with at least one novel androgen receptor axis targeted agent (ARATA).

[0394] Prior orchiectomy and/or ongoing androgen-deprivation therapy and a castrate level of serum testosterone (<50 ng/dL or <1.7 nmol/L).

[0395] Prior treatment with at least 1 but no more than 2 taxane regimens or been deemed ineligible for or refused taxane therapy on consultation with their physician.

[0396] Dose Expansion Group B only: prior treatment with at least one cycle of ^{177}Lu PSMA therapy greater than 6 weeks from start of study treatment.

[0397] At least one PSMA-avid distant metastatic lesion confirmed by PSMA-PET/CT with a permitted PSMA-targeting imaging agent. A PSMA-avid distant metastatic lesion is defined as physiological uptake greater than that of adjacent tissue in lesions of any size in any organ system. The participant should have no PSMA negative lesions, defined as physiological uptake equal to or lower than that of adjacent tissue in any lymph node with a short axis of at least 2.5 cm, in any solid organ lesion with a short axis of at least 1.0 cm, or in any bone lesion with a soft-tissue component of at least 1.0 cm in the short axis, confirmed by PSMA-PET/CT with a permitted PSMA-targeting imaging agent within 28 days before start of study treatment.

[0398] Dose Expansion Groups only: At least one measurable lesion per RECIST 1.1 criteria.

[0399] Adequate bone marrow, renal and hepatic function, as assessed by the following laboratory requirements within 28 days before start of study treatment:

[0400] a. Absolute neutrophil count (ANC) $\geq 1.5 \times 10^9/\text{L}$

[0401] b. Hemoglobin ≥ 9.0 g/dL

[0402] c. Platelet count $\geq 100 \times 10^9/\text{L}$

[0403] d. Serum creatinine $\leq 1.5 \times \text{ULN}$ and estimated glomerular filtration rate (eGFR) ≥ 60 mL/min/1.73 m², according to the Modified Diet in Renal Disease (MDRD) abbreviated formula.

[0404] e. Total bilirubin $\leq 1.5 \times$ the upper limit of normal (ULN) (except if confirmed history of Gilbert's disease, in which case direct bilirubin must be normal)

[0405] f. ALT and AST $< 2.5 \times \text{ULN}$ ($\leq 5 \times \text{ULN}$ for participants with liver involvement)

[0406] Male participants must agree to use highly effective contraception during treatment and for 6 months after the last study treatment and refrain from donating sperm during this period.

[0407] Contraceptive use should be consistent with local regulations regarding methods of contraception for those participating in clinical studies.

[0408] Capable of giving signed informed consent which includes compliance with the requirements and restrictions listed in the Informed Consent Form (ICF) and in this protocol.

Exclusion Criteria

[0409] Participants are excluded from the study if any of the following criteria apply:

[0410] Urinary obstruction, or where there is deemed a substantial risk of urinary obstruction due to pelvic disease.

[0411] Untreated CNS metastases (therapy includes surgery, radiotherapy, gamma knife) or treated CNS metastases which are not adequately controlled. Treated CNS metastases are permitted provided they are neurologically stable, asymptomatic, and corticosteroids are not required for to maintain neurologic integrity.

[0412] Symptomatic medullary cord compression, or clinical or radiologic findings indicative of impending cord compression.

[0413] Diffuse bone or bone-marrow involvement i.e., a "superscan": defined as bone scintigraphy in which there is diffuse and intense activity throughout the axial skeleton due to metastatic disease. Participants with borderline/ambiguous assessment may be included on a case-by-case basis on agreement with the Investigator and Sponsor.

[0414] Participants known to be affected by genetic defects linked to radiation hypersensitivity.

[0415] History of Myelodysplastic syndrome (MDS), treatment-related acute myeloid leukemia (t-AML) or features suggestive of MDS/AML.

[0416] A known additional malignancy that has required active treatment within the past 2 years before start of study treatment, except for adequately treated basal or squamous cell carcinoma of the skin, or carcinoma in situ that has undergone curative therapy.

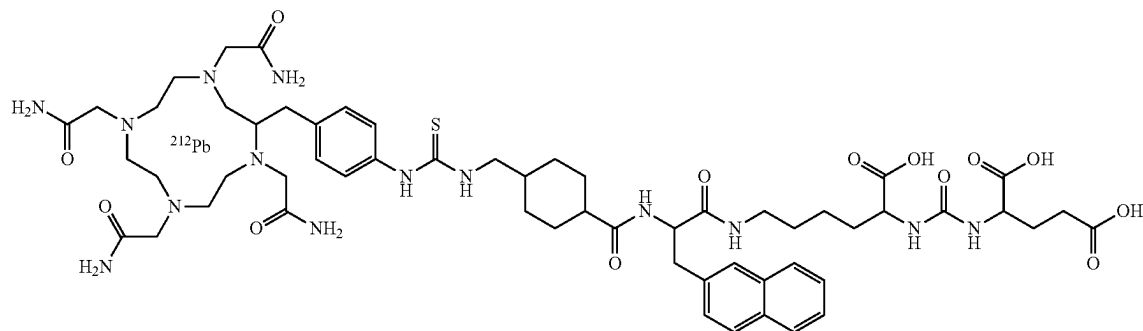
- [0417] Infections of CTCAE v5.0 Grade 2 not responding to therapy or active clinically serious infections of CTCAE Grade >2.
- [0418] Human immunodeficiency virus (HIV) infection with any of the following: a CD4+ T-cell count of <350 cells/uL; a history of AIDS defining opportunistic infection within the past 12 months; established on stable antiretroviral therapy for less than 4 weeks; an HIV viral RNA level >400 copies/mL; requiring either antiretroviral therapy prophylactic antimicrobials expected to cause significant drug-drug interactions or overlapping toxicities.
- [0419] Active hepatitis B virus (HBV) or hepatitis C virus (HCV) infection requiring treatment. Participants with chronic HBV or HCV infection are eligible provided the infection has been controlled with appropriate treatment and the viral load is below the limit of quantification.
- [0420] Serious, infected non-healing wound, ulcer, or bone fracture.
- [0421] Major surgery, open biopsy, or significant trauma within 4 weeks before start of study treatment.
- [0422] Impaired cardiac function or clinically significant cardiac disease, including: congestive heart failure (CHF) (New York Heart Association (NYHA) Class II, III or IV); unstable angina, new-onset angina (within 3 months of start of study treatment), myocardial infarction less than 6 months before the start of study treatment; clinically significant arrhythmia; uncontrolled hypertension (systolic blood pressure >140 mmHg and/or diastolic blood pressure >100 mmHg).
- [0423] Known allergies, hypersensitivity, or intolerance to the study treatment (including excipients).
- [0424] Prior systemic anticancer therapy (e.g., chemotherapy, immunotherapy or biological therapy (including monoclonal antibodies)) or investigational therapy within 4 weeks before the start of study treatment, with the exception of Luteinizing Hormone-releasing Hormone (LHRH) or Gonadotropin-releasing Hormone (GnRH). Start of study treatment is allowed in shorter timeframes provided 5 half-lives of the prior agent(s) have elapsed.
- [0425] Previous PSMA-targeted radiopharmaceutical treatment except:
- [0426] a. Participants who received only a micro-dose of ^{212}Pb -Compound 1 in a Phase 0 study.
- [0427] b. Participants in Dose Expansion Group B must have had prior treatment with ^{177}Lu -PSMA; this must have ended at least 6 weeks before the start of study treatment.
- [0428] Prior treatment with any other systemic radiopharmaceutical therapy within 6 months before the start of study treatment (e.g., agents based on ^{177}Lu , ^{225}Ac , ^{223}Ra , etc.).
- [0429] Prior definitive radiotherapy completed less than 6 weeks before the start of study treatment; palliative radiotherapy within this time is allowed provided (i) no more than 10% of the participants' bone marrow is irradiated, and (ii) it does not encompass all potential target/measurable lesions for participants in dose expansion.
- [0430] Prior external beam radiotherapy (EBRT) where dose to a critical organ (e.g., kidneys, heart, lung) has exceeded the EBRT normal organ tolerance dose limit.
- [0431] Hemi-body irradiation within 6 months of the start of study treatment.
- [0432] Previous high-dose chemotherapy needing hematopoietic-stem-cell-rescue.
- [0433] Previous autologous or allogeneic stem-cell transplantation.
- [0434] Ongoing toxicity CTCAE Grade >2 due to prior anti-cancer therapy that is significant or not stabilized (including Grade >2 dry mouth [i.e. xerostomia]). Chronic toxic effects Grade <2 where no further resolution is expected do not require exclusion (e.g., chemotherapy-induced neuropathy, fatigue, alopecia, anorexia, etc.).
- [0435] Live vaccines within 4 weeks before the start of study treatment; vaccines for COVID-19 are allowed.
- [0436] Biological response modifiers, such as granulocyte colony-stimulating factor (G-CSF), granulocyte macrophage colony-stimulating factor (GM-CSF) and erythropoietin within 3 weeks before start of study treatment. Chronic use of erythropoietin is allowed, provided there is no dose modification within 3 weeks before the start of study treatment.
- [0437] Systemic corticosteroids >10 mg of prednisone/prednisolone per day or herbal products that may decrease PSA levels (e.g., saw palmetto) within 4 weeks before the start of study treatment.

Example 13: Experimental Metastasis Models of Prostate Cancer

[0438] Experimental metastasis models of prostate cancer involve injecting prostate cancer cells directly into the blood stream of a model animal, such as mice, to study the ability of cancer cells to grow into distant organs. Intravenous, intracardiac, intraileiac artery, intrasplenic, intratibial, intraperitoneal, or intra-bone marrow injection of prostate cancer cells may lead to establishment of metastasis in distant organs. Exemplary prostate cancer cells used in experimental metastasis models include LNCaP, PC3 (including PSMA transfected PC3-PIP cells), DU145, VCaP, MDA PCa 2b, LAPC4, and 22Rv1. Intracardiac injection allows for dissemination of cancer cells and is commonly used to model bone, liver, lung, and lymph node metastasis. As an example, metastatic prostate cancer may be established in mice via intracardiac injection of 10^6 - 10^7 prostate cancer cells, such as PC3 or PC3-PIP. After some time, the mice are intravenously injected with a ^{212}Pb radiopharmaceutical conjugate disclosed herein, such as ^{212}Pb -Compound 1, and sacrificed up to 150 days later. It is anticipated that untreated mice will develop distant organ metastases in bone, viscera (e.g., lungs, liver, heart, etc.), lymph node(s), or a combination thereof. The location of metastatic lesions may be detected using any appropriate method known in the art, such as bioluminescent imaging. It is expected that treatment with the ^{212}Pb radiopharmaceutical conjugate disclosed herein, such as ^{212}Pb -Compound 1, will prevent or reduce the formation of distant metastases, and in particular prevent or reduce bone lesions. Further, it is expected that the ^{212}Pb radiopharmaceutical conjugate disclosed herein, such as ^{212}Pb -Compound 1, will prevent or reduce the formation of said metastases to a greater extent than non- ^{212}Pb radiopharmaceuticals, such as ^{177}Lu -PSMA-617.

We claim:

1. A method of treating a prostate cancer in a human subject in need thereof, comprising administering to the human subject a radiopharmaceutical conjugate, wherein the radiopharmaceutical conjugate comprises



or a pharmaceutically acceptable salt thereof,

wherein the radiopharmaceutical conjugate is administered to the subject about once every week, about once every 2 weeks, about once every 3 weeks, about once every 4 weeks, about once every 5 weeks, or about once every 6 weeks.

2. The method of claim 1, wherein the radiopharmaceutical conjugate is administered to the subject about once every 3 weeks.

3. The method of claim 1, wherein the radiopharmaceutical conjugate is administered to the subject about once every 4 weeks.

4. The method of claim 1, wherein the radiopharmaceutical conjugate is administered to the subject about once every 5 weeks.

5. The method of claim 1, wherein the radiopharmaceutical conjugate is administered 2 times (i.e., for 2 cycles), 3 times, (i.e., for 3 cycles), 4 times, (i.e., 4 cycles), 5 times (i.e., 5 cycles), 6 times (i.e., 6 cycles), 7 times (i.e., 7 cycles), 8 times (i.e., 8 cycles), 9 times (i.e., 9 cycles), 10 times (i.e., 10 cycles), 11 times (i.e., 11 cycles), or 12 times (i.e., 12 cycles).

6. The method of claim 1, wherein the radiopharmaceutical conjugate is administered 4 times, 5 times, or 6 times.

7. The method of claim 1, wherein the radiopharmaceutical conjugate is administered in an amount between 50 kBq to 500 MBq per dose.

8. The method of claim 1, wherein the radiopharmaceutical conjugate is administered in an amount of about 75 MBq per dose to about 300 MBq per dose.

9. A method of preventing or reducing the formation of a metastatic bone lesion in a human subject in need thereof, comprising administering to the subject a radiopharmaceutical composition comprising

(a) a compound of Formula (I):



or a pharmaceutically acceptable salt thereof, wherein:

TL is a PSMA targeting ligand comprising a urea;

L is a bivalent linking moiety; and

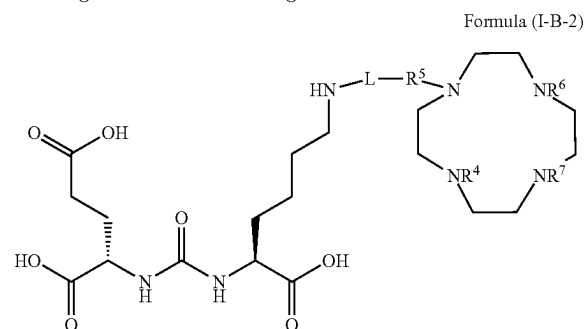
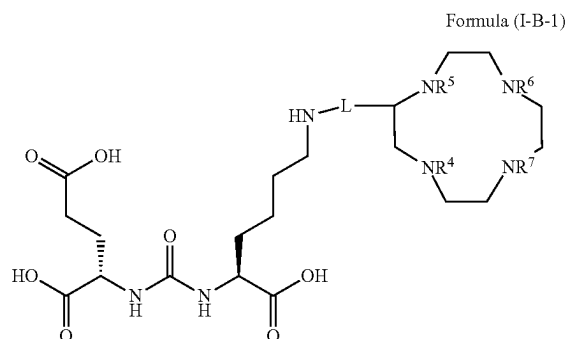
R^M is a metal chelator; and

(b) a radionuclide bound to the metal chelator, wherein the radionuclide is ^{212}Pb ;

wherein the human subject has prostate cancer; and

wherein the radiopharmaceutical conjugate is administered to the human subject in an amount between 50 kBq to 500 MBq per dose.

10. The method of claim 9, wherein the radiopharmaceutical conjugate comprises a structure of Formula (I-B-1) or Formula (I-B-2):



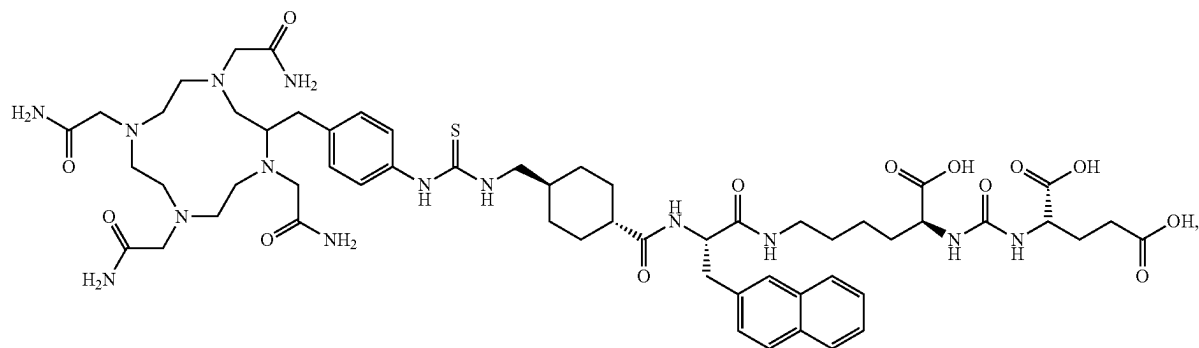
or a pharmaceutically acceptable salt thereof, wherein:

R^4 , R^5 , R^6 , and R^7 are each independently selected from the group consisting of $-\text{C}_{1-3}\text{alkylene}-\text{C}(=\text{O})\text{OR}^3$, and $-\text{C}_{1-3}\text{alkylene}-\text{C}(=\text{O})\text{N}(\text{R}^3)_2$, wherein each $\text{C}_{1-3}\text{alkylene}$ is optionally substituted with $-\text{C}(=\text{O})\text{OH}$; and

each R^3 is independently selected from the group consisting of H and $\text{C}_{1-6}\text{alkyl}$;

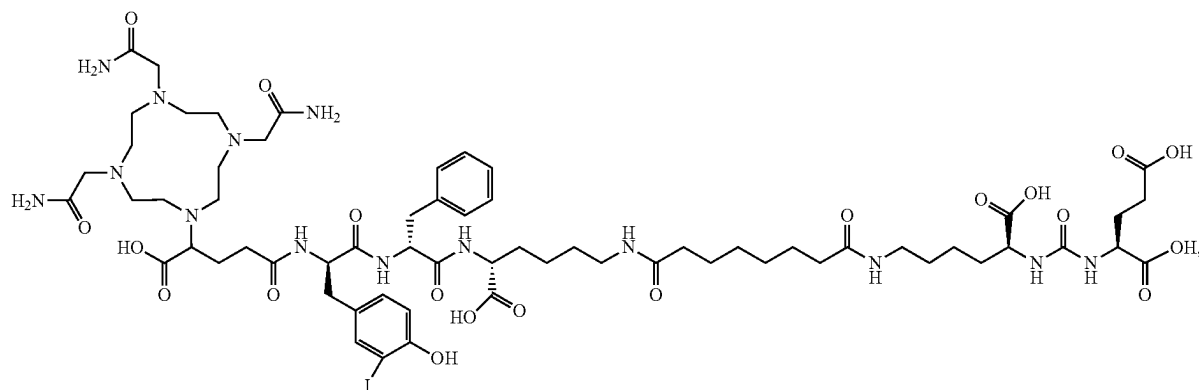
provided that R^5 in Formula (I-B-2) is $-\text{C}_{1-3}\text{alkylene}-\text{C}(=\text{O})-$, wherein the $\text{C}_{1-3}\text{alkylene}$ is optionally substituted with $-\text{C}(=\text{O})\text{OH}$.

11. The method of claim 9, wherein the compound of Formula (I) is



or a pharmaceutically acceptable salt thereof.

12. The method of claim 9, wherein the compound of Formula (I) is



or a pharmaceutically acceptable salt thereof.

13. The method of claim 9, wherein the prostate cancer is local and administering the radiopharmaceutical conjugate to the subject prevents or reduces skeletal metastases.

14. The method of claim 9, wherein the prostate cancer is metastatic, but does not comprise bone lesions, and administering the radiopharmaceutical conjugate to the subject prevents or reduces further skeletal metastases.

15. The method of claim 9, wherein the radiopharmaceutical conjugate is administered in an amount of about 75 MBq per dose to about 300 MBq per dose.

16. The method of claim 9, wherein the radiopharmaceutical conjugate is administered in an amount of about 75 MBq per dose, about 100 MBq per dose, about 150 MBq per dose, about 200 MBq per dose, about 250 MBq per dose, or about 300 MBq per dose.

17. The method of claim 9, wherein the radiopharmaceutical conjugate is administered about once every week, about once every 2 weeks, about once every 3 weeks, about once every 4 weeks, about once every 5 weeks, or about once every 6 weeks.

18. The method of claim 9, wherein the radiopharmaceutical conjugate is administered 2 times (i.e., for 2 cycles), 3 times, (i.e., for 3 cycles), 4 times, (i.e., 4 cycles), 5 times

(i.e., 5 cycles), 6 times (i.e., 6 cycles), 7 times (i.e., 7 cycles), 8 times (i.e., 8 cycles), 9 times (i.e., 9 cycles), 10 times (i.e., 10 cycles), 11 times (i.e., 11 cycles), or 12 times (i.e., 12 cycles).

19. A method of treating a prostate cancer in a human subject in need thereof, comprising:

administering to the human subject a radiopharmaceutical conjugate, wherein the radiopharmaceutical conjugate comprises,

(a) a compound of Formula (I):

TL-L-R^M

Formula (I)

or a pharmaceutically acceptable salt thereof, wherein:

TL is a PSMA targeting ligand comprising a urea;

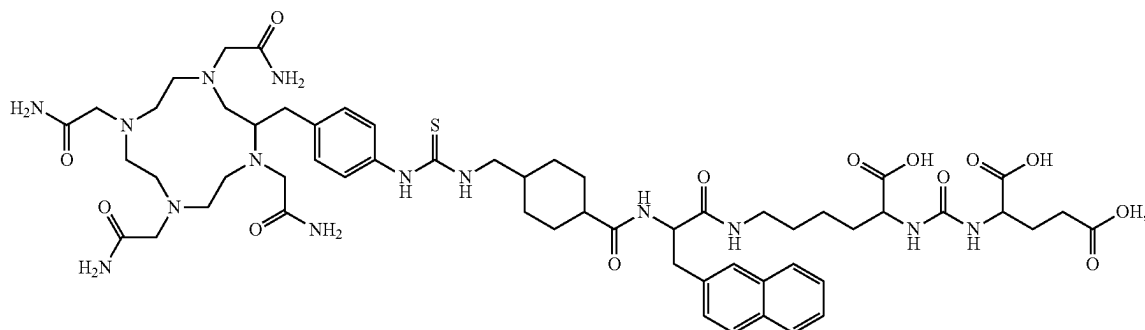
L is a bivalent linking moiety; and

R^M is a metal chelator; and

(b) a radionuclide bound to the metal chelator, wherein the radionuclide is ²¹²Pb; and

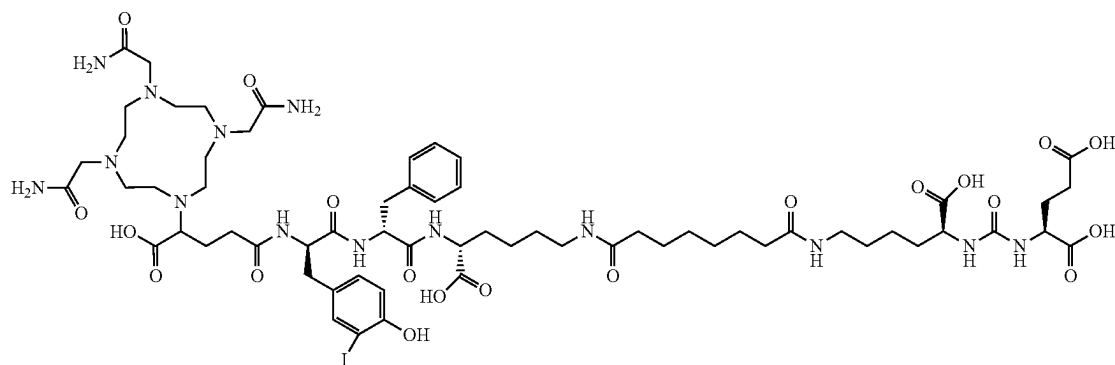
wherein the radiopharmaceutical conjugate is administered in combination with folate.

20. The method of claim 19, wherein the compound has the structure of



or a pharmaceutically acceptable salt thereof.

21. The method of claim 19, wherein the compound has the structure of



or a pharmaceutically acceptable salt thereof.

22. The method of claim 19, wherein the folate is administered prior to, or concurrently with, the radiopharmaceutical conjugate.

23. The method of claim 19, wherein the folate is administered concurrently with the radiopharmaceutical conjugate.

24. The method of claim 19, wherein the folate is administered up to 12 hours prior to administering the radiopharmaceutical conjugate.

25. The method of claim 19, wherein the folate is administered in an amount between 100 µg and 25 mg.

26. The method of claim 19, wherein the folate is folic acid, dihydrofolate (DHF), tetrahydrofolate (THF), 5, 10-methylenetetrahydrofolate (5, 10-MTHF), or 5-methyltetrahydrofolate (5-MTHF).

27. The method of claim 19, wherein the radiopharmaceutical conjugate is administered in an amount between 50 kBq to 500 MBq per dose.

28. The method of claim 19, wherein the radiopharmaceutical conjugate is administered in an amount of about 75 MBq per dose to about 300 MBq per dose.

29. The method of claim 19, wherein the radiopharmaceutical conjugate is administered about once every week, about once every 2 weeks, about once every 3 weeks, about once every 4 weeks, about once every 5 weeks, or about once every 6 weeks.

30. The method of claim 19, wherein the radiopharmaceutical conjugate is administered 2 times (i.e., for 2 cycles), 3 times (i.e., for 3 cycles), 4 times (i.e., 4 cycles), 5 times (i.e., 5 cycles), 6 times (i.e., 6 cycles), 7 times (i.e., 7 cycles), 8 times (i.e., 8 cycles), 9 times (i.e., 9 cycles), 10 times (i.e., 10 cycles), 11 times (i.e., 11 cycles), or 12 times (i.e., 12 cycles).

* * * * *